

LLM Sensitivity of Generative Agents: Evidence from an Epidemic Model

Ross Williams

Industrial and Systems Engineering, Virginia Tech; Falls Church, VA 22043, USA.

Email: rossfw@vt.edu

Abstract

Generative agent-based modeling (GABM) replaces the rule-based decision-making of classical agent-based models with LLM-mediated decisions, enabling researchers to simulate human behavior that resists mathematical specification. The original epidemic GABM and its follow-on prompt-sensitivity study both used GPT-3.5, leaving the generalizability of GABM findings to other LLMs untested. We address this with a frozen-task probe study: 100 personas, 40 community-infection levels, and 5 repetitions across 21 LLM configurations from OpenAI, Anthropic, and Google, totaling 420,000 binary stay-home decisions during a simulated epidemic. Most LLMs agree on the direction of persona effects, but magnitudes vary several-fold by provider (Gemini » Anthropic » OpenAI). The model attributes a researcher would typically use to choose an LLM (size, generation, training cutoff, release date) give no insight into where any given configuration’s behavior lands. The reasoning text accompanying each decision encodes substantive predictive signal beyond the decision itself, and reveals an other-focused-vs-self-focused framing of the stay-home choice. We argue that LLM choice is itself a parameterization of agent behavior: GABM with general-purpose LLMs is most defensible for qualitative claims about agent behavior and weakest for the quantitative claims modelers most often want to make. LLM sensitivity should become a routine evidentiary expectation in GABM publications.

One-Sentence Summary: Across 21 LLM configurations from three providers, generative agents agree on the direction of persona effects but diverge on magnitude, making the LLM itself a parameter of any GABM result.

Keywords: Generative Artificial Intelligence, Large Language Models, Agent Based Model, Generative Agent Based Model, Computational Social Science, Sensitivity Analysis, LLM-based Agents, Epidemic Modeling

1. Introduction

Generative agent-based modeling (GABM) was introduced as a paradigm for incorporating human behavior into simulations of complex social systems [1]. Where a traditional agent-based model relies on the modeler to write decision rules and tune their parameters, a GABM connects each agent to a large language model (LLM) and lets the LLM decide. This sidesteps a long-standing barrier: human behavior in many domains resists clean mathematical specification, and a modeler’s hand-coded rules typically capture only a thin slice of what real people consider when making a decision. The original GABM application showed that agents driven this way produce realistic individual decisions and emergent population-level dynamics in an epidemic context, including multiple infection waves and an endemic period [1]. The original epidemic GABM [1] used GPT-3.5, as did the follow-on prompt-sensitivity study [2]. Both papers relied on a single LLM, leaving the generalizability of their findings to other LLMs untested. Since those publications, the space of available LLMs has expanded dramatically: dozens of models span three major providers (OpenAI, Anthropic, Google), multiple generations within each provider, distinct training-data eras, and configurable parameters such as reasoning level. Therefore, characterizing whether GABM findings generalize across LLMs is especially important now, as each LLM configuration is a plausible decision engine for a generative agent.

The answer determines what GABM can credibly claim. If LLM choice does not materially shape simulated behavior, LLMs are interchangeable and GABM as currently practiced is a robust technique. If LLM choice does shape simulated behavior, the LLM itself becomes a parameter of the model: any GABM result is implicitly conditional on which LLM was used, complicating the method’s claim that generative-agent decision-making removes behavioral parameterization.

We address the question with a frozen-task probe study. 100 personas (covering Big Five trait combinations and a population-clock-weighted age distribution), 40 community-infection levels (0% to 7%), and 5 independent repetitions are held constant across 21 LLM configurations from three providers, yielding 420,000 binary decisions. The 21 configurations span six lenses for comparison across LLMs: provider, reasoning level, model size within a family, model evolution across releases, knowledge cutoff, and release date. Each response includes a free-text reasoning passage alongside the binary decision, allowing both quantitative comparison of decision distributions and content analysis of how the agents reason. We analyze the responses through three progressively deeper analyses: aggregate mobility curves at the configuration level, persona-level random-effects logistic regressions that test how each persona attribute shapes the binary decision, and content analysis of the reasoning passages.

The investigation is organized around fourteen research questions, grouped by the three analyses. Table 2 lists each question alongside the section in which its results are presented and the figures that bear on it.

Table 2. The fourteen research questions investigated in this paper, grouped by the three analyses described in §3.7. Each row lists the section in which the question is answered and the paper figures that bear on it.

RQ	Research question	Results	Figures
<i>Mobility Curve Analysis</i> — <i>does the choice of LLM affect aggregate mobility behavior?</i>			
1	Does LLM provider affect mobility curves?	§4.1.1	Figs 3, 4
2	Does LLM reasoning level affect mobility curves?	§4.1.2	Fig 5
3	Does LLM size affect mobility curves?	§4.1.3	Fig 6
4	Does LLM evolution affect mobility curves?	§4.1.4	Fig 7
5	Does LLM knowledge cutoff affect mobility curves?	§4.1.5	Figs 8, 9
6	Does LLM release date affect mobility curves?	§4.1.6	Figs 10, 11
<i>Persona Analysis</i> — <i>how does each LLM weigh agent attributes, and do LLMs agree on what matters?</i>			
7	What is the direction of each persona predictor’s effect on the stay-home decision?	§4.2.1	Fig 12
8	Which persona predictors most strongly drive the stay-home decision?	§4.2.2	Fig 13
9	Are personality effects larger than infection effects?	§4.2.3	Fig 14
10	Do LLMs agree on which agent personas are most cautious?	§4.2.4	Fig 15
<i>Response Analysis</i> — <i>what does the reasoning passage reveal, and how heterogeneous are responses?</i>			
11	How often do LLMs mention trait, infection, and age concepts in their reasoning?	§4.3.1	Fig 16
12	When a trait is mentioned, does the mention amplify or contradict the trait’s behavioral effect?	§4.3.2	Fig 17
13	Which concept categories are cited in stay-home vs. go-out reasoning?	§4.3.3	Figs 18, 19
14	How agent-specific is each LLM’s reasoning text?	§4.3.4	Figs 20, 21

Across the 21 configurations, most LLMs agree on the direction of persona effects but disagree several-fold on magnitude. None of the model attributes a researcher would typically use to choose an LLM (size, generation, training cutoff, release date) gives any insight into where the configuration’s behavior lands. The reasoning text encodes substantive predictive signal beyond the decision itself. We argue these findings reframe what GABM with general-purpose LLMs can credibly claim: the method is well-suited to qualitative claims about agent behavior, but the LLM choice itself functions as a parameterization for any quantitative claim.

3. Methods

3.1 Simulation setup

The probe is a simplified extension of the generative agent-based model of Williams et al. [1]. Two simplifications make each response cleanly attributable to the agent’s persona and the perceived infection level. First, each query is treated as an independent, one-shot decision: where Williams et al. [1] simulated disease spread within an agent population over many days, the present probe samples each agent and infection level directly, with no inter-agent contagion. Second, agent health is held constant — each agent is told “You feel healthy and well.” — rather than evolving with a disease state, so behavioral variation reflects persona and the perceived community-infection rate alone. The town setting (Dewberry Hollow), occupational role, and disease (the fictional *Catasat* virus) carry over from Williams et al. [1].

3.2 Agent population

The probe uses 100 frozen personas generated by the procedure of Williams et al. [1]. The five Big Five trait labels are each sampled independently with 50/50 probability between the dimension’s two opposing labels: extraverted or introverted (Extraversion); agreeable or antagonistic (Agreeableness); conscientious or unconscientious (Conscientiousness); emotionally stable or neurotic (Neuroticism); and open to experience or closed to experience (Openness). Ages are drawn between 18 and 65 from a probability distribution that mirrors the 2023 U.S. working-age population [2]. Names and gender are paired by sampling without replacement from the top male and female U.S. names, balanced to produce 50 female and 50 male agents. The full procedure is run with a fixed random seed (42) for reproducibility. Sample demographics are summarized in Figure 1.

Agent Characteristics

Gender



Extraversion



Agreeableness



Conscientiousness



Emotional Stability



Openness



Age Distribution

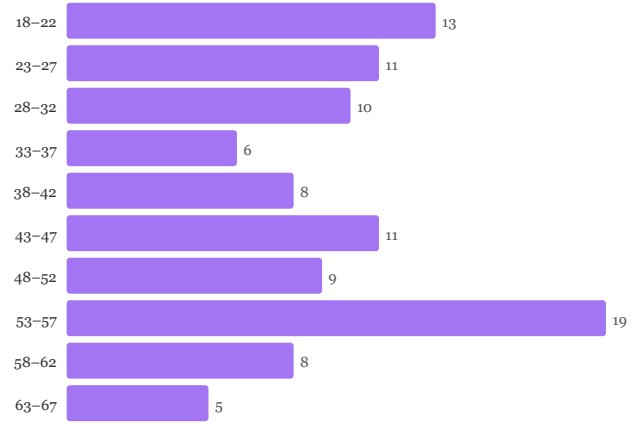


Figure 1. Sample demographics of the 100 frozen personas: age distribution, gender split, and the frequency of each Big Five trait label.

3.3 Probe design

Each agent is queried at 40 infection levels with 5 repetitions for each combination of agent and infection level. Infection levels are sampled densely in the low range, with 0.1 percentage-point steps from 0% to 3.5%, then larger steps at 4%, 5%, 6%, and 7%. Each LLM configuration therefore produces $100 \times 40 \times 5 = 20,000$ responses; across 21 configurations the full corpus is 420,000 responses.

3.4 LLM configurations

The 21 configurations were chosen to span six lenses for comparison across LLMs: reasoning level, model size within a family, model evolution within a family across time, provider, knowledge cutoff, and release date. To support the reasoning-level lens, GPT-5.2 and Gemini 3 Flash Preview were each run at four reasoning levels (off, low, medium, high), with each level treated as a separate configuration; all other models were run with reasoning off (or, for OpenAI’s o3, with reasoning required by the API). Data collection across the 21 configurations cost approximately \$840 in total API charges (~\$464 OpenAI, ~\$263 Anthropic, ~\$113 Google). Across providers, the configurations span 9 from OpenAI, 5 from Anthropic, and 7 from Google; the full list is given in Table 1.

Sampling temperature is set to 0 for non-reasoning configurations where the API allows it; reasoning-capable models use a temperature of 1 (required by Anthropic; recommended by Google). The OpenAI o-series accepts no temperature parameter.

Table 1. The 21 LLM configurations used in this study. Each model with multiple reasoning levels contributes one configuration per level. Pinned model versions, pricing, and additional metadata are listed in Appendix [TBD].

Provider	Model	Reasoning levels	Release date	Knowledge cutoff
OpenAI	GPT-3.5 Turbo	off	Jan 2024	Sep 2021
OpenAI	GPT-4o	off	Nov 2024	Oct 2023
OpenAI	GPT-4.1	off	Apr 2025	Jun 2024
OpenAI	GPT-5.1	off	Nov 2025	Sep 2024
OpenAI	GPT-5.2	off, low, medium, high	Dec 2025	Aug 2025
OpenAI	o3	required	Apr 2025	Jun 2024
Anthropic	Claude 3 Haiku	off	Mar 2024	Aug 2023
Anthropic	Claude Sonnet 4.0	off	May 2025	Mar 2025
Anthropic	Claude Sonnet 4.5	off	Sep 2025	Jan 2025
Anthropic	Claude Haiku 4.5	off	Oct 2025	Feb 2025
Anthropic	Claude Opus 4.5	off	Nov 2025	Aug 2025
Google	Gemini 2.0 Flash	off	Feb 2025	Jun 2024
Google	Gemini 2.5 Flash Lite	off	Jun 2025	Jan 2025
Google	Gemini 2.5 Flash	off	May 2025	Jan 2025
Google	Gemini 3 Flash Preview	off, low, medium, high	Dec 2025	Jan 2025

3.5 Prompt template

Every query uses the same template, identical across the 21 configurations. Each query substitutes four variables into the template: the agent’s first name, the agent’s age in years, the agent’s comma-separated Big Five trait list, and the day’s perceived community-infection rate. The full template is Figure 2.


```
You are {name}. You are {age} years old.  
You are a person who is {traits_str}.  
You live in the town of Dewberry Hollow. You have a job and go to the office  
for work every day.  
You feel healthy and well.  
You know about the Catasat virus spreading across the country. It is an  
airborne virus causing an infectious disease that spreads from human to  
human. The deadliness of the virus is unknown.  
You check the newspaper and find that {infection_pct}% of Dewberry Hollow's  
population were diagnosed with new infections of the Catasat virus yesterday.  
Should you stay at home for the entire day? Please provide your reasoning.
```

```
If the answer is "Yes", please state your reasoning as  
"Reasoning: [explanation]."  
If the answer is "No", please state your reasoning as  
"Reasoning: [explanation]."
```

```
The format should be as follows:  
Reasoning:  
Response:
```

```
Example:  
Reasoning: You are tired.  
Response: Yes
```

```
You have to provide Response in a single word; it must be either  
"Yes" or "No".
```

Figure 2. The prompt template. The placeholders in braces are filled per query: the name placeholder takes the agent’s first name, the age placeholder the agent’s age in years, the trait placeholder the agent’s comma-separated Big Five trait list (with “and” before the final item), and the infection-rate placeholder the day’s perceived community-infection rate. For agent Luis (age 54; introverted, agreeable, conscientious, emotionally stable, and closed to experience) at infection level 3.2%, the first two lines and the newspaper line read “You are Luis. You are 54 years old. You are a person who is introverted, agreeable, conscientious, emotionally stable, and closed to experience.” and “You check the newspaper and find that 3.2% of Dewberry Hollow’s population were diagnosed with new infections of the Catasat virus yesterday.”

3.6 Response collection and parsing

Each response is required to be a free-text reasoning passage followed by a single-word yes/no decision, in the format specified at the end of the prompt template (Figure 2). A parser extracts the two components and rejects any response that does not conform; 0 of 420,000 responses required rejection. The reasoning text is preserved for the downstream textual analysis.

3.7 Analytical methods

We perform three analyses on the 420,000 responses, each progressively deeper into the data: (1) Mobility Curve Analysis summarizes each LLM configuration’s aggregate behavior by plotting mobility against infection level and compares curves across configurations with multilinear regression; (2) Persona Analysis examines the agent-level predictors of the stay-home decision (Big Five trait labels, gender, age, and infection level) via two logistic regressions; and (3) Response Analysis investigates the content of the reasoning passage along three dimensions: which concepts the LLM invokes, via mention counts of curated keyword and concept lists; whether mentioning a concept amplifies its behavioral effect, via a third logistic regression that incorporates mention flags as predictors; and how agent-specific the reasoning is, via embedding-based cosine similarity.

3.7.1 Mobility Curve Analysis

For each LLM configuration, the stay-home rate at a given infection level is computed by averaging 500 binary decisions — 5 repetitions for each of the 100 agents. The mobility curve plots mobility (the complement of the stay-home rate, $\text{mobility} = 1 - \text{stay-home rate}$) on the y -axis against the 40 sampled infection levels (0 to 3.5% in 0.1 percentage-point steps, plus 4%, 5%, 6%, and 7%) on the x -axis. To discern differences between mobility curves, we employ multilinear regression in the same style as Williams et al. [3]. For each set of configurations being compared (e.g., the four GPT-5.2 reasoning levels), we fit two models. Both take mobility as the dependent variable and use the infection level, the infection level squared, and one dummy variable per non-baseline configuration as predictors; the dummy is set to 1 if the data is from the configuration being tested and 0 if it is from the baseline. The first model, a *baseline-shift* model, stops there; a statistically significant dummy coefficient indicates that the configuration produces a level shift in the mobility curve relative to the baseline. The second model, a *baseline-plus-slope-shift* model, additionally includes interactions between each dummy and the infection level; a statistically significant interaction indicates that the configuration’s slope of response to infection level differs from the baseline’s. Per-comparison tables for both models are reported in Appendix A.

3.7.2 Persona Analysis

To examine how agent traits, gender, age, and infection level systematically influence the stay-home decision, two logistic regressions are fit per LLM configuration on the 20,000 individual responses, in the same style as Williams et al. [1]. The dependent variable is a binary variable of the agent’s decision to stay home (equal to 1 if the agent stays home and 0 otherwise). The fixed-effect logit absorbs each agent’s average tendency to stay home through agent dummies, isolating how the decision responds to changes in infection level. It also provides a pseudo- R^2 for goodness-of-fit. The random-effect logit replaces the agent dummies with a single random intercept per agent, freeing up the model to estimate trait, gender, age, and infection effects. The reference categories are *female*, *introverted*, *antagonistic*, *unconscientious*, *neurotic*, and *closed to experience*; age is in raw years. Per-configuration coefficient tables for both specifications are reported in Appendix B.1; diagnostics — calibration, residual Q-Q, random-intercept normality, BLUPs versus predictors, and residual fit by infection and age — are in Appendix C.

3.7.3 Response Analysis

Each query asks the agent to provide its reasoning before stating its yes/no decision, so the reasoning text likely shapes the decision rather than merely justifying it after the fact. From this text we measure four things in turn: (1) how often LLMs mention specific traits in their reasoning, via keyword counting; (2) whether those mentions amplify the corresponding behavioral effect, via the mention-flag logit; (3) how broader concept categories correlate with the decision direction, via keyword counting; and (4) how agent-specific the reasoning text is, via embedding cosine similarity.

For each Big Five trait label, a keyword set covers the obvious word-form variants of each pole. For example, an extraversion mention is flagged when any of {extroverted, extrovert, extraverted, extravert, extraversion, extroversion, introverted, introvert, introversion} appears in the response. Two additional flags cover infection (keywords like {infection, infected, cases, diagnosed} or a numeric percentage pattern) and age (keywords like {years old, young, age, old}). All flags are binary per response and case-insensitive.

These mention flags then enter a third logistic regression — the *mention-flag logit* — which extends the random-effects logit from §3.7.2 with one binary indicator per Big Five trait label, plus one for infection and one for age. A flag enters the model only when its overall mention rate falls within [15%, 85%]; outside this window the flag carries no usable variation. The mention-flag coefficients answer whether mentioning a concept amplifies or dampens its behavioral effect. Per-configuration coefficient tables are in Appendix B.2; diagnostics are in Appendix C.

A separate, broader concept dictionary covers five themes: *Work*, *Community*, *Self Health*, *Virus Properties*, and *Virus Prevalence*. These categories were drawn from the prompt template (Figure 2), which mentions going to the office for work, feeling healthy and well, an airborne virus that spreads from human to human, a daily new-infection rate, and the town of Dewberry Hollow. Keywords for each concept were then selected via manual inspection of random response samples across all 21 configurations. Like the trait flags, concept-category flags are binary per response and case-insensitive.

Lastly, for agent individuation, every reasoning passage is embedded with OpenAI’s text-embedding-3-large model (3,072 dimensions). Two questions follow, both restricted to agents whose 5 repetitions all returned the same decision (the “5/5 agreement” subset). The first: how similar are an agent’s own 5 responses? *Within-agent* similarity is computed as the mean of the 10 pairwise cosines among an agent’s 5 reps, then averaged across all qualifying 5/5 agreements for each LLM. The second: how similar are two different agents who reach the same decision at the same infection level? *Across-agent* similarity is computed bucket by bucket (one bucket per infection level and decision direction): every pair of qualifying agents in the bucket contributes the mean of the 25 rep-pair cosines between them, and the result is averaged across all such pairs.

4. Results

This section presents the results of the three analyses introduced in §3.7. Within each subsection, results are presented in the paper order described in Table 2.

4.1 Mobility Curve Results

The 21 mobility curves vary dramatically: average stay-home rates span nearly the full 0–100% range across the 21 configurations. The six subsections below examine variation across LLMs from six different angles. §4.1.1 examines the role of the LLM provider; §4.1.2 the role of reasoning level; §4.1.3 the role of model size within a provider family; §4.1.4 the role of model evolution within a family across time; §4.1.5 the role of knowledge cutoff; and §4.1.6 the role of release date.

4.1.1 Cross-provider differences (RQ1) — Does LLM provider affect mobility curves?

The mobility-curve landscape within each provider is itself wide. Figure 3 overlays all configurations from each of the three providers, showing the spread of mobility responses across the configurations included in this study. Within OpenAI alone, mobility ranges from near-100% stay-home (GPT-3.5 Turbo, GPT-5.2 off) to near-0% stay-home (GPT-4o); within Anthropic, from Claude 3 Haiku at the cautious end to Claude Haiku 4.5 at the mobile end; within Google, the spread is narrower but still spans the lower half of the mobility range. There is no single “Anthropic look” or “OpenAI look” to a mobility curve once different models within each provider are considered.

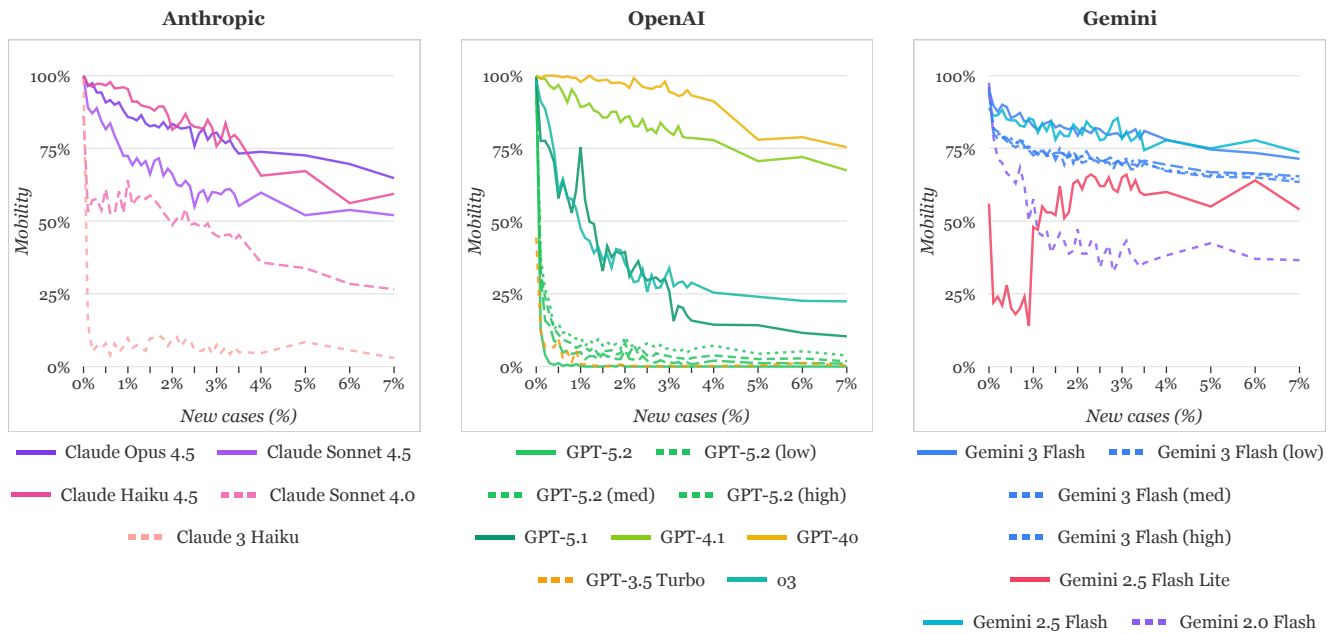


Figure 3. Within-provider model variation. All configurations are overlaid within each provider panel.

To make a controlled cross-provider comparison, we hold model size (flagship within each provider, as of the December 2025 data-collection window) and reasoning level (off) constant. Figure 4 shows the resulting flagship mobility curves: GPT-5.2 (OpenAI), Claude Opus 4.5 (Anthropic), and Gemini 3 Flash Preview (Google). All three start near 100% mobility at 0% infection, but the curves diverge sharply as soon as any community infection is reported. GPT-5.2 becomes cautious almost immediately — its mobility drops to near zero by 0.3% infection and stays there across the rest of the range. Claude Opus 4.5 and Gemini 3 Flash Preview, in contrast, behave as relatively mobile agents across the full range; their mobility hovers in the 75–90% band and declines only gradually as infection rises. The multilinear regression confirms this pattern: GPT-5.2 differs significantly from the other two flagships, while Claude Opus 4.5 and Gemini 3 Flash Preview are not statistically distinguishable from each other (Appendix A).

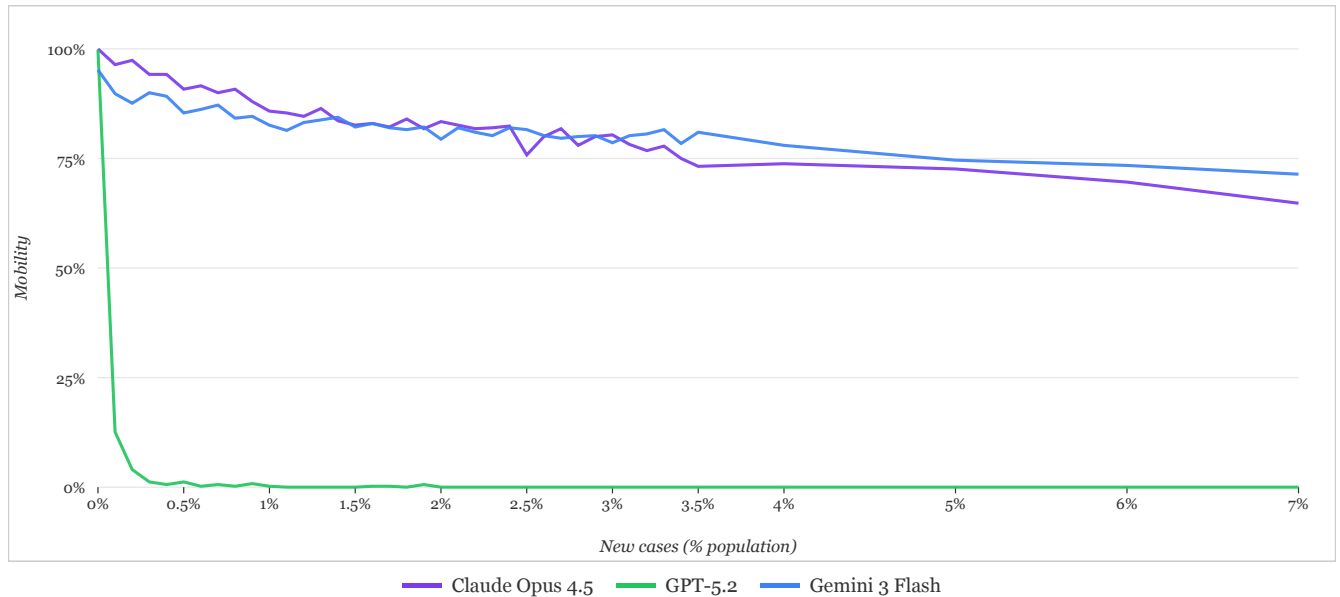


Figure 4. Cross-provider flagship comparison. One flagship model per provider, all evaluated with reasoning off: GPT-5.2 (OpenAI), Claude Opus 4.5 (Anthropic), Gemini 3 Flash Preview (Google).

The two figures together suggest that the choice of LLM matters along multiple dimensions. Within each provider, model variants span a wide mobility range; across providers, even matched flagships occupy distinct regions of the mobility space. For a researcher building a generative-agent simulation, neither “pick any provider” nor “pick the flagship” produces interchangeable behavior.

4.1.2 Reasoning level (RQ2) — Does LLM reasoning level affect mobility curves?

Reasoning level is a configurable LLM parameter controlling how much hidden deliberation the model performs before producing its final response (see §2.2). Two configurations span the full reasoning-level ladder (off, low, medium, high): GPT-5.2 (OpenAI) and Gemini 3 Flash Preview (Google). Figure 5 places these two reasoning ladders side by side, with each panel showing how the same model’s mobility curve responds to changes in reasoning level.

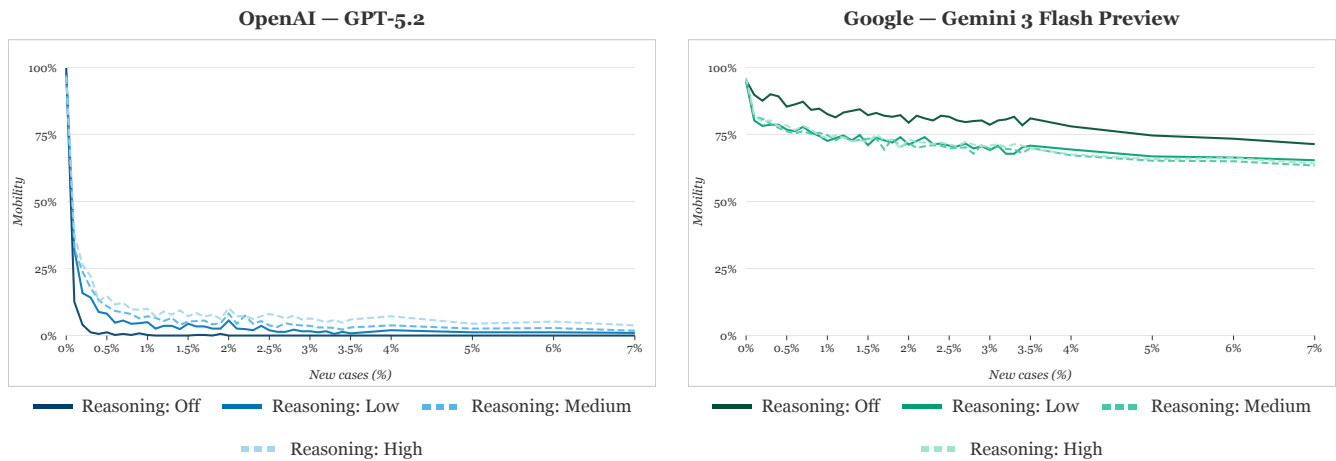


Figure 5. Effect of reasoning level on mobility, for the two configurations sampled across all four reasoning levels: GPT-5.2 (left) and Gemini 3 Flash Preview (right).

For GPT-5.2, increasing reasoning shifts the mobility curve modestly upward — mobility rises by roughly 9 percentage points from reasoning off to reasoning high. For Gemini 3 Flash Preview, the direction is opposite: enabling any non-zero reasoning level shifts the curve downward by roughly 9 percentage points relative to reasoning off, with little additional change between low, medium, and high. The multilinear regression in Appendix A reports which configurations differ significantly.

Reasoning is a real but minor lever, and its sign differs between the two LLMs tested. Both shifts are roughly 9 percentage points and small compared to the cross-provider gap reported in §4.1.1. The result complicates any general claim that more reasoning makes a generative agent uniformly more or less cautious.

4.1.3 Model size (RQ3) — Does LLM size affect mobility curves?

Model size refers to a provider’s relative scale tier within a single family of releases (see §2.2). Two providers in the data have multiple model sizes within a single generation: Anthropic’s Claude 4.5 family (Haiku [small], Sonnet [middle], Opus [large]) and Google’s Gemini 2.5 family (Flash Lite [small], Flash [middle]). All five configurations are evaluated with reasoning off. Figure 6 places the two size ladders side by side.

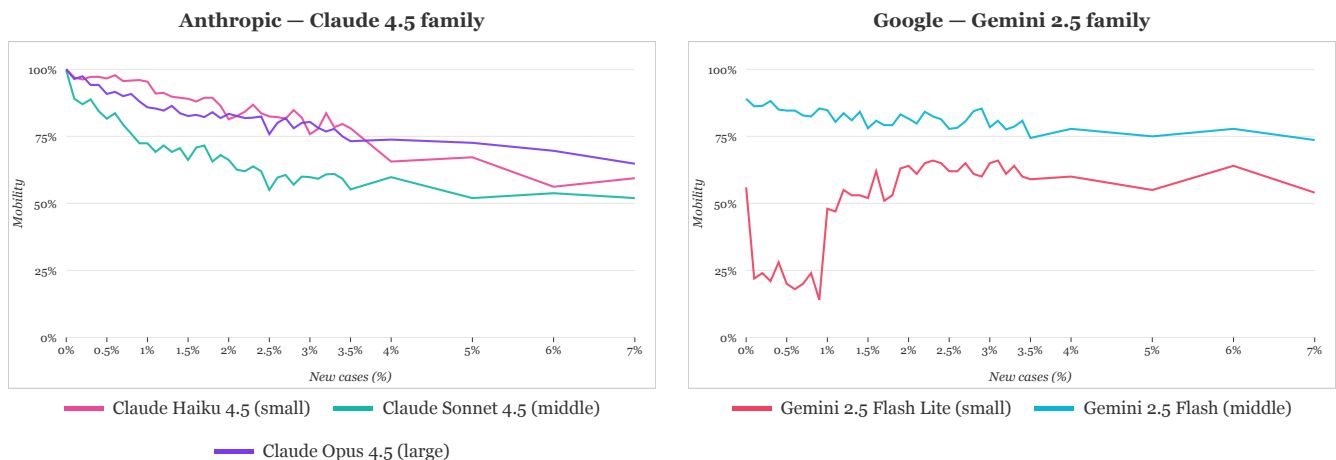


Figure 6. Effect of model size on mobility, for the two providers with multiple sizes in a single generation: Anthropic’s Claude 4.5 family (left) and Google’s Gemini 2.5 family (right). All configurations evaluated with reasoning off.

For the Anthropic Claude 4.5 family, mobility does not change monotonically with size. Going from Haiku (small) to Sonnet (middle) shifts the curve substantially downward — Sonnet sits well below Haiku across the full infection range. But going from Sonnet to Opus (large) reverses direction: Opus tracks close to Haiku rather than continuing further down. For the Google Gemini 2.5 family, Gemini 2.5 Flash sits well above Gemini 2.5 Flash Lite across most of the infection range. Flash Lite itself shows an unusual within-curve pattern — its mobility starts moderate at 0% infection, dips between roughly 0% and 1%, then recovers above 50% from 1.5% onward. The multilinear regression in Appendix A reports which configurations differ significantly.

Across both providers’ size ladders, the data shows substantial variation but no consistent direction of effect attributable to size.

4.1.4 Model evolution (RQ4) — Does LLM evolution affect mobility curves?

Model evolution refers to the sequence of major releases within a single provider’s family over time, where each successive release typically incorporates new training data, refined alignment, and architectural improvements (see §2.2). All three providers contribute model evolution data: Anthropic from Claude Sonnet 4.0 to Claude Sonnet 4.5; OpenAI from GPT-3.5 Turbo through GPT-4o, GPT-4.1, GPT-5.1, and GPT-5.2;

and Google from Gemini 2.0 Flash through Gemini 2.5 Flash to Gemini 3 Flash Preview. All configurations are evaluated with reasoning off. Figure 7 places the three evolution sequences side by side.

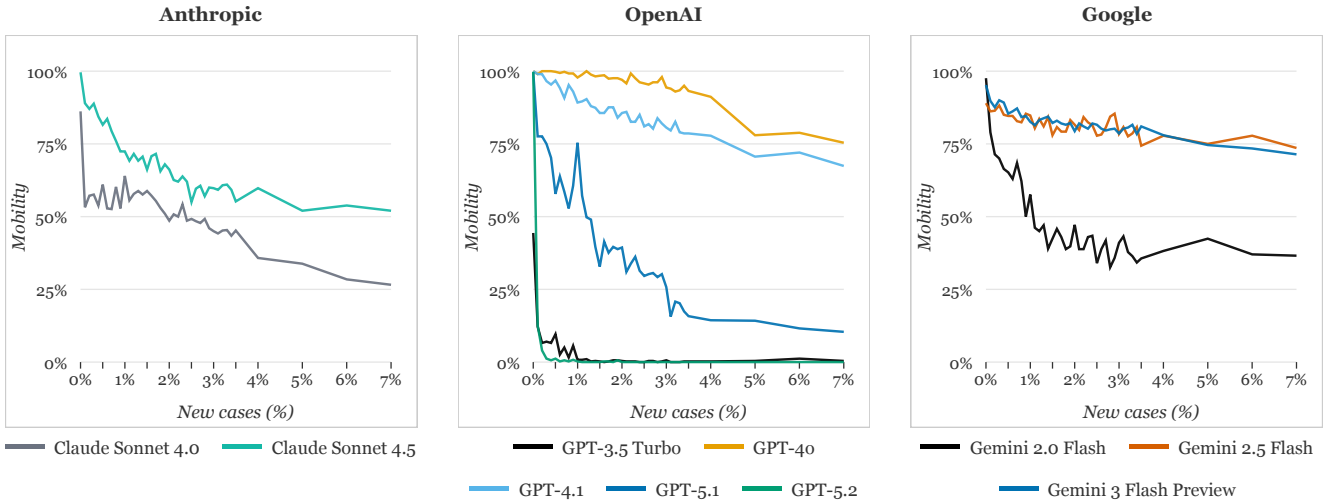


Figure 7. Effect of model evolution on mobility, one panel per provider: Anthropic, OpenAI, and Google. Each panel shows the evolution sequence within that provider, all at reasoning off.

The Anthropic sequence has only two points: Claude Sonnet 4.5 sits modestly above Claude Sonnet 4.0 across the infection range. The OpenAI sequence shows wide swings between successive releases — GPT-3.5 Turbo and GPT-5.2 sit near the cautious end of the mobility range; GPT-4.0 sits at the mobile end; GPT-4.1 and GPT-5.1 sit at intermediate positions. Going from one release to the next does not move the OpenAI curve in a consistent direction. The Google sequence shows a large upward shift from Gemini 2.0 Flash to Gemini 2.5 Flash, then almost no further change between Gemini 2.5 Flash and Gemini 3 Flash Preview. The multilinear regression in Appendix A reports which configurations differ significantly.

Across the three providers, model evolution does not produce a consistent direction of effect on mobility, and the within-provider patterns differ from each other.

4.1.5 Knowledge cutoff (RQ5) — Does LLM knowledge cutoff affect mobility curves?

Knowledge cutoff is the date through which the model’s training data extends; models with later cutoffs have been exposed to more recent text from the public internet (see §2.2). The 21 configurations span knowledge cutoffs from September 2021 (GPT-3.5 Turbo) to August 2025 (Claude Opus 4.5 and GPT-5.2). Figure 8 plots each configuration on a knowledge-cutoff timeline, showing the four era groupings used in the analysis below: pre-2024, mid-2024 (Jun–Sep), early 2025 (Jan–Mar), and late 2025 (Aug). Figure 9 then shows the mobility curves within each era.

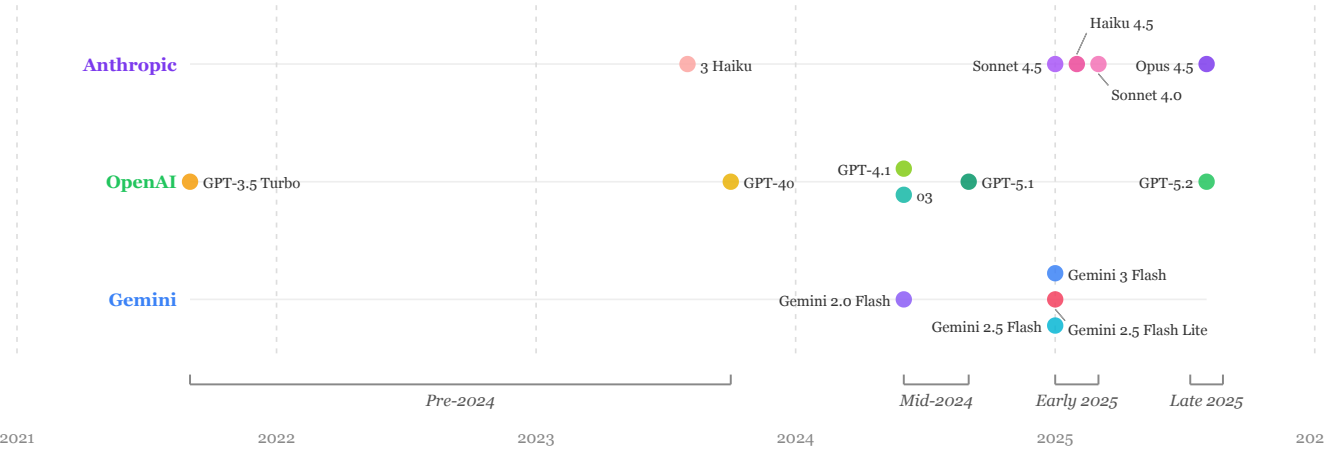


Figure 8. Knowledge-cutoff timeline. Each base configuration (reasoning = off; o3 at required) is plotted by knowledge cutoff date, grouped by provider; bracketed segments mark the four era groupings used in Figure 9. Reasoning ladders for GPT-5.2 and Gemini 3 Flash Preview are collapsed to a single point at their reasoning = off rung, since reasoning level is orthogonal to knowledge cutoff.

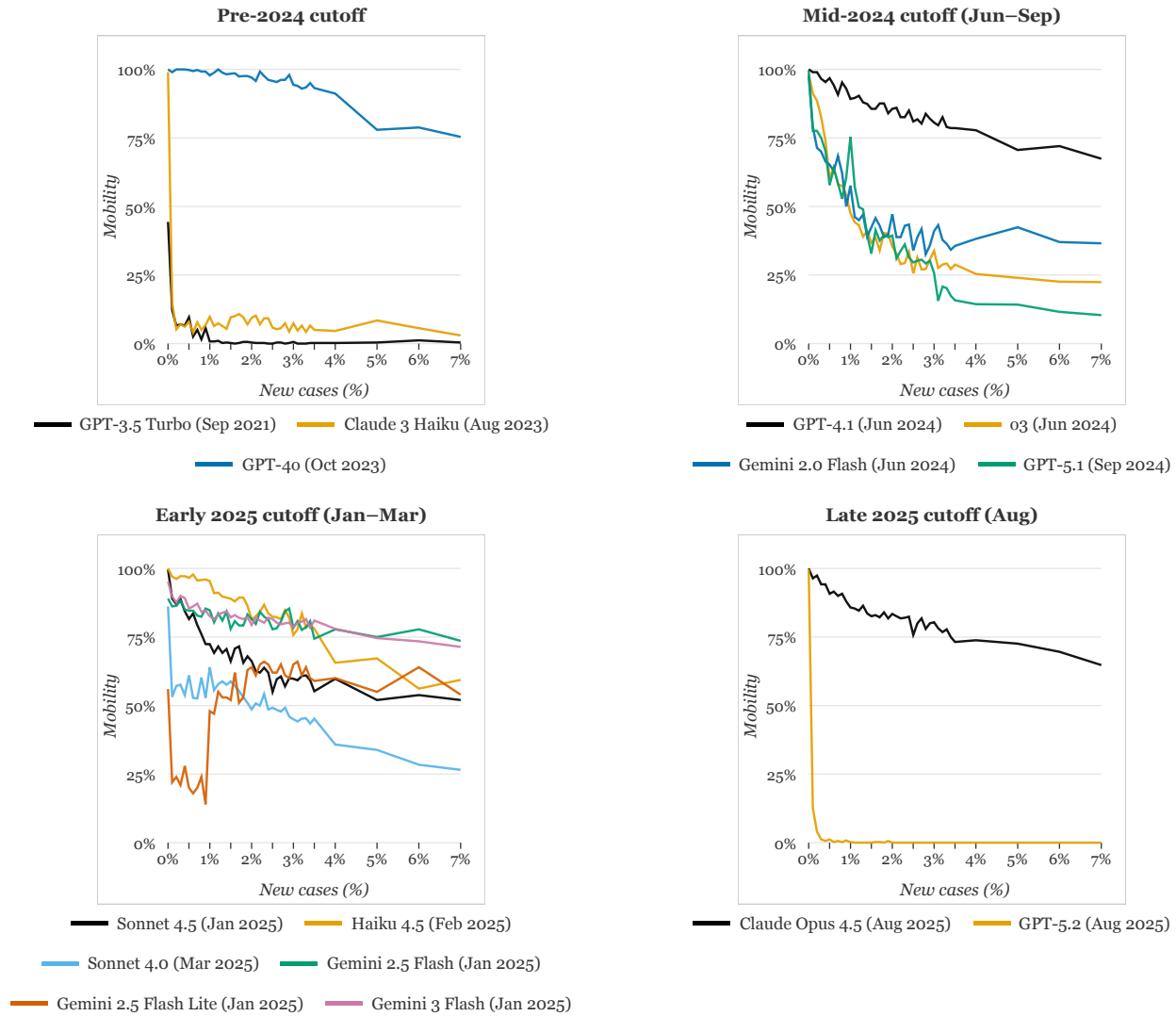


Figure 9. Mobility curves within each knowledge-cutoff era: pre-2024 (top left), mid-2024 (top right), early 2025 (bottom left), and late 2025 (bottom right). All configurations evaluated with reasoning off; o3 uses reasoning = required (its only available mode).

Within each era panel, the curves do not cluster by knowledge cutoff. Models with the same or near-identical training-data cutoff date land at very different positions on the mobility axis: in the pre-2024 panel, GPT-3.5 Turbo and GPT-4o sit at opposite ends of the mobility range despite cutoffs only two years apart; in the mid-2024 panel, four configurations with cutoffs in a 3-month window (Jun–Sep 2024) span nearly the full mobility range; in the early-2025 panel, six configurations with cutoffs within a 3-month window again span the full range; in the late-2025 panel, Claude Opus 4.5 and GPT-5.2 share an August 2025 cutoff but produce distinctly different curves. The multilinear regression in Appendix A reports which configurations differ significantly.

Knowledge cutoff does not predict mobility curve placement in this data. Whatever drives the variation across LLMs reported in §4.1.1, it is not the recency of training data alone.

4.1.6 Release date (RQ6) — Does LLM release date affect mobility curves?

Release date is the date the model was made publicly available via API; it is generally independent of the model’s knowledge cutoff (see §2.2). The 21 configurations span release dates from January 2024 (GPT-3.5 Turbo) to December 2025 (GPT-5.2 and Gemini 3 Flash Preview).

Figure 10 plots each configuration on a release-date timeline, showing the three era groupings used in the analysis below: legacy (pre-2025), early 2025 (Feb–Jun), and late 2025 (Sep–Dec). Figure 11 then shows the mobility curves within each era.

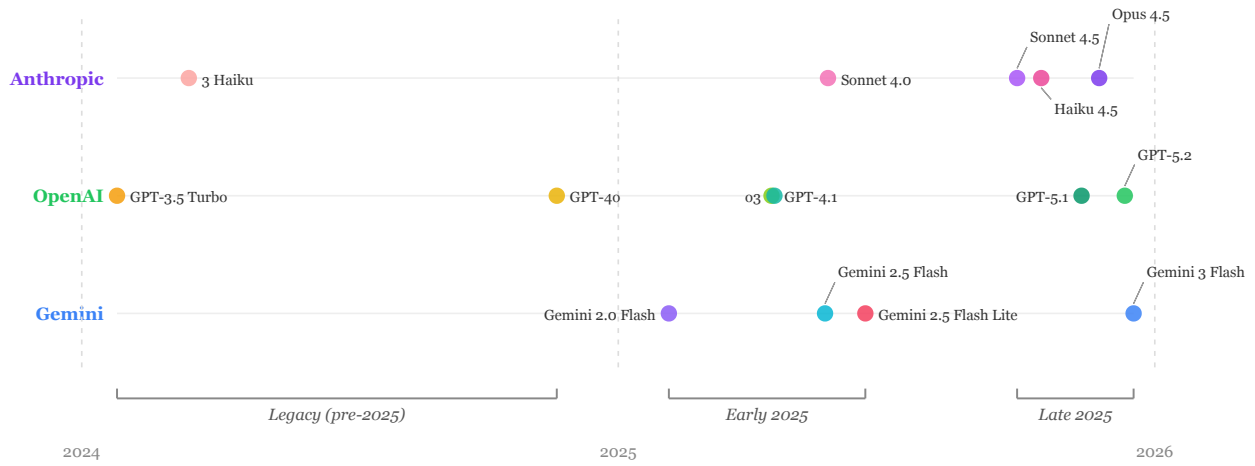


Figure 10. Release-date timeline. Each base configuration (reasoning = off; o3 at required) is plotted by API release date, grouped by provider; bracketed segments mark the three era groupings used in Figure 11. Reasoning ladders for GPT-5.2 and Gemini 3 Flash Preview are collapsed to a single point at their reasoning = off rung, since reasoning level is orthogonal to release date.

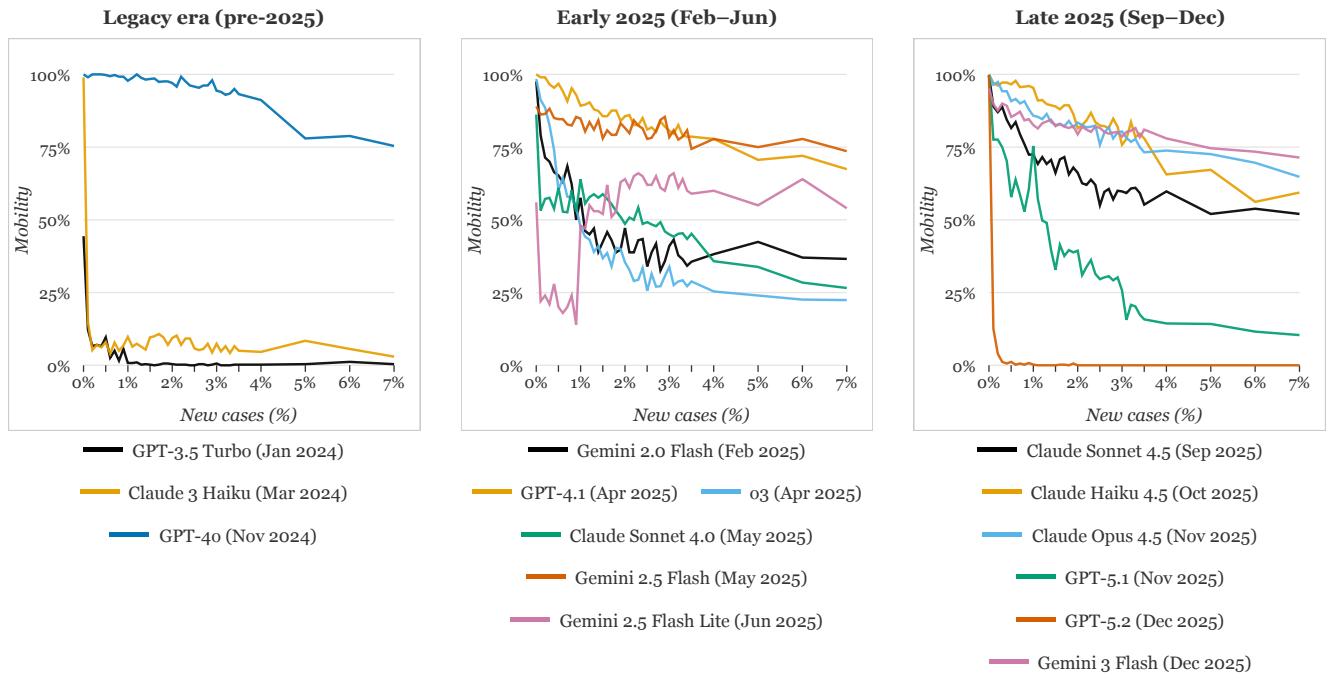


Figure 11. Mobility curves within each release-date era: legacy era / pre-2025 (left), early 2025 (middle), and late 2025 (right). All configurations evaluated with reasoning off; o3 uses reasoning = required (its only available mode).

As with knowledge cutoff, configurations sharing a release-date era do not cluster on the mobility axis. The legacy panel shows GPT-3.5 Turbo and GPT-4o at opposite ends of the range despite release dates within a year of each other. The early-2025 panel shows six configurations released within a five-month window spanning the full mobility range. The late-2025 panel shows six configurations released within a four-month window also spanning a wide range. The multilinear regression in Appendix A reports which configurations differ significantly. Release date does not predict mobility curve placement in this data, and does not explain the variation across LLMs observed in §4.1.1.

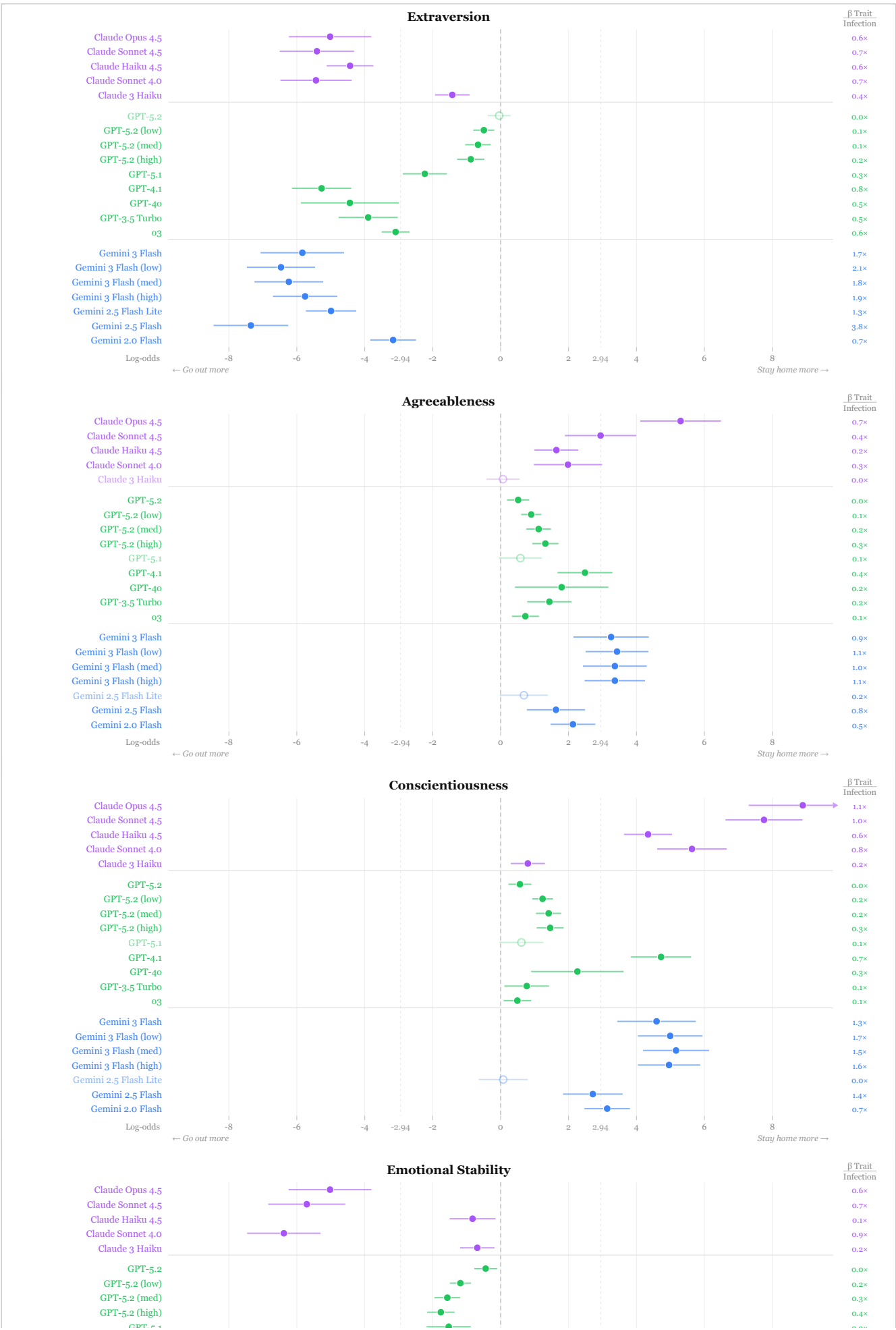
4.2 Persona Results

The four subsections below examine the agent-level predictors of the stay-home decision and how the 21 LLMs vary in their treatment of those predictors. §4.2.1 examines the direction of each persona predictor's effect (RQ7); §4.2.2 ranks persona predictors by their relative importance within and across LLMs (RQ8); §4.2.3 the magnitude of all persona effects combined relative to the infection-level effect (RQ9); and §4.2.4 agreement across LLMs on which agent personas are most cautious (RQ10).

4.2.1 Trait effect directions (RQ7) — What is the direction of each persona predictor's effect on the stay-home decision?

Agent traits enter the random-effects logit (§3.7.2) as predictors: five Big Five trait dummies, a male dummy, and age in years. Each Big Five dummy equals 1 for the named pole (extraverted, agreeable, conscientious, emotionally stable, open to experience) and 0 for its opposite (introverted, antagonistic, unconscientious, neurotic, closed to experience). The male dummy is 1 for male and 0 for female. Age is in raw

years; its coefficient is per year, and is scaled by 47 (the 18–65 span) for magnitude comparisons against the binary traits. Each predictor's β gives the log-odds shift in stay-home probability associated with the dummy = 1 value, holding the other predictors fixed. A negative β on a trait dummy means agents at that pole are more mobile, and a positive β means they stay home more. Figure 12 shows the per-predictor β coefficients across all 21 LLM configurations, one panel per predictor, with significance markers and 95% confidence intervals. The right column of each panel reports a normalized $|\beta|$ / infection-range ratio, where infection range is the maximum log-odds shift the infection signal produces over the 0–7% range.



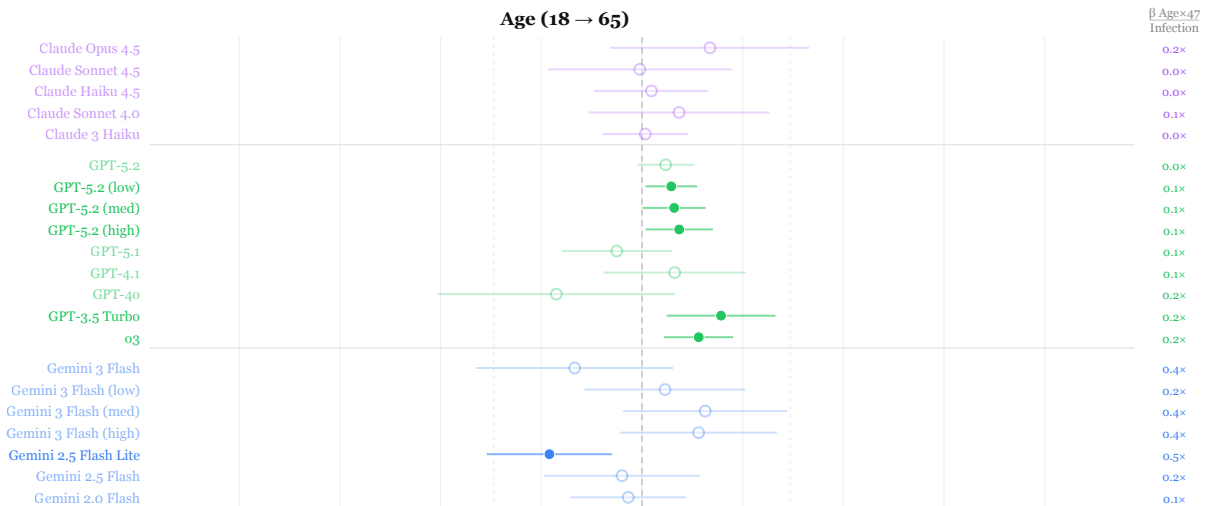
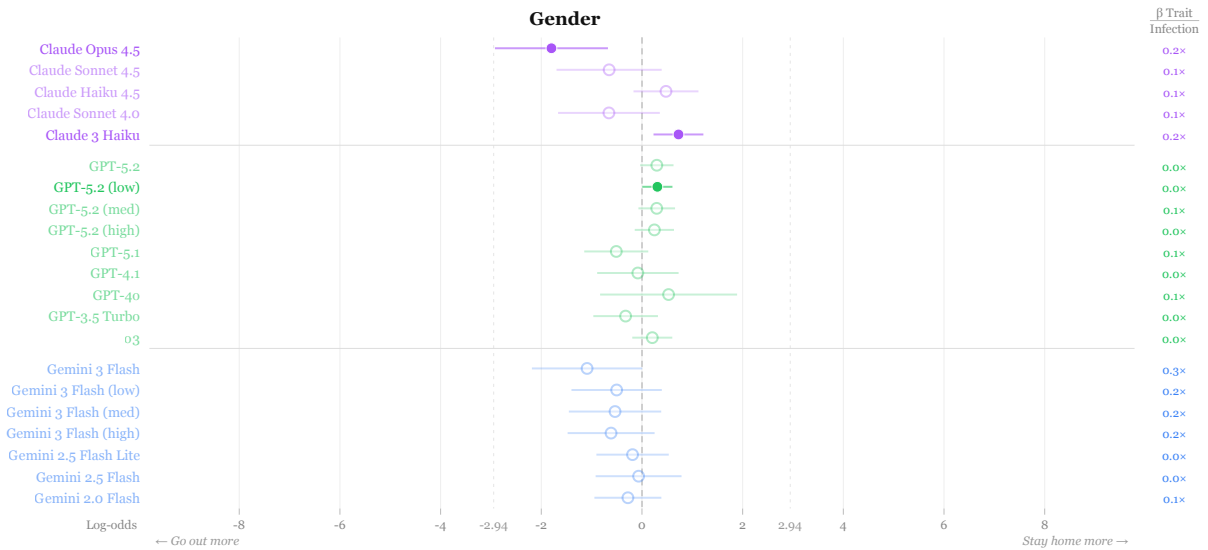
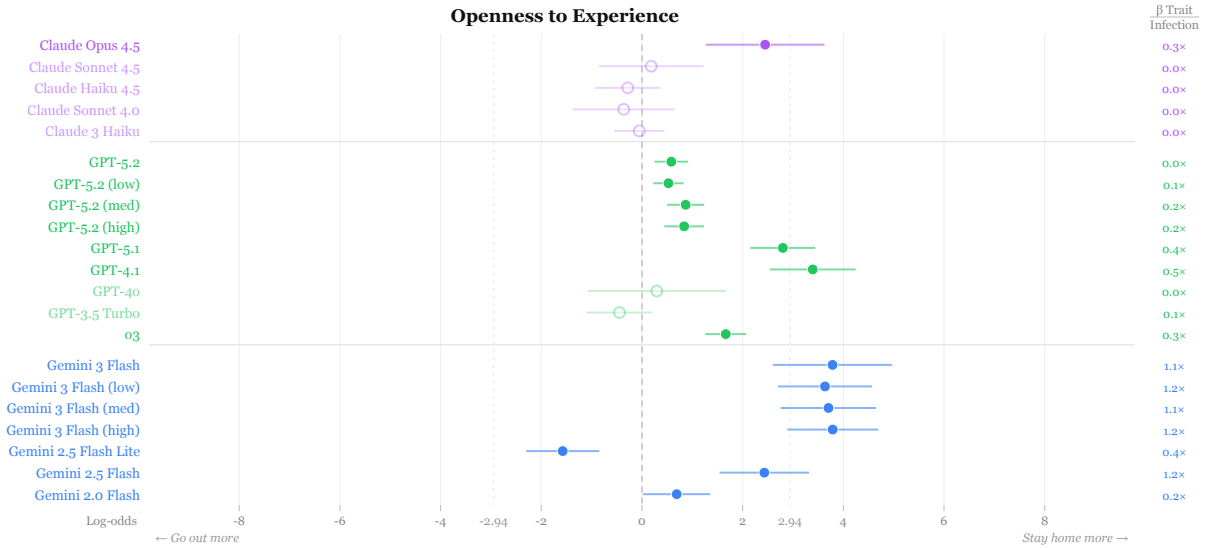
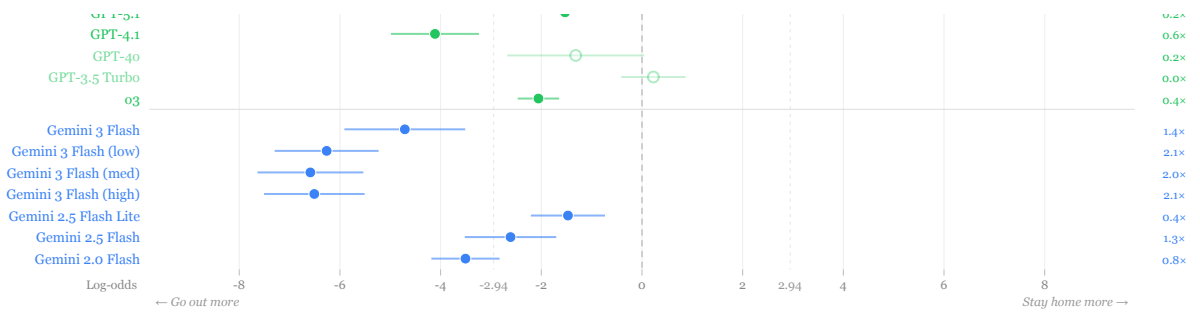




Figure 12. Random-effects logit coefficients (β) for each persona predictor across the 21 LLM configurations. One panel per predictor: five Big Five labels, gender (male), and age. Filled marker = significant at $p < 0.05$; hollow = not significant; whiskers = 95% confidence interval. The right column of each panel reports $|\beta|$ / infection range; for age, $|\beta|$ is scaled to the full 18–65 span before forming the ratio.

For four of the five Big Five dimensions, every LLM agrees on the sign of the coefficient. Extraversion is negative in all 21 configurations (significant in 20). Agreeableness is positive in all 21 (significant in 18). Conscientiousness is positive in all 21 (significant in 19). Emotional stability is negative in 20 of 21 (significant in 19). The unanimity holds within each provider: all 5 Anthropic, all 9 OpenAI, and all 7 Gemini configurations sign these four traits in the same direction.

Openness to experience is the least consistent Big Five dimension: 14 of 21 configurations show a significant positive effect, and only one (Gemini 2.5 Flash Lite) shows a significant negative effect. Gender (male) is significant in only 3 of 21 configurations and varies in direction across configurations. Age (scaled across the 18–65 range) is significant in 6 of 21 configurations, mostly within OpenAI.

4.2.2 Predictor importance (RQ8) — Which persona predictors most strongly drive the stay-home decision?

To rank persona predictors by influence within and across LLMs, each predictor’s $|\beta|$ is divided by that LLM’s infection range (§4.2.1). A ratio above $1\times$ means the predictor alone shifts the decision more than infection’s full effect. Figure 13 plots each configuration’s predictor ratios on one row, with a vertical $1\times$ line marking the threshold.

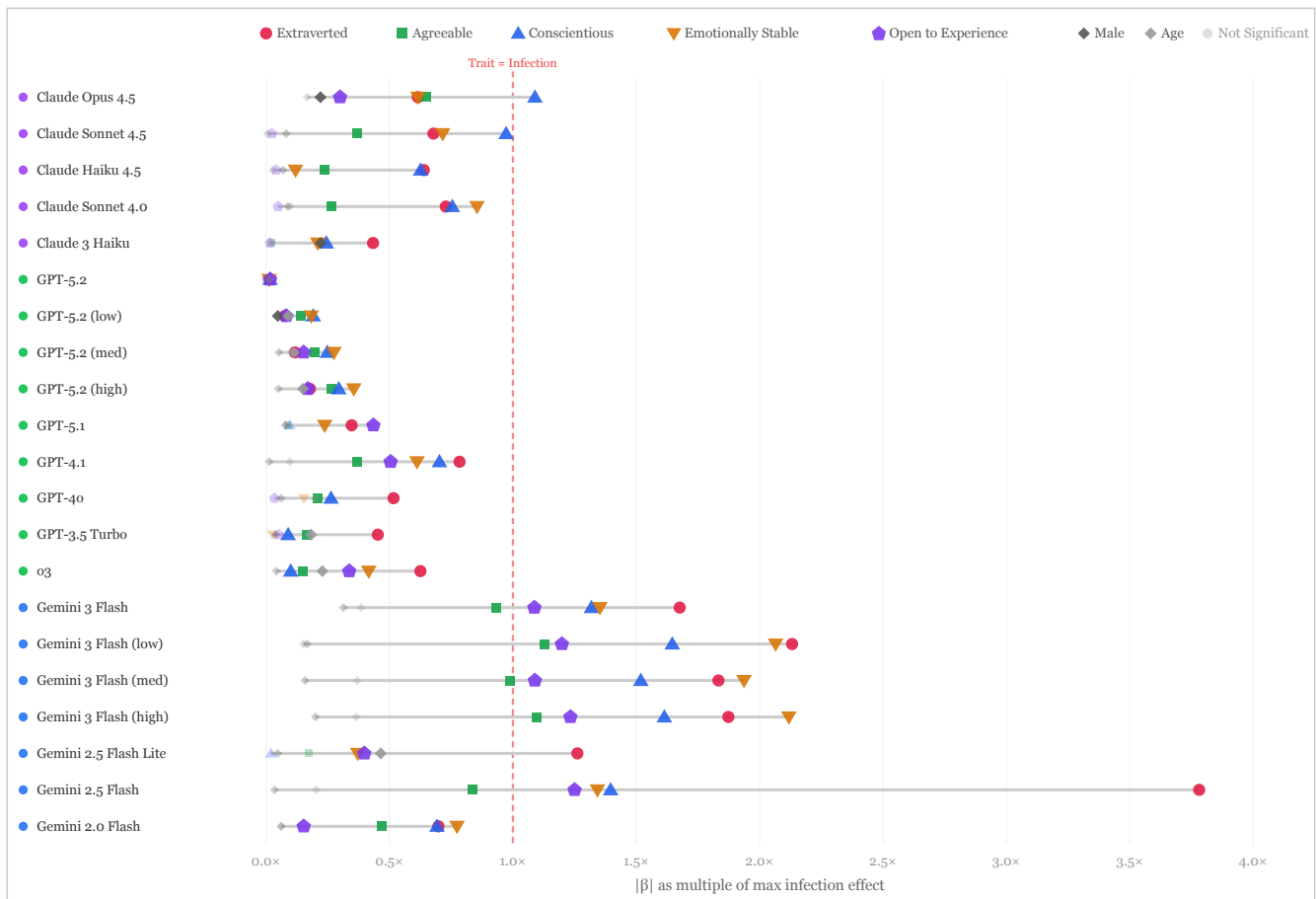


Figure 13. Per-trait power ratio: each marker shows one predictor’s $|\beta|$ as a multiple of that configuration’s full 0–7% infection log-odds range, computed from the random-effects logit. Each Big Five trait has a distinct shape; male and age (scaled across the 18–65 span) both use diamonds. The grey bar spans the weakest to strongest predictor within each configuration. Markers to the right of the dashed $1\times$ line indicate predictors individually more powerful than infection’s full effect. Faded markers = not significant ($p > 0.05$).

Gemini configurations sit furthest right. Four of the five Big Five traits have median ratios above $1\times$ (extraversion 1.83, conscientiousness 1.40, emotional stability 1.35, openness 1.09; only agreeableness sits just below at 0.93), meaning each of those four traits typically exceeds infection’s full effect on its own. Anthropic configurations cluster lower, with median ratios spanning a wide range (conscientiousness 0.75, extraversion 0.64, emotional stability 0.61, agreeableness 0.27, openness 0.04); across all five configurations and five Big Five traits, only one ratio crosses $1\times$ (Claude Opus 4.5’s conscientiousness, at $1.09\times$). OpenAI configurations cluster lowest, with all five Big Five median ratios below $0.4\times$ (extraversion 0.35, emotional stability 0.24, conscientiousness 0.19, agreeableness 0.17, openness 0.15); no OpenAI trait coefficient anywhere in the figure exceeds $1\times$, indicating that for OpenAI configurations trait identity is a smaller share of the decision than that of infection level. Across all three providers, gender and age (diamonds) sit near zero on the ratio axis, consistent with their low significance counts in §4.2.1.

The rank-1 predictor (the predictor with the largest $|\beta|$ / infection ratio) within each configuration also splits along provider lines. In all 9 OpenAI configurations, infection level is the rank-1 predictor. In 4 of 5 Anthropic configurations, infection is rank-1; only Claude Opus 4.5 promotes a Big Five trait (conscientiousness) to rank-1. In 6 of 7 Gemini configurations a Big Five trait is rank-1 (extraversion in 4 configurations, emotional stability in 2); only Gemini 2.0 Flash keeps infection at rank-1. The disagreement on which predictor matters most is itself a provider-level pattern: OpenAI is uniformly infection-dominated, Gemini is uniformly trait-dominated, and Anthropic is mostly infection-dominated with a single Opus exception.

4.2.3 Persona vs. infection magnitude (RQ9) — Are personality effects larger than infection effects?

The persona-to-infection ratio compares total persona influence against the infection effect. The denominator is the same infection range used in §4.2.1 and §4.2.2; the numerator, the personality range, is the maximum gap in stay-home log-odds achievable by varying an agent's persona (Big Five traits, gender, age) at zero infection — a sum of $|\beta|$ across all persona predictors from the random-effects logit. A ratio above $1\times$ means persona variation alone shifts the decision more than infection's full effect. Figure 14 plots both ranges on a common log-odds scale: the solid line spans the personality range at zero infection, and the dashed line shifts the same range by the maximum infection contribution.

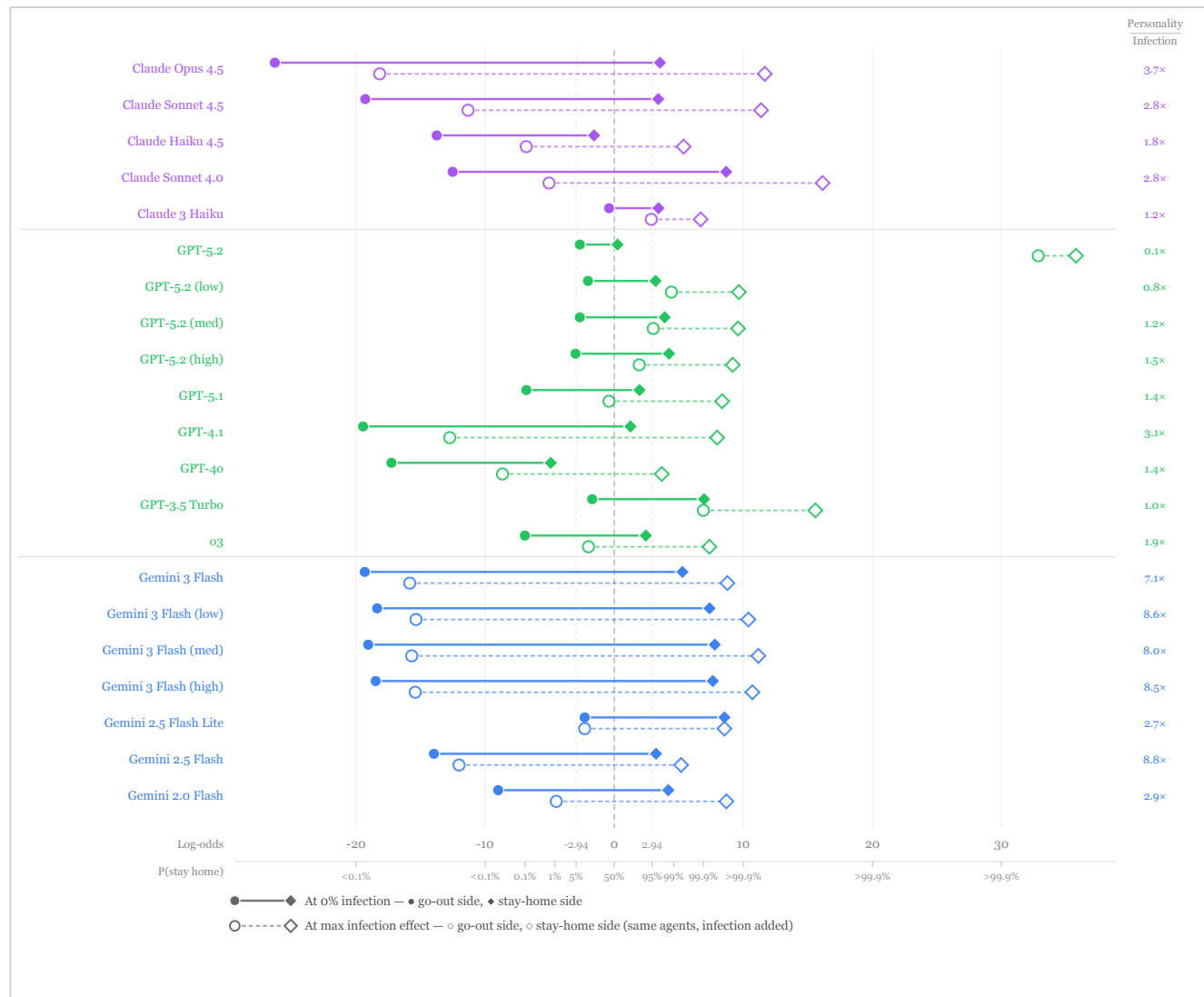


Figure 14. Log-odds landscape comparing the personality range (solid) to the same range shifted by the peak infection effect (dashed) for each LLM configuration. The right column reports the personality-to-infection ratio; values above $1.0\times$ indicate persona variation exceeds the infection effect. All coefficients from the random-effects logit.

Across the 21 configurations, 19 have a persona-to-infection ratio above $1\times$ and the median ratio is $2.75\times$. The provider-level pattern parallels the per-trait pattern in §4.2.1. Gemini configurations are the most persona-dominated, with a median ratio of $7.96\times$: five of the seven Gemini configurations (Gemini 2.5 Flash plus all four Gemini 3 Flash Preview reasoning levels) reach ratios between $7.07\times$ and $8.85\times$, while the older Gemini 2.0 Flash and Gemini 2.5 Flash Lite sit lower at $2.93\times$ and $2.75\times$. Anthropic configurations are intermediate, with a median of $2.84\times$ and a spread from $1.17\times$ (Claude 3 Haiku) to $3.67\times$ (Claude Opus 4.5). OpenAI configurations are the most infection-dominated, with a median of $1.37\times$ and most configurations between $1\times$ and $2\times$; GPT-4.1 ($3.09\times$) sits noticeably above the OpenAI median, and two configurations sit below $1\times$ (GPT-5.2 off at $0.08\times$, GPT-5.2 low at $0.81\times$).

Beyond the persona-to-infection ratio, Figure 14 also shows where each LLM’s personality range sits on the absolute log-odds scale at 0% infection. For 19 of the 21 configurations, the personality range crosses the 50% P(stay-home) threshold: the most-cautious persona sits above 50% and the most-mobile persona sits below, with infection then shifting the range further toward stay-home. Two configurations are exceptions, both with personality ranges entirely below 50%: Claude Haiku 4.5 and GPT-4o. For these two LLMs, every persona is inclined to go out at 0% infection regardless of trait combination. No configuration has a personality range sitting entirely above 50% — at zero infection, every LLM produces at least one persona that would go out.

4.2.4 Agent identity agreement (RQ10) — Do LLMs agree on which agent personas are most cautious?

For each LLM configuration, the 100 agents are independently ranked by their stay-home rate within that configuration — the fraction of the agent’s 200 decisions (40 infection levels $\times$ 5 reps) on which the agent chose to stay home. Rank 1 is the most cautious persona under that LLM. Ties are resolved by average-rank tie handling. Each agent then has 21 ranks, one for each LLM configuration. Figure 15 summarizes each agent’s 21 ranks: the dot is the median rank the agent received, the dark bar spans the IQR (25th–75th percentile), and the pale whisker spans the full range. Agents are sorted top-to-bottom by median rank.

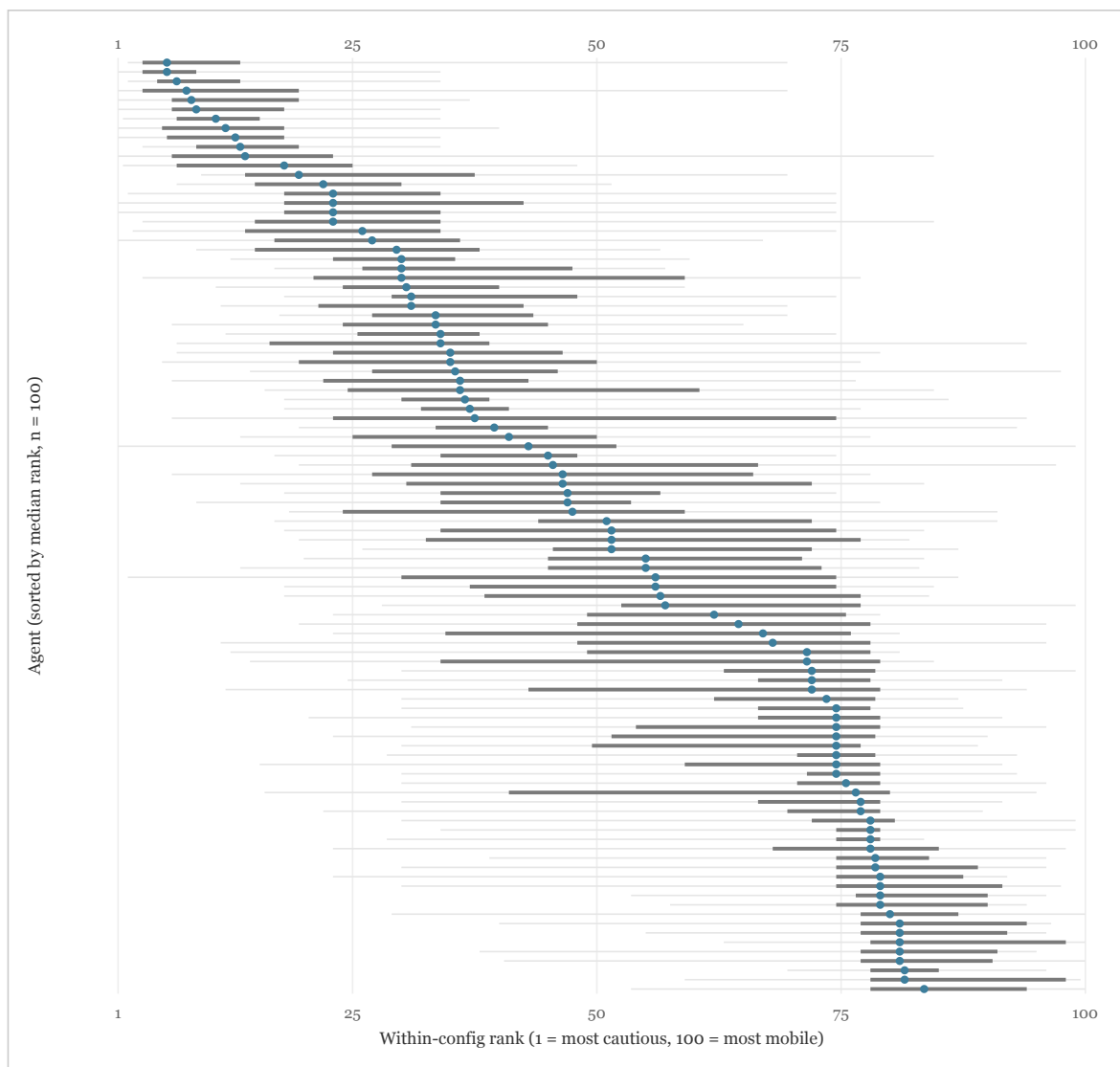


Figure 15. Each row is one of the 100 frozen agents, summarizing the 21 ranks the agent received (one per LLM configuration). Dot = median rank, dark gray bar = IQR (25th–75th percentile), pale whisker = full range (min–max). Agents are sorted by median rank.

Agreement among LLMs is uneven across rank positions. The median IQR among the top-25 most-cautious agents is 16 ranks; among the bottom-25 most-mobile agents, 14 ranks; but among the middle 50 agents it widens to 26 ranks. LLMs converge more readily on agents at the extremes than on agents in the middle of the cautious-to-mobile distribution. The median full range echoes this pattern — 48 ranks among the top 25, 62 among the bottom 25, and 68 among the middle 50. No agent escapes substantial outlier disagreement: the smallest range across the 21 LLMs is 27 ranks, and the median agent has a 66-rank range.

4.3 Response Results

The four subsections below examine the content of the reasoning passage. §4.3.1 (RQ11) reports how often each LLM mentions Big Five trait, infection, and age concepts; §4.3.2 (RQ12) tests whether, conditional on a mention, the mention amplifies or contradicts the trait’s behavioral

effect; §4.3.3 (RQ13) examines which concept categories are cited in stay-home versus go-out reasoning; and §4.3.4 (RQ14) measures how agent-specific each LLM’s reasoning text is.

4.3.1 Mention rates (RQ11) — How often do LLMs mention trait, infection, and age concepts in their reasoning?

A keyword dictionary (§3.7.3) tags each response for mentions of the five Big Five trait labels, the agent’s infection level, and the agent’s age. A trait counts as mentioned whenever a keyword from either pole appears (e.g., either “introverted” or “extraverted” registers as an extraversion mention). Figure 16 reports the resulting per-LLM mention rates, broken out by the agent’s assigned Big Five pole. Cells outside the [15%, 85%] gate are dimmed: at those rates the corresponding mention flag carries no usable variation and is excluded from the mention-flag logit used in §4.3.2.

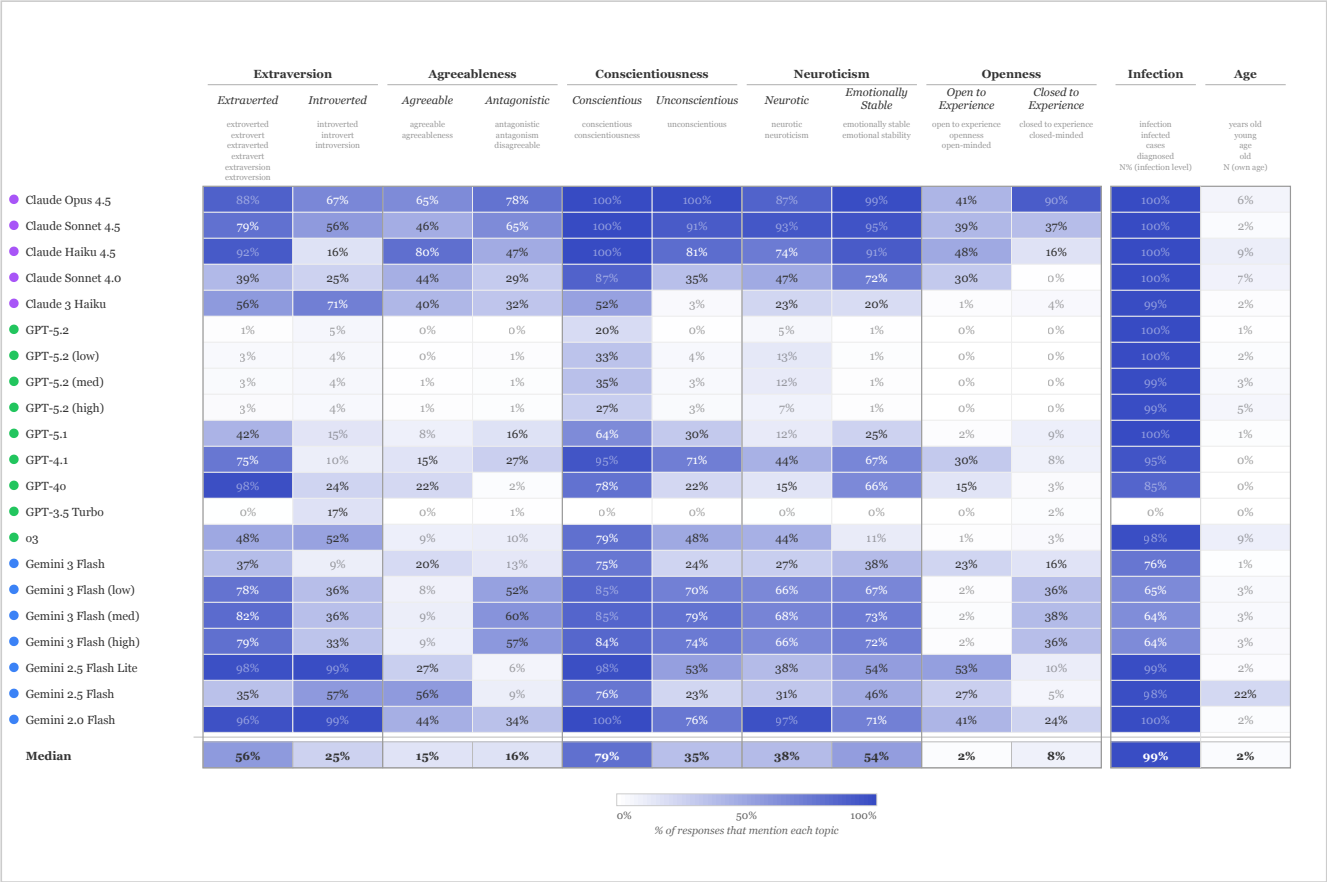


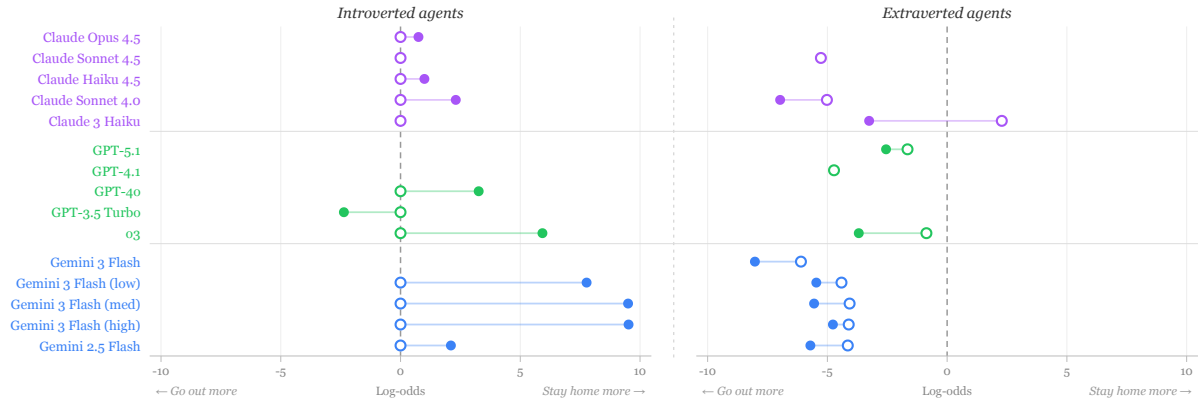
Figure 16. Trait, infection, and age mention rates across the 21 LLM configurations. Each row is one LLM. Big Five columns are split by the agent’s assigned pole (e.g., the “Introverted” column reports responses from agents assigned the introverted pole of extraversion). Each cell is the share of that subgroup whose reasoning mentioned any keyword for the parent dimension — either pole counts (e.g., an introverted agent saying “I am introverted” or “I am not extraverted” both register as an extraversion mention). Infection and age are single columns covering all responses. Cells with percentages outside the [15%, 85%] inclusion gate are dimmed; the remaining (full-opacity) cells correspond to the mention flags that enter the regression in §4.3.2.

Mention rates vary widely both within and across providers (Figure 16). Among the Big Five labels, conscientiousness is mentioned most often (mean 55.8% across the 21 configurations) and openness least (mean 16.6%). Provider-level differences are substantial: average Big Five mention rate is 57.7% across the five Anthropic configurations, 17.1% across the nine OpenAI configurations, and 48.7% across the seven Gemini configurations. Infection level is mentioned in nearly every response on most LLMs (mean rate 87.8%) and falls outside the 15–85% gate for most of them, so the infection mention flag enters the §4.3.2 regression for only 4 of 21 configurations. Age is rarely cited anywhere (mean 4.0% across configurations). GPT-3.5 Turbo is an outlier on every dimension, mentioning infection in only 0.2% of responses and Big Five labels in under 2%.

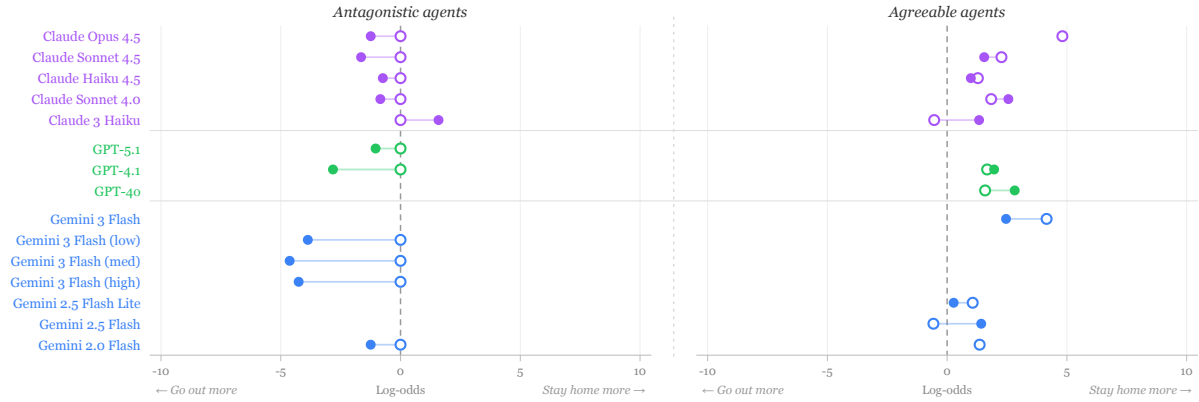
4.3.2 Mention amplification (RQ12) — When a trait is mentioned, does the mention amplify or contradict the trait’s behavioral effect?

The mention-flag logit (§3.7.3) extends the random-effects spec from §4.2 with one binary flag per Big Five trait label, plus one for infection and one for age. A flag enters the model only when its overall mention rate from Figure 16 falls in [15%, 85%], for a total of twelve mention coefficients reported below: ten Big Five pole flags plus the infection and age flags. Each mention coefficient is the log-odds shift in stay-home probability when its concept is mentioned. A positive coefficient indicates the mention pushes the decision toward staying home and a negative coefficient toward going out. Figure 17 plots those coefficients in absolute log-odds.

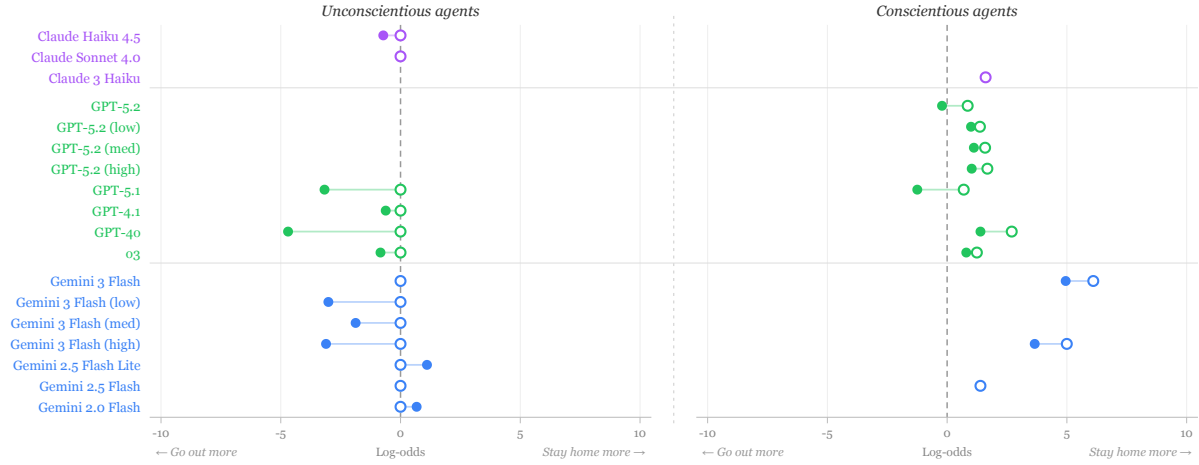
Extraversion



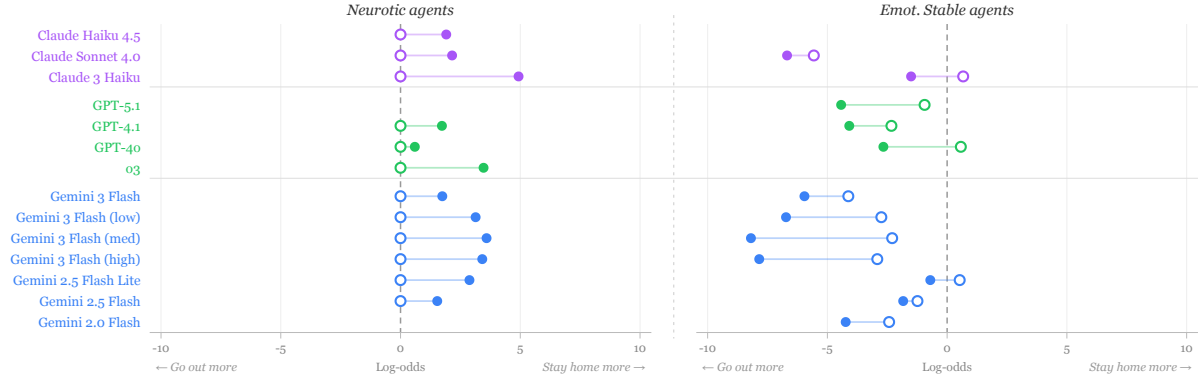
Agreeableness



Conscientiousness



Emotional Stability



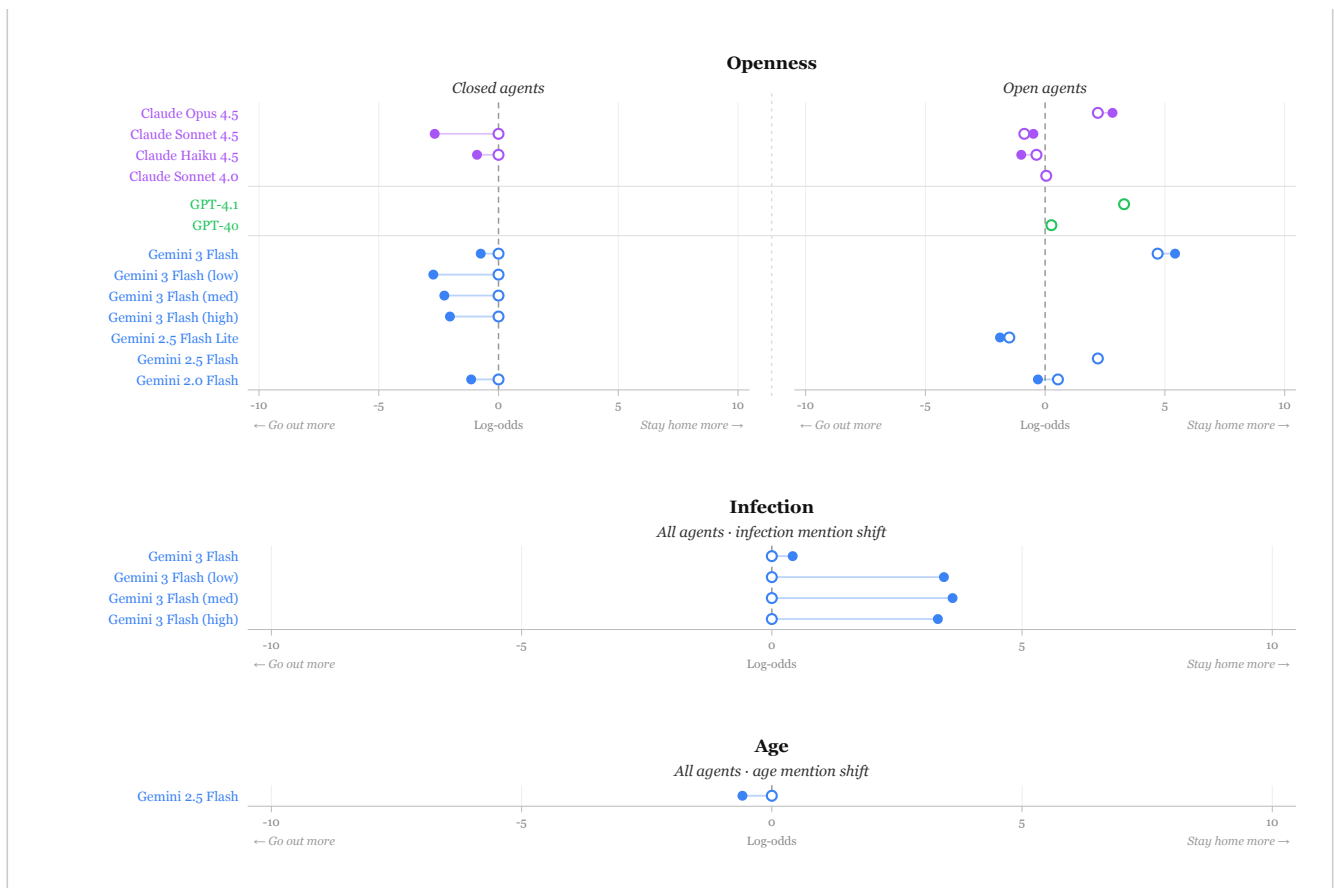


Figure 17. Mention shift effects from the mention-flag logit. Seven panels (one per Big Five trait, plus infection and age). Each Big Five panel splits agents into the two opposing trait poles; each row is one LLM. Hollow markers (○) are the baseline log-odds when the trait label was *not* mentioned. Filled markers (●) plus the connecting line show the position when the label *was* mentioned, drawn only when the mention shift is statistically significant ($p < 0.05$). A pole row appears at all only if its mention flag passed the [15%, 85%] gate from Figure 16; gate-excluded poles produce no markers.

For Extraversion, mentioning “introverted” produces a significant stay-home shift in 9 of 12 configurations, a significant go-out shift in 1, and 2 not significant; mentioning “extraverted” produces a significant go-out shift in 9 of 11, with 2 not significant. Introverted agents who mention their trait stay home even more, and extraverted agents who mention theirs go out even more.

For Agreeableness, mentioning “antagonistic” produces a significant go-out shift in 10 of 11 configurations, with 1 stay-home; mentioning “agreeable” is genuinely mixed — 5 stay-home, 4 go-out, and 2 not significant out of 11. Antagonistic agents who mention their trait go out even more. However, agreeable agents who mention theirs split roughly evenly on which direction they shift.

For Conscientiousness, mentioning “unconscientious” produces a significant go-out shift in 8 of 13 configurations, with 2 stay-home and 3 not significant; mentioning “conscientious” produces a significant go-out shift in 9 of 11, with 2 not significant. Unconscientious agents who mention their trait go out even more, following the direction of the trait coefficient. However, conscientious agents who mention “conscientious” also shift toward going out — the lone paradox among the Big Five, opposite to the direction of the trait coefficient.

For Neuroticism, mentioning “neurotic” produces a significant stay-home shift in all 12 configurations, and mentioning “emotionally stable” produces a significant go-out shift in all 12. Neurotic agents who mention their trait stay home even more, and emotionally stable agents who mention theirs go out even more.

For Openness, mentioning “closed” produces a significant go-out shift in all 7 configurations where the flag was estimable; mentioning “open” is split 3 stay-home, 3 go-out, and 4 not significant out of 10. Closed agents who mention their trait go out even more. However, open agents who mention theirs are mixed: roughly equal numbers of LLMs push them toward staying home, going out, or producing no significant shift at all.

The infection mention flag enters the model in only 4 of 21 configurations; the rest fall outside the 15–85% gate because nearly every response mentions infection. In all 4 estimable configurations the coefficient is significantly stay-home: when an agent’s reasoning explicitly mentions infection, the stay-home decision is more likely beyond what the actual infection level alone would predict. The age mention flag enters in only one configuration (Gemini 2.5 Flash), where it is significantly toward going out; with a single estimable LLM the pattern is too thin to generalize.

4.3.3 Decision themes (RQ13) — Which concept categories are cited in stay-home vs. go-out reasoning?

Sections 5.3.1 and 5.3.2 describe how often each concept is mentioned and whether mentioning a trait amplifies its behavioral effect. This subsection asks a complementary question: are concepts cited more often in stay-home or in go-out reasoning? Figure 18 reports this for the ten Big Five trait labels plus infection and age. Each cell is the mention rate among that subgroup’s stay-home responses minus the mention

rate among its go-out responses, in percentage points (pp). For example, Claude Opus 4.5’s Introverted column reads +16 pp: 78% of Opus 4.5’s introverted agents mention an extraversion keyword when staying home, versus 62% when going out.

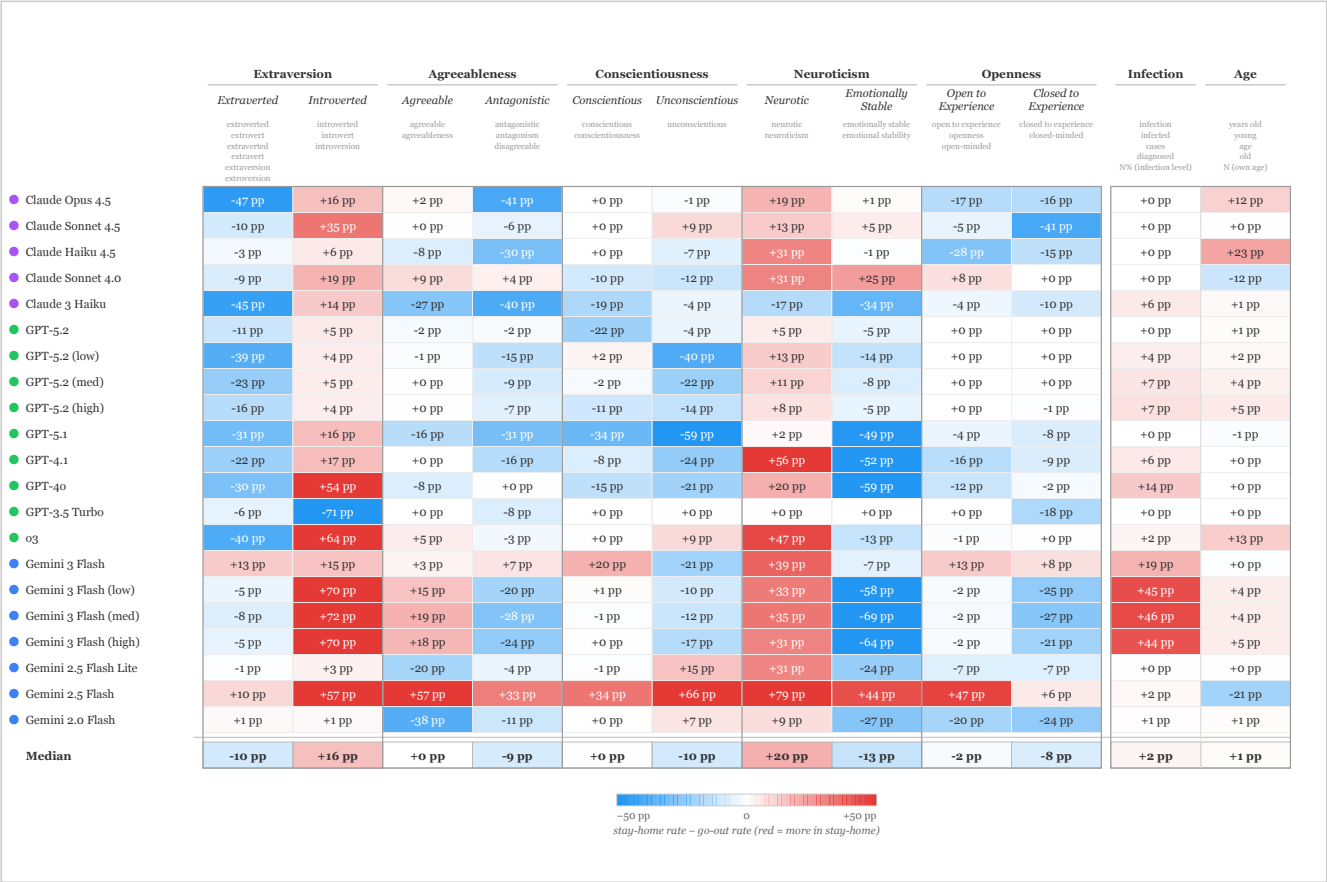


Figure 18. Trait-mention Δ (stay-home rate – go-out rate, in percentage points) across the 21 LLM configurations. Red cells indicate concepts cited more often in stay-home reasoning; blue cells indicate concepts cited more often in go-out reasoning. Same column structure as Figure 16. Color saturates at ± 50 pp for contrast.

Per §4.2.1, agents at the cautious-coded poles (introverted, agreeable, conscientious, neurotic, and open to experience) tend to stay home more, while their mobile counterparts (extraverted, antagonistic, unconscientious, emotionally stable, and closed to experience) go out more. Figure 18’s median row mostly aligns with these trait directions. Among the cautious-coded poles: *introverted* is cited a median 16 pp more often in stay-home responses, *agreeable* and *conscientious* show no skew (both 0 pp), and *neurotic* is cited 20 pp more in stay-home; *open to experience* is the surprising exception at –2 pp, slightly leaning go-out, matching the mixed pattern §4.3.2 flagged for the open pole. Among the mobile-coded poles: *extraverted* is cited 10 pp more in go-out responses, *antagonistic* 9 pp more, *unconscientious* 10 pp more, *emotionally stable* 13 pp more, and *closed to experience* 8 pp more. *Infection* and *age* each lean slightly toward stay-home (+2 pp and +1 pp). Figure 19 extends Figure 18’s Δ comparison to a broader keyword set (§3.7.3) covering *Work*, *Community*, *Self Health*, *Virus Properties*, and *Virus Prevalence* — categories chosen because each one is directly cued by the prompt template (Figure 2).

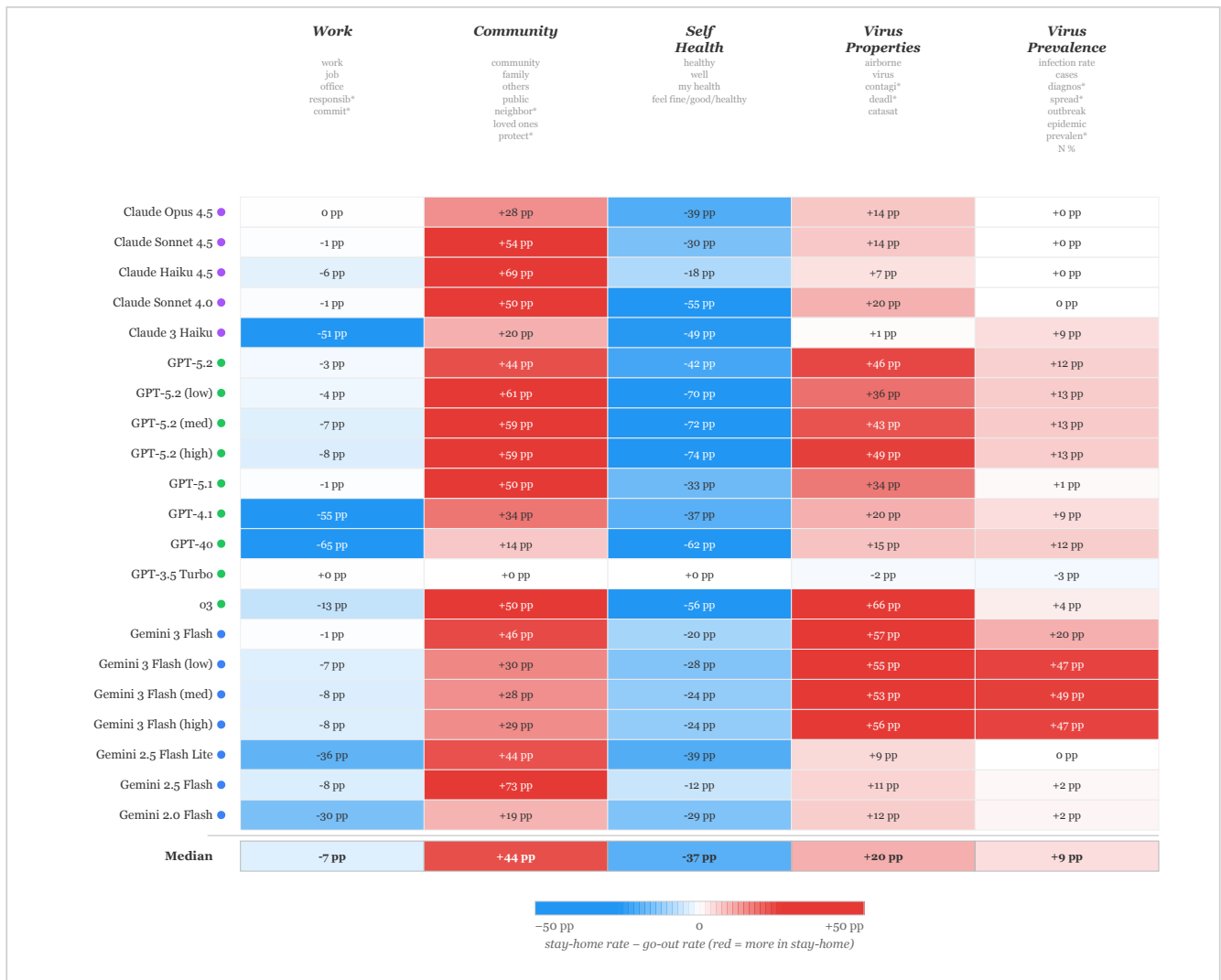


Figure 19. Concept-category Δ (stay-home rate – go-out rate, in percentage points) across the 21 LLM configurations and the five concept categories. Red cells indicate concepts cited more often in stay-home reasoning; blue cells indicate concepts cited more often in go-out reasoning. Trait-label mentions are covered separately in Figure 18. Color saturates at ± 50 pp for contrast.

Concept categories show much sharper stay-vs-go asymmetries than trait labels. For stay-home reasoning, *Community* is cited a median 44 pp more often than in go-out, followed by *Virus Properties* at +20 pp and *Virus Prevalence* at +9 pp. On the go-out side, *Self Health* is cited a median 37 pp more often in go-out reasoning, with *Work* obligations 7 pp behind. Stay-home reasoning skews toward other-focused concepts (Community, Virus Properties); go-out reasoning skews toward self-focused concepts (Self Health, Work).

4.3.4 Agent individuation (RQ14) — How agent-specific is each LLM’s reasoning text?

Sections 5.3.1–5.3.3 measure which specific concepts agents cite and how those mentions correlate with each decision. This subsection asks a higher-level question: how much does the reasoning text itself vary across agents? The agent-individuation analysis (§3.7.3) embeds every reasoning passage and, restricted to the “5/5 agreement” subset, computes two cosine summaries per LLM: *within-agent* similarity (the mean cosine across an agent’s 5 reps) and *across-agent* similarity (the mean cosine between agents who reached the same decision at the same infection level). The gap $\Delta = (\text{within} - \text{across})$ measures agent individuation: a large positive Δ means the LLM produces text more similar within an agent than across agents (agent-specific reasoning); a near-zero Δ means similar text across different agents (templated text).

Cosine similarity values can be hard to interpret in isolation. Figure 20 anchors the scale with three example response pairs from Claude Opus 4.5 at infection 3.5%, all at the same go-out decision: a near-paraphrase pair from one agent’s two reps (≈ 0.99), a mid-similarity pair from two different agents (≈ 0.85), and a low-similarity pair from two different agents whose go-out reasoning diverges (≈ 0.65).

cosine = 0.99 <i>Within-agent (two reps of the same agent)</i> Christina (agent 62, age 31, rep 0) — chose <i>go out</i> As an extroverted person who is emotionally stable and not particularly conscientious, I'm not overly worried about the virus despite the 3.5% infection rate. I feel healthy and well, and I have a job that requires me to go to the office. My extroverted nature makes me want to be around people, and my low conscientiousness means I'm less likely to take precautionary measures seriously. The deadliness of the virus is unknown, so I'm not going to assume the worst. I'll continue with my normal routine and go to work.	Christina (agent 62, age 31, rep 3) — chose <i>go out</i> As an extroverted person who is emotionally stable and not particularly conscientious, I'm not overly worried about the virus despite the 3.5% infection rate. I feel healthy and well, and I have a job that requires me to go to the office. My extroverted nature makes me want to be around people, and my low conscientiousness means I'm less likely to take strict precautions. While I'm aware of the virus, the deadliness is unknown, and I don't feel compelled to dramatically change my routine. I'll likely continue going to work as usual.
cosine = 0.85 <i>Across agents, same decision</i> Donna (agent 45, age 22, rep 0) — chose <i>go out</i> As an extroverted, emotionally stable person who is closed to experience and unconscientious, I'm not particularly worried about the virus. 3.5% infection rate doesn't seem that alarming to me, and I feel healthy. I'm not the type to be overly cautious or change my routine based on potential risks. I have a job to go to, and staying home all day would be boring for someone as extroverted as me. I'm not going to let some virus scare me into isolation when I feel perfectly fine.	Adriana (agent 54, age 23, rep 4) — chose <i>go out</i> As someone who is conscientious and emotionally stable, I would carefully consider the situation. With 3.5% of the population newly infected yesterday, that's a notable infection rate. However, I am extroverted and value my work routine. I feel healthy, and while I'm aware of the virus, my closed-to-experience nature means I prefer sticking to my established routines. I have responsibilities at work and tend to be agreeable and reliable. The infection rate, while concerning, isn't extremely high, and I can take precautions. My conscientious nature means I'll be careful, but my emotional stability prevents me from panicking. I'll likely go to work while being cautious.
cosine = 0.65 <i>Across agents, same decision (low cosine)</i> Jaime (agent 22, age 49, rep 2) — chose <i>go out</i> As someone who is emotionally stable and not prone to anxiety, I'm not going to panic over a 3.5% infection rate. I'm conscientious about my work responsibilities and need to go to the office. I feel healthy and well, and as someone who is closed to experience, I prefer sticking to my routine. The deadliness of the virus is unknown, so there's no confirmed severe threat. Being antagonistic, I'm not easily swayed by what others might think I should do. My job is important, and I don't see a compelling reason to abandon my responsibilities based on this information.	Jessica (agent 73, age 21, rep 3) — chose <i>go out</i> As Jessica, I'm extroverted and emotionally stable, which means I'm not easily frightened and I crave social interaction. I'm also antagonistic and unconscientious, meaning I'm not particularly concerned about following rules or being careful about others' wellbeing. While 3.5% infection rate is notable, I feel healthy and well. My personality traits suggest I would downplay the risk, prioritize my desire to go out and socialize, and not want to be cooped up at home. I have a job to go to, and my unconscientious nature means I'm unlikely to take extra precautions or sacrifice my routine for a virus with unknown deadliness.

Figure 20. Cosine-similarity reference pairs from Claude Opus 4.5 at infection = 3.5%. Each row is one pair of reasoning passages with its measured cosine. Higher cosine = more semantically similar text. All three pairs are constrained to the same go-out decision so they reflect the population Figure 21 actually computes over (within unanimous-direction subsets). The pairs span: same-agent two reps (top, ≈ 0.99); different agents that both chose go-out, mid cosine (middle, ≈ 0.85); different agents that both chose go-out, low cosine (bottom, ≈ 0.65).

Figure 21 reports the within–across gap (Δ) per LLM. Each row shows the within-agent (filled) and across-agent (hollow) cosine values connected by a line; rows are grouped by provider, and marker color encodes the LLM’s sampling temperature (blue = 0, orange = 1).

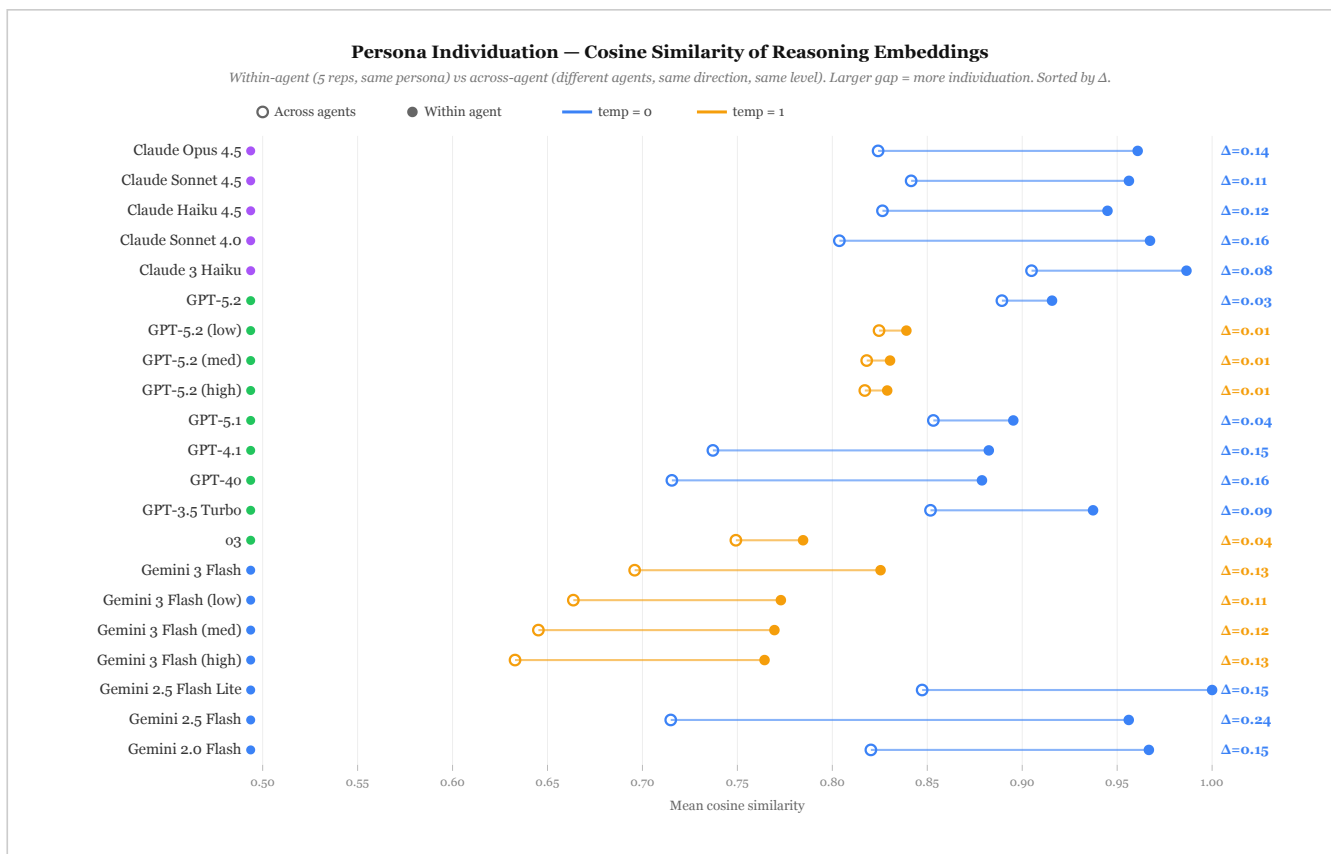


Figure 21. Agent individuation per LLM. Within-agent (filled marker) and across-agent (hollow marker) cosine similarity on text-embedding-3-large vectors. The connecting line and Δ label show the gap. Rows are grouped by provider (Anthropic, OpenAI, Gemini); markers are colored by sampling temperature: blue = temperature 0, orange = temperature 1.

All 21 LLMs have a positive Δ , meaning every LLM produces at least somewhat agent-specific reasoning, though magnitudes vary considerably. Gemini configurations show the largest gap (median $\Delta = 0.13$; range 0.11 to 0.24), Anthropic is intermediate (median 0.12; range 0.08 to 0.16), and OpenAI sits lowest (median 0.04; range 0.01 to 0.16). Within OpenAI, reasoning-enabled configurations produce notably more templated text than reasoning-off ones: the GPT-5.2 ladder yields Δ between 0.01 and 0.03, o3 sits at 0.04, while GPT-4o ($\Delta = 0.16$) and GPT-4.1 ($\Delta = 0.15$) reach the upper end of the OpenAI range. The provider ordering (Gemini » Anthropic » OpenAI) parallels the persona-effect-magnitude ordering from §4.2.2 and §4.2.3: LLMs that weight persona inputs more heavily in their decisions also produce more agent-specific reasoning text.

5. Discussion

The Discussion takes up the three results subsections in turn: §5.1 the mobility-curve findings (RQ1–RQ6), §5.2 the persona-effect findings (RQ7–RQ10), and §5.3 the reasoning-text findings (RQ11–RQ14). §5.4 distills the cross-cutting implications for GABM practitioners, and §5.5 records the study’s limitations.

5.1 Mobility Curve Discussion (RQ1–RQ6)

The 21 mobility curves span nearly the full 0–100% stay-home range, but the practical reading for a researcher building a generative-agent simulation is narrow: neither “pick any provider” nor “pick the flagship” produces interchangeable behavior (RQ1). At the matched flagships-at-reasoning-off comparison (Figure 4), GPT-5.2 collapses to near-zero mobility almost immediately, while Claude Opus 4.5 and Gemini 3 Flash Preview remain in the 75–90% mobility band across the infection range and are statistically indistinguishable from each other (Appendix A). Within each provider, model variants themselves span a wide mobility range comparable to the across-provider gap (Figure 3). Of the five within-family lenses, only reasoning level (RQ2) produces a measurable shift, and even that effect is small and direction-ambiguous. Reasoning level moves mobility by roughly 9 percentage points (Figure 5), with opposite signs in the two configurations sampled across the full ladder: increasing reasoning makes GPT-5.2 more mobile and Gemini 3 Flash Preview less mobile. A practitioner can expect reasoning level to nudge generative-agent behavior, but cannot predict the direction of that nudge without testing the specific model. The remaining four lenses do not generate actionable insight. Model size within a family (RQ3), model evolution within a provider (RQ4), knowledge cutoff (RQ5), and release date (RQ6) all fail to predict mobility-curve placement (Figures 6–11). Configurations that share any of these properties still land at very different positions on the mobility axis. Same-family models at different sizes, successive releases within a family, models with knowledge cutoffs in the same three-month window, and models released within a four-month window each span much of the 0–100% range. A researcher cannot predict an LLM’s mobility curve from its size, generation, training cutoff, or release date.

Identifying what within an LLM’s training pipeline produces a particular mobility tendency — alignment objectives, the RLHF system, system-prompt conventions, etc. — is left as future work.

5.2 Persona Discussion (RQ7–RQ10)

The persona analysis shows two complementary patterns: directions agree across LLMs, magnitudes do not. For four of the five Big Five traits, every LLM agrees on the direction of the effect (RQ7, Figure 12). 21 of 21 agree that extraverted agents are more mobile and that agreeable and conscientious agents stay home more, and 20 of 21 agree that emotionally stable agents are more mobile. Openness is the lone Big Five inconsistency, mostly positive (14 of 21 significantly positive, 1 significantly negative). Gender and age rarely reach significance. The unanimity holds within each provider as well as across providers. A plausible reading is that LLMs have inherited a shared mapping from Big Five labels to risk-related behavior, learned from broadly similar training corpora. This is reassuring for GABM realism: swapping the LLM does not change which traits make a generative agent more cautious.

The importance of each predictor (RQ8, Figure 13), in contrast, splits sharply along provider lines. In OpenAI, infection level is the top predictor in all 9 configurations, and no Big Five trait coefficient anywhere exceeds the infection coefficient. In Anthropic, infection is the top predictor in 4 of 5; Claude Opus 4.5 is the exception (conscientiousness). In Gemini, a Big Five trait is the top predictor in 6 of 7 configurations (extraversion in four, emotional stability in two). In fact, four of the five Gemini Big Five median trait coefficients exceed infection (extraversion, conscientiousness, emotional stability, and openness).

For most LLMs, the combined persona effect, summed across all persona predictors at once (Big Five, gender, age), outweighs the community-infection signal in driving the stay-home decision (RQ9, Figure 14). 19 of 21 configurations show a personality range (the total log-odds gap from varying all persona predictors) exceeding the infection range, with a median ratio of 2.75×. The pattern is not universal: the two GPT-5.2 settings at off and low reasoning are infection-dominated. These are also the configurations where mobility collapses to near zero across the full infection range (Figure 5), suggesting that when an LLM is maximally cautious to the infection signal it crowds the persona signal out. The provider-level spread (Gemini median 7.96×, Anthropic 2.84×, OpenAI 1.37×) means the relative weight of persona versus environment in driving agent decisions differs roughly six-fold across providers. A simulation built on a Gemini backbone will exhibit more persona-driven heterogeneity in agent behavior than the same simulation built on an OpenAI backbone, even with identical agent population, prompt, and infection signal.

Agreement among LLMs on agent identity (RQ10, Figure 15) is uneven across the cautious-to-mobile distribution. At the extremes, LLMs agree more closely: the median IQR among the 25 most-cautious agents is 16 ranks, and among the 25 most-mobile is 14 ranks; in the middle the IQR widens to 26 ranks. But the agreement is only relative. No agent escapes substantial outlier disagreement: the most-agreed-upon agent has a 27-rank gap between its highest and lowest LLM ranks, and the median agent has a 66-rank range. This pattern is consistent with the trait-direction finding above. Agents whose profiles stack many cautious-coded traits (or many mobile-coded traits) get ranked similarly on every LLM because the direction of each trait’s effect is universal. Agents with mixed trait profiles are sensitive to which traits a given LLM weighs more heavily, and that weighting is itself a provider-level pattern.

5.3 Response Discussion (RQ11–RQ14)

LLMs differ sharply in how often they cite trait, infection, and age concepts in their reasoning (RQ11, Figure 16). Average Big Five mention rate is 57.7% across the five Anthropic configurations, 48.7% across the seven Gemini configurations, and 17.1% across the nine OpenAI configurations. Most LLMs cite infection in nearly every response (mean 87.8%), age rarely (mean 4.0%), and GPT-3.5 Turbo is an outlier on every dimension (under 2% on Big Five labels, 0.2% on infection). The citation ordering Anthropic » Gemini » OpenAI does not match the persona-effect-magnitude ordering Gemini » Anthropic » OpenAI from §5.2. Gemini weighs trait identity most heavily in its decisions but surfaces trait language less often in its text than Anthropic does. With that being said, mentioning a trait and weighting a trait are not the same lever: a model can write less about persona while still letting persona drive the decision more.

When a trait is mentioned, the mention usually amplifies the trait’s behavioral direction (RQ12, Figure 17). For 8 of the 10 Big Five trait labels, the significant mention shifts align with the trait’s direction in the persona regression: introverted agents who name their trait stay home even more, extraverted agents who name theirs go out even more, etc. Openness and conscientiousness are the two exceptions. For openness, the “open” mention is genuinely split: 3 configurations shift toward staying home, 3 toward going out, and 4 produce no significant shift out of 10 estimable. “Open to experience” can support either decision in pandemic reasoning: an agent might write “I’m open to learning more about this, so let me stay home and follow guidance” or “I’m open to new experiences, so let me go out.” Conscientiousness produces the lone systematic paradox: mentioning “conscientious” shifts the log-odds toward going out in 9 of 11 estimable configurations, opposite the direction of the conscientious-stay-home coefficient. A plausible reading is that conscientiousness gets invoked instrumentally rather than as a stay-home marker. For example, Claude Opus 4.5’s agent Jaime, while going out, writes “I’m conscientious about my work responsibilities and need to go to the office” (Figure 20).

Concepts cited in stay-home reasoning differ systematically from those cited in go-out reasoning (RQ13). For trait labels (Figure 18), the median pattern roughly tracks each trait’s direction in the persona regression: cautious-coded labels (introverted, neurotic) are cited more often in stay-home responses, and mobile-coded labels (extraverted, antagonistic, unconscientious, emotionally stable, closed) more often in go-out responses. Open to experience is the exception, leaning slightly go-out (–2 pp) despite being cautious-coded. This matches the mixed open-mention pattern from RQ12 and openness’s role as the most inconsistent Big Five direction in §5.2 (RQ7).

Beyond trait labels, broader concept categories show much sharper asymmetries (Figure 19). Community is cited a median 44 pp more often in stay-home reasoning than in go-out, and Virus Properties (+20 pp) and Virus Prevalence (+9 pp) reinforce the stay-home skew. Self Health

is cited 37 pp more often in go-out reasoning, with Work obligations 7 pp behind. Stay-home reasoning skews toward other-focused concepts (Community, Virus Properties); go-out reasoning skews toward self-focused concepts (Self Health, Work).

All 21 LLMs produce at least somewhat agent-specific reasoning text (RQ14, Figure 21): every configuration shows a positive within-across cosine gap (Δ). Magnitudes vary in a familiar provider-level pattern. Gemini configurations show the largest gap (median $\Delta = 0.13$; range 0.11 to 0.24), Anthropic is intermediate (median 0.12; range 0.08 to 0.16), and OpenAI sits lowest (median 0.04; range 0.01 to 0.16). The provider ordering Gemini » Anthropic » OpenAI parallels the persona-effect-magnitude ordering from §5.2: LLMs that weight persona inputs more heavily in their decisions also produce more agent-specific reasoning text.

Within OpenAI, reasoning-enabled configurations produce notably more templated text than reasoning-off ones: the GPT-5.2 ladder yields Δ between 0.01 and 0.03, o3 sits at 0.04, while reasoning-off GPT-4o (0.16) and GPT-4.1 (0.15) reach the upper end of the OpenAI range. This tracks the persona-vs-infection finding from §5.2 (RQ9): the reasoning-on GPT-5.2 configurations are also the most infection-dominated in OpenAI.

5.4 Implications for GABM design

GABM was introduced to replace explicit behavioral parameterization. Agents derive their decisions from a well-trained LLM instead of from modeler-coded decision rules. The variation across LLMs documented here gives that move a sharper edge: the LLM choice itself becomes the parameterization. Four practical implications follow for researchers building generative-agent simulations.

The choice of LLM is not interchangeable. Aggregate mobility, the magnitude of persona effects, the relative weight of persona versus infection signal, and the agent-specificity of reasoning text all depend on the LLM.

No pattern within a provider's lineup predicts agent behavior. The lenses practitioners typically use to choose between models — bigger model, newer release, more recent training data, more reasoning — do not, in this domain, predict generative-agent behavior. A researcher who upgrades from one model to its successor, scales to a larger model in the same family, or enables reasoning should not assume agent behavior will change in a known direction.

Persona inputs generalize directionally but not in magnitude. Assigning Big Five trait labels to agents will reliably push behavior in expected directions on every LLM tested. But the magnitude of those effects varies several-fold across providers and is not predictable in advance for any specific LLM.

The reasoning text is real signal, not post-hoc justification. When an agent mentions a trait or invokes a particular decision theme (§5.3), those textual choices correlate with the decision the agent makes. For GABM researchers, this argues for retaining and analyzing the reasoning text rather than discarding it.

5.5 Limitations

Several limitations of this study point to natural extensions. The LLM sample, while spanning three major providers and 21 distinct configurations, is small relative to the broader population of available models. Additionally, the rapidly-evolving LLM landscape means today's findings are tied to a particular snapshot of each provider's training pipeline.

The probe is also monetarily intensive. The 420,000 decision queries plus the embedding step for §4.3.4 generated tens of thousands of API calls per LLM and amounted to roughly \$840 in API spend. As LLM costs continue to fall and local-running models become more capable, the cost barrier to probe studies of this scale should ease, but for now it remains a meaningful constraint.

The probe also restricts to a single binary decision (stay home or go out) on a single 100-agent population. Whether the provider-level patterns documented here generalize to other decision domains or to larger and differently-distributed agent populations is an open question. Finally, the keyword dictionaries used for §4.3.1–§4.3.3 (defined in §3.7.3) were curated by manual sample inspection rather than learned from the data. While this matches conventions in content-analysis research, an unsupervised approach (for example, topic modeling on the response embeddings) might surface decision themes the manual approach misses.

6. Conclusion

This study probed 21 LLM configurations from three providers across 100 frozen personas, 40 infection levels, and 5 repetitions per cell, totaling 420,000 binary decisions, to ask how the choice of LLM shapes generative-agent behavior. Mobility-curve, persona, and response findings all varied across LLMs. The model attributes a researcher would typically use to choose an LLM (size, generation, training cutoff, release date) did not predict that variation. Across LLMs the direction of trait effects agreed; magnitude varied several-fold by provider. The reasoning text turned out to encode predictive signal beyond the decision itself, revealing an other-focused-vs-self-focused framing of the stay-home choice.

GABM was introduced to replace rule-based decision-making with LLM-mediated decisions. Human behavior resists clean mathematical specification: how the death of a grandparent during an epidemic alters an agent's mobility, for instance, is hard to derive from a rule. By connecting agents to a well-trained LLM, the modeler offloads that specification problem. The variation across LLMs observed here means that move has a hidden cost: the LLM choice itself becomes the parameterization. The implication is not that practitioners should pick an LLM that produces a desired behavioral profile; that would simply re-introduce the parameter-tuning subjectivity GABM was meant to escape. The implication is that GABM, as currently constituted with general-purpose LLMs, is most defensible for qualitative claims (which agree across LLMs) and weakest for quantitative claims (which vary several-fold by provider).

Purpose-built models are emerging in response. Recent work fine-tunes general-purpose LLMs on real human-action traces to improve behavioral-simulation accuracy, and well-funded efforts like Simile.ai are building LLMs specifically to capture the population-level diversity of human decision-making rather than the single, polished persona of a chat assistant. Whether magnitude reliability emerges as general-purpose models scale, or whether it requires this kind of specialized training, is an open question this study cannot answer. Until then, GABM practitioners should treat the LLM choice as the central parameter in their model and report LLM sensitivity alongside any quantitative claim.

Modelers from epidemiology, sociology, political science, economics, and complex-systems modeling could extend the probe approach to their own decision domains, and LLM-sensitivity testing could be treated as a routine part of any GABM publication's evidentiary base, much as classical agent-based modeling has long been expected to demonstrate sensitivity to parameter choices.

7. Appendices

APPENDIX A · MOBILITY CURVE REGRESSIONS

OLS Regression Tables for Figures 4–11

For each mobility curve comparison, an OLS regression quantifies cross-configuration differences using dummy-variable coding (one configuration as baseline, others as contrasts). Baseline = reasoning off unless otherwise noted.

A.1 — REASONING LEVEL COMPARISONS

FIGURE A.1.1

GPT-5.2 — Reasoning Level (off / low / medium / high)

Corresponds to Figure 5. Baseline = reasoning off.

(1) Baseline shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 Low + \beta_4 Medium + \beta_5 High$

(2) Baseline + slope shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 Low + \beta_4 NC \times Low + \beta_5 Medium + \beta_6 NC \times Medium + \beta_7 High + \beta_8 NC \times High$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+20.70^{***}	+18.25^{***}
New Cases	-13.67^{***}	-12.51^{***}
New Cases ²	+1.66^{***}	+1.66^{***}
Low	+3.62	+6.43
New Cases × Low	—	-1.32
Medium	+5.94[*]	+9.43
New Cases × Med	—	-1.64
High	+8.46^{**}	+11.98[*]
New Cases × High	—	-1.66
R ²	0.317	0.322
N	160	160

Baseline = reasoning off. Highlighted rows = reasoning level tests. ^{***} p<0.001 ^{**} p<0.01 ^{*} p<0.05

FIGURE A.1.2

Gemini 3 Flash Preview — Reasoning Level (off / low / medium / high)

Corresponds to Figure 5. Baseline = reasoning off.

(1) Baseline shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 Low + \beta_4 Medium + \beta_5 High$

(2) Baseline + slope shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 Low + \beta_4 NC \times Low + \beta_5 Medium + \beta_6 NC \times Medium + \beta_7 High + \beta_8 NC \times High$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+90.28^{***}	+90.15^{***}
New Cases	-5.15^{***}	-5.09^{***}
New Cases ²	+0.42^{***}	+0.42^{***}
Low	-9.20^{***}	-9.42^{***}
New Cases × Low	—	+0.11
Medium	-9.71^{***}	-9.22^{***}
New Cases × Med	—	-0.23
High	-8.85^{***}	-8.59^{***}
New Cases × High	—	-0.12
R ²	0.845	0.846
N	160	160

Baseline = reasoning off. Highlighted rows = reasoning level tests. ^{***} p<0.001 ^{**} p<0.01 ^{*} p<0.05

A.2 — MODEL SIZE COMPARISONS

FIGURE A.2.1

Anthropic — Model Size (Haiku · Sonnet · Opus 4.5)

Corresponds to Figure 6. Baseline = Claude Haiku 4.5.

(1) Baseline shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{Sonnet 4.5} + \beta_4 \text{Opus 4.5}$
(2) Baseline + slope shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{Sonnet 4.5} + \beta_4 NC \times \text{Sonnet 4.5} + \beta_5 \text{Opus 4.5} + \beta_6 NC \times \text{Opus 4.5}$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+101.73***	+103.23***
New Cases	-9.84***	-10.55***
New Cases ²	+0.69***	+0.69***
Sonnet 4.5	-17.58***	-18.11***
New Cases × Sonnet 4.5	—	+0.25
Opus 4.5	-2.47**	-6.45***
New Cases × Opus 4.5	—	+1.87***
R ²	0.922	0.933
N	120	120

Baseline = Claude Haiku 4.5. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

FIGURE A.2.2

Gemini — Model Size (2.5 Flash Lite · 2.5 Flash)

Corresponds to Figure 6. Baseline = Gemini 2.5 Flash Lite.

(1) Baseline shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{2.5 Flash}$
(2) Baseline + slope shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{2.5 Flash} + \beta_4 NC \times \text{2.5 Flash}$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+38.78***	+29.92***
New Cases	+9.38***	+13.55***
New Cases ²	-1.18***	-1.18***
2.5 Flash	+30.92***	+48.64***
New Cases × 2.5 Flash	—	-8.34***
R ²	0.702	0.809
N	80	80

Baseline = Gemini 2.5 Flash Lite. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

A.3 — MODEL EVOLUTION

FIGURE A.3.1

Gemini — Evolution (2.0 Flash → 2.5 Flash → 3 Flash)

Corresponds to Figure 7. Baseline = Gemini 2.0 Flash.

(1) Baseline shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{2.5 Flash} + \beta_4 \text{3 Flash}$
(2) Baseline + slope shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{2.5 Flash} + \beta_4 NC \times \text{2.5 Flash} + \beta_5 \text{3 Flash} + \beta_6 NC \times \text{3 Flash}$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+60.76***	+66.82***
New Cases	-9.01***	-11.86***
New Cases ²	+0.91***	+0.91***
2.5 Flash	+33.63***	+23.72***
New Cases × 2.5 Flash	—	+4.66***
3 Flash	+34.39***	+26.11***
New Cases × 3 Flash	—	+3.90***
R ²	0.885	0.914
N	120	120

Baseline = Gemini 2.0 Flash. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

FIGURE A.3.2

Anthropic — Evolution (Sonnet 4.0 → Sonnet 4.5)

Corresponds to Figure 7. Baseline = Claude Sonnet 4.0.

(1) **Baseline shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 Sonnet\ 4.5$
(2) **Baseline + slope shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 Sonnet\ 4.5 + \beta_4 NC \times Sonnet\ 4.5$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+67.82***	+67.13***
New Cases	-10.27***	-9.94***
New Cases ²	+0.74***	+0.74***
Sonnet 4.5	+16.91***	+18.29***
New Cases × Sonnet 4.5	—	-0.65
R ²	0.871	0.872
N	80	80

Baseline = Claude Sonnet 4.0. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

FIGURE A.3.3

OpenAI — Evolution (GPT-3.5 Turbo → 4o → 5.1 → 5.2)

Corresponds to Figure 7. Baseline = GPT-3.5 Turbo.

(1) **Baseline shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 GPT-4o + \beta_4 GPT-5.1 + \beta_5 GPT-5.2$
(2) **Baseline + slope shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 GPT-4o + \beta_4 NC \times GPT-4o + \beta_5 GPT-5.1 + \beta_6 NC \times GPT-5.1 + \beta_7 GPT-5.2 + \beta_8 NC \times GPT-5.2$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+19.63***	+13.11***
New Cases	-11.41***	-8.34***
New Cases ²	+1.08***	+1.08***
GPT-4o	+93.05***	+96.06***
New Cases × GPT-4o	—	-1.41
GPT-5.1	+38.28***	+59.85***
New Cases × GPT-5.1	—	-10.15***
GPT-5.2	+0.28	+1.81
New Cases × GPT-5.2	—	-0.72
R ²	0.925	0.949
N	160	160

Baseline = GPT-3.5 Turbo. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

A.4 — CROSS-PROVIDER FLAGSHIP

FIGURE A.4.1

Flagship Models — Claude Opus 4.5 vs GPT-5.2 vs Gemini 3 Flash

Corresponds to Figure 4. Baseline = Claude Opus 4.5.

(1) **Baseline shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 GPT-5.2 + \beta_4 Gemini\ 3\ Flash$
(2) **Baseline + slope shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 GPT-5.2 + \beta_4 NC \times GPT-5.2 + \beta_5 Gemini\ 3\ Flash + \beta_6 NC \times Gemini\ 3\ Flash$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+94.09***	+96.89***
New Cases	-7.47***	-8.78***
New Cases ²	+0.70**	+0.70**
GPT-5.2	-80.01***	-84.09***
New Cases × GPT-5.2	—	+1.92
Gemini 3 Flash	-0.82	-5.13
New Cases × Gemini 3 Flash	—	+2.02
R ²	0.950	0.952
N	120	120

Baseline = Claude Opus 4.5. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

A.5 — KNOWLEDGE CUTOFF ERA COMPARISONS

FIGURE A.5.1

Knowledge Cutoff — Pre-2024 Era

Corresponds to Figure 9. Baseline = GPT-3.5 Turbo (Sep 2021 cutoff).

(1) **Baseline shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{ Claude 3 Haiku} + \beta_4 \text{ GPT-4o}$

(2) **Baseline + slope shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{ Claude 3 Haiku} + \beta_4 NC \times \text{Claude 3 Haiku} + \beta_5 \text{ GPT-4o} + \beta_6 NC \times \text{GPT-4o}$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+11.12***	+9.67***
New Cases	-5.46***	-4.78**
New Cases ²	+0.47*	+0.47*
Claude 3 Haiku	+6.62**	+7.98*
New Cases × Claude 3 Haiku	—	-0.64
GPT-4o	+93.06***	+96.06***
New Cases × GPT-4o	—	-1.41
R ²	0.958	0.958
N	120	120

Baseline = GPT-3.5 Turbo. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

FIGURE A.5.2

Knowledge Cutoff — Mid-2024 (Jun–Sep)

Corresponds to Figure 9. Baseline = GPT-4.1 (Jun 2024 cutoff).

(1) **Baseline shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 o3 + \beta_4 \text{ Gemini 2.0 Flash} + \beta_5 \text{ GPT-5.1}$

(2) **Baseline + slope shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 o3 + \beta_4 NC \times o3 + \beta_5 \text{ Gemini 2.0 Flash} + \beta_6 NC \times \text{Gemini 2.0 Flash} + \beta_7 \text{ GPT-5.1} + \beta_8 NC \times \text{GPT-5.1}$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+115.43***	+107.82***
New Cases	-20.68***	-17.10***
New Cases ²	+2.08***	+2.08***
o3	-43.17***	-32.07***
New Cases × o3	—	-5.22***
Gemini 2.0 Flash	-37.95***	-34.28***
New Cases × Gemini 2.0 Flash	—	-1.73
GPT-5.1	-44.75***	-29.10***
New Cases × GPT-5.1	—	-7.36***
R ²	0.896	0.928
N	160	160

Baseline = GPT-4.1. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

FIGURE A.5.3

Knowledge Cutoff — Early 2025 (Jan–Mar)

Corresponds to Figure 9. Baseline = Claude Sonnet 4.0 (Jan 2025 cutoff).

(1) **Baseline shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{ Haiku 4.5} + \beta_4 \text{ Sonnet 4.0} + \beta_5 \text{ Gemini 2.5 Flash} + \beta_6 \text{ Gemini 2.5 Flash Lite} + \beta_7 \text{ Gemini 3 Flash}$

(2) **Baseline + slope shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{ Haiku 4.5} + \beta_4 NC \times \text{Haiku 4.5} + \beta_5 \text{ Sonnet 4.0} + \beta_6 NC \times \text{Sonnet 4.0} + \beta_7 \text{ Gemini 2.5 Flash} + \beta_8 NC \times \text{Gemini 2.5 Flash} + \beta_9 \text{ Gemini 2.5 Flash Lite} + \beta_{10} \text{ Gemini 2.5 Flash Lite} \times \text{Gemini 2.5 Flash Lite} + \beta_{11} \text{ Gemini 3 Flash} + \beta_{12} \text{ Gemini 3 Flash} \times \text{Gemini 3 Flash}$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+73.16***	+80.68***
New Cases	-2.15	-5.69***
New Cases ²	-0.09	-0.09
Haiku 4.5	+17.57***	+18.11***
New Cases × Haiku 4.5	—	-0.25
Sonnet 4.0	-16.91***	-18.29***
New Cases × Sonnet 4.0	—	+0.65
Gemini 2.5 Flash	+13.52***	+4.14
R ²	0.718	0.868
N	240	240

Baseline = Claude Sonnet 4.5. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

Variable	(1) Baseline shift	(2) Baseline + slope shift
New Cases × Gemini 2.5 Flash	—	+4.41***
Gemini 2.5 Flash Lite	-17.39***	-44.50***
New Cases × Gemini 2.5 Flash Lite	—	+12.76***
Gemini 3 Flash	+14.28***	+6.53*
New Cases × Gemini 3 Flash	—	+3.65***
R ²	0.718	0.868
N	240	240

Baseline = Claude Sonnet 4.5. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

FIGURE A.5.4

Knowledge Cutoff — Late 2025 (Aug)

Corresponds to Figure 9. Baseline = Claude Opus 4.5 (Aug 2025 cutoff).

(1) Baseline shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 GPT-5.2$

(2) Baseline + slope shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 GPT-5.2 + \beta_4 NC \times GPT-5.2$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+96.20***	+98.24***
New Cases	-9.22***	-10.18***
New Cases ²	+0.94**	+0.94**
GPT-5.2	-80.01***	-84.09***
New Cases × GPT-5.2	—	+1.92
R ²	0.938	0.939
N	80	80

Baseline = Claude Opus 4.5. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

A.6 — RELEASE DATE ERA COMPARISONS

FIGURE A.6.1

Release Date — Legacy Era (pre-2025)

Corresponds to Figure 11. Baseline = GPT-3.5 Turbo (Jan 2024 release).

(1) Baseline shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{Claude 3 Haiku} + \beta_4 GPT-4o$

(2) Baseline + slope shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{Claude 3 Haiku} + \beta_4 NC \times \text{Claude 3 Haiku} + \beta_5 GPT-4o + \beta_6 NC \times GPT-4o$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+11.12***	+9.67***
New Cases	-5.46***	-4.78**
New Cases ²	+0.47*	+0.47*
Claude 3 Haiku	+6.62**	+7.98*
New Cases × Claude 3 Haiku	—	-0.64
GPT-4o	+93.06***	+96.06***
New Cases × GPT-4o	—	-1.41
R ²	0.958	0.958
N	120	120

Baseline = GPT-3.5 Turbo. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

FIGURE A.6.2

Release Date — Early 2025 (Feb–Jun)

Corresponds to Figure 11. Baseline = Gemini 2.0 Flash (Feb 2025 release).

(1) Baseline shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 GPT-4.1 + \beta_4 o3 + \beta_5 \text{Sonnet 4.0} + \beta_6 \text{Gemini 2.5 Flash} + \beta_7 \text{Gemini 2.5 Flash Lite}$

(2) Baseline + slope shift: $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 GPT-4.1 + \beta_4 NC \times GPT-4.1 + \beta_5 o3 + \beta_6 NC \times o3 + \beta_7 \text{Sonnet 4.0} + \beta_8 NC \times \text{Sonnet 4.0} + \beta_9 \text{Gemini 2.5 Flash} + \beta_{\text{undefined}} NC \times \text{Gemini 2.5 Flash} + \beta_{\text{undefined}} \text{Gemini 2.5 Flash Lite} + \beta_{\text{undefined}} NC \times \text{Gemini 2.5 Flash Lite}$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+58.99***	+64.94***
New Cases	-7.11***	-9.91***
New Cases ²	+0.58**	+0.58***
GPT-4.1	+37.95***	+34.28***
New Cases × GPT-4.1	—	+1.73
o3	-5.22	+2.21
New Cases × o3	—	-3.49**
Sonnet 4.0	+3.20	+1.29
New Cases × Sonnet 4.0	—	+0.90
Gemini 2.5 Flash	+33.63***	+23.72***
New Cases × Gemini 2.5 Flash	—	+4.66***
Gemini 2.5 Flash Lite	+2.72	-24.92***
New Cases × Gemini 2.5 Flash Lite	—	+13.01***
R ²	0.701	0.836
N	240	240

Baseline = Gemini 2.0 Flash. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

FIGURE A.6.3

Release Date — Late 2025 (Sep–Dec)

Corresponds to Figure 11. Baseline = Claude Sonnet 4.5 (Sep 2025 release).

- (1) **Baseline shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{Haiku 4.5} + \beta_4 \text{Opus 4.5} + \beta_5 \text{GPT-5.1} + \beta_6 \text{GPT-5.2} + \beta_7 \text{Gemini 3 Flash}$
- (2) **Baseline + slope shift:** $Mobility = \beta_0 + \beta_1 NC + \beta_2 NC^2 + \beta_3 \text{Haiku 4.5} + \beta_4 NC \times \text{Haiku 4.5} + \beta_5 \text{Opus 4.5} + \beta_6 NC \times \text{Opus 4.5} + \beta_7 \text{GPT-5.1} + \beta_8 NC \times \text{GPT-5.1} + \beta_9 \text{GPT-5.2} + \beta_{10} \text{undefined NC} \times \text{GPT-5.2} + \beta_{11} \text{undefined Gemini 3 Flash} + \beta_{12} \text{undefined NC} \times \text{Gemini 3 Flash}$

Variable	(1) Baseline shift	(2) Baseline + slope shift
Constant	+86.15***	+87.10***
New Cases	-11.90***	-12.34***
New Cases ²	+1.03***	+1.03***
Haiku 4.5	+17.57***	+18.11***
New Cases × Haiku 4.5	—	-0.25
Opus 4.5	+15.10***	+11.66***
New Cases × Opus 4.5	—	+1.62
GPT-5.1	-26.90***	-14.39***
New Cases × GPT-5.1	—	-5.89***
GPT-5.2	-64.91***	-72.44***
New Cases × GPT-5.2	—	+3.54**
Gemini 3 Flash	+14.28***	+6.53*
New Cases × Gemini 3 Flash	—	+3.65***
R ²	0.927	0.950
N	240	240

Baseline = Claude Sonnet 4.5. Highlighted rows = model comparison tests. *** p<0.001 ** p<0.01 * p<0.05

APPENDIX B • LOGISTIC REGRESSIONS

Per-Model Logistic Regression Tables — All 21 Configurations

Full coefficient tables for each of the 21 LLM configurations under two regression specifications. **B.1** shows the fixed-effects and random-effects logits used in Figure 12 (infection + traits + demographics). **B.2** shows the random-effects logit with mention flags used in Figure 17 (adds yes/no flags for whether the response mentioned each trait label or context category). Diagnostic validation for both specifications is provided in Appendix C.

B.1 — FIXED-EFFECTS & RANDOM-EFFECTS LOGITS (FIGURE 12)

FIGURE B.1.1

Logistic Regression — Claude Opus 4.5

Claude Opus 4.5

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-27.220	1759.242	[-3475.33, 3420.90]	< 10 ⁻⁶		-14.973	1.441	[-17.80, -12.15]	< 10 ⁻⁶	***
Infection Rate (%)	2.513	0.097	[2.32, 2.70]	12.338	***	2.507	0.096	[2.32, 2.70]	12.268	***
Infection Rate ² (%)	-0.191	0.013	[-0.22, -0.17]	0.826	***	-0.192	0.013	[-0.22, -0.17]	0.825	***
Male						-1.798	0.573	[-2.92, -0.68]	0.166	**
Extraverted						-5.019	0.619	[-6.23, -3.81]	0.007	***
Agreeable						5.299	0.606	[4.11, 6.49]	200.189	***
Conscientious						8.896	0.812	[7.30, 10.49]	7299.211	***
Emotionally Stable						-5.021	0.622	[-6.24, -3.80]	0.007	***
Open to Experience						2.447	0.601	[1.27, 3.63]	11.550	***
Age (years)						0.029	0.021	[-0.01, 0.07]	1.029	
AIC			4,456.7					4,505.16		
BIC			5,262.86					4,592.1		
Pseudo R ²			0.7662					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								3.6088		
RE SD (σ_u)								1.8997		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.2

Logistic Regression — Claude Sonnet 4.5

Claude Sonnet 4.5

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-26.448	1087.121	[-2157.21, 2104.31]	< 10 ⁻⁶		-7.465	1.196	[-9.81, -5.12]	0.001	***
Infection Rate (%)	3.091	0.087	[2.92, 3.26]	22.006	***	3.075	0.086	[2.91, 3.24]	21.640	***
Infection Rate ² (%)	-0.297	0.012	[-0.32, -0.27]	0.743	***	-0.296	0.012	[-0.32, -0.27]	0.744	***
Male						-0.652	0.533	[-1.70, 0.39]	0.521	
Extraverted						-5.408	0.558	[-6.50, -4.31]	0.004	***
Agreeable						2.944	0.536	[1.89, 3.99]	18.982	***
Conscientious						7.755	0.579	[6.62, 8.89]	2334.136	***
Emotionally Stable						-5.707	0.579	[-6.84, -4.57]	0.003	***
Open to Experience						0.185	0.530	[-0.85, 1.22]	1.203	
Age (years)						-0.001	0.020	[-0.04, 0.04]	0.999	
AIC			6,092.94					6,239.4		
BIC			6,899.1					6,326.34		
Pseudo R ²			0.7652					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								4.4111		
RE SD (σ_u)								2.1003		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.3

Logistic Regression — Claude Haiku 4.5

Claude Haiku 4.5

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-24.721	1085.174	[-2151.66, 2102.22]	< 10 ⁻⁶		-8.287	0.762	[-9.78, -6.79]	0.000	***
Infection Rate (%)	1.697	0.070	[1.56, 1.83]	5.456	***	1.695	0.069	[1.56, 1.83]	5.444	***
Infection Rate ² (%)	-0.100	0.010	[-0.12, -0.08]	0.905	***	-0.101	0.009	[-0.12, -0.08]	0.904	***
Male						0.477	0.330	[-0.17, 1.12]	1.612	
Extraverted						-4.432	0.351	[-5.12, -3.74]	0.012	***
Agreeable						1.642	0.329	[1.00, 2.29]	5.167	***
Conscientious						4.341	0.362	[3.63, 5.05]	76.753	***
Emotionally Stable						-0.826	0.345	[-1.50, -0.15]	0.438	*
Open to Experience						-0.282	0.333	[-0.93, 0.37]	0.754	
Age (years)						0.004	0.012	[-0.02, 0.03]	1.004	
AIC			7,077.54					7,182.64		
BIC			7,883.7					7,269.58		
Pseudo R ²			0.5842					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								1.6820		
RE SD (σ_u)								1.2969		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.4

Logistic Regression — Claude Sonnet 4.0

Claude Sonnet 4.0

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-1.049	0.183	[-1.41, -0.69]	0.350	***	0.030	1.193	[-2.31, 2.37]	1.031	
Infection Rate (%)	0.922	0.063	[0.80, 1.05]	2.515	***	0.941	0.062	[0.82, 1.06]	2.562	***
Infection Rate ² (%)	0.025	0.011	[0.00, 0.05]	1.026	*	0.018	0.010	[-0.00, 0.04]	1.018	.
Male						-0.656	0.515	[-1.67, 0.35]	0.519	
Extraverted						-5.436	0.535	[-6.48, -4.39]	0.004	***
Agreeable						1.983	0.512	[0.98, 2.99]	7.266	***
Conscientious						5.635	0.523	[4.61, 6.66]	279.926	***
Emotionally Stable						-6.381	0.550	[-7.46, -5.30]	0.002	***
Open to Experience						-0.361	0.517	[-1.38, 0.65]	0.697	
Age (years)						0.016	0.019	[-0.02, 0.05]	1.016	
AIC			7,398.81					7,617.99		
BIC			8,204.97					7,704.92		
Pseudo R ²			0.7404					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								5.1588		
RE SD (σ_u)								2.2713		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.5

Logistic Regression — Claude 3 Haiku

Claude 3 Haiku

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	2.083	0.461	[1.18, 2.99]	8.026	***	1.735	0.560	[0.64, 2.83]	5.668	**
Infection Rate (%)	1.554	0.060	[1.44, 1.67]	4.728	***	1.525	0.059	[1.41, 1.64]	4.593	***
Infection Rate² (%)	−0.180	0.010	[−0.20, −0.16]	0.835	***	−0.177	0.009	[−0.20, −0.16]	0.838	***
Male						0.726	0.253	[0.23, 1.22]	2.066	**
Extraverted						−1.422	0.259	[−1.93, −0.92]	0.241	***
Agreeable						0.071	0.250	[−0.42, 0.56]	1.074	
Conscientious						0.801	0.257	[0.30, 1.30]	2.228	**
Emotionally Stable						−0.686	0.259	[−1.19, −0.18]	0.504	**
Open to Experience						−0.053	0.255	[−0.55, 0.45]	0.948	
Age (years)						0.001	0.009	[−0.02, 0.02]	1.001	
AIC			7,473.09					7,627.32		
BIC			8,279.25					7,714.26		
Pseudo R²			0.4168					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								1.3450		
RE SD (σ_u)								1.1597		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.6

Logistic Regression — GPT-5.2

GPT-5.2

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	−0.939	0.688	[−2.29, 0.41]	0.391		−2.366	0.387	[−3.12, −1.61]	0.094	***
Infection Rate (%)	21.387	0.937	[19.55, 23.22]	> 10 ⁶	***	19.070	0.809	[17.48, 20.66]	> 10 ⁶	***
Infection Rate² (%)	−2.850	0.133	[−3.11, −2.59]	0.058	***	−2.548	0.118	[−2.78, −2.32]	0.078	***
Male						0.295	0.169	[−0.04, 0.63]	1.343	.
Extraverted						−0.041	0.171	[−0.38, 0.29]	0.959	
Agreeable						0.517	0.167	[0.19, 0.85]	1.677	**
Conscientious						0.569	0.172	[0.23, 0.91]	1.767	***
Emotionally Stable						−0.440	0.172	[−0.78, −0.10]	0.644	*
Open to Experience						0.584	0.171	[0.25, 0.92]	1.793	***
Age (years)						0.010	0.006	[−0.00, 0.02]	1.010	
AIC			1,606.04					1,571.44		
BIC			2,412.2					1,658.38		
Pseudo R²			0.7438					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								0.2531		
RE SD (σ_u)								0.5031		

Notes

20,000 observations per configuration (100 agents $\times$ 40 infection levels $\times$ 5 repetitions). Computed in R using `glm()` and `lme4::glmer(optimizer = bobqvga)`.

Logistic Regression – GPT-5.2 (low reasoning)

GPT-5.2 (low)

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	0.181	0.337	[-0.48, 0.84]	1.199		-0.581	0.342	[-1.25, 0.09]	0.559	.
Infection Rate (%)	3.191	0.088	[3.02, 3.36]	24.303	***	3.112	0.085	[2.95, 3.28]	22.473	***
Infection Rate ² (%)	-0.384	0.013	[-0.41, -0.36]	0.681	***	-0.374	0.013	[-0.40, -0.35]	0.688	***
Male						0.306	0.153	[0.01, 0.61]	1.358	*
Extraverted						-0.493	0.157	[-0.80, -0.19]	0.611	**
Agreeable						0.902	0.151	[0.60, 1.20]	2.464	***
Conscientious						1.236	0.156	[0.93, 1.54]	3.440	***
Emotionally Stable						-1.184	0.159	[-1.50, -0.87]	0.306	***
Open to Experience						0.526	0.154	[0.22, 0.83]	1.692	***
Age (years)						0.012	0.006	[0.00, 0.02]	1.012	*
AIC						5,845.4				
BIC						6,651.56				
Pseudo R ²						0.4253				
N						20,000				
Groups (agents)						100				
RE Variance (σ^2_u)						0.3970				
RE SD (σ_u)						0.6300				

20,000 observations per configuration (100 agents $\times$ 40 infection levels $\times$ 5 repetitions). Computed in R using `glm()` and `lme4::glmer(optimizer = bobqvga)`.

Logistic Regression – GPT-5.2 (medium reasoning)

GPT-5.2 (med)

Predictor	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit					
Intercept	0.280	0.319	[−0.35, 0.91]	1.323	
Infection Rate (%)	2.743	0.074	[2.60, 2.89]	15.526	***
Infection Rate² (%)	−0.321	0.011	[−0.34, −0.30]	0.725	***
Male					
Extraverted					
Agreeable					
Conscientious					
Emotionally Stable					
Open to Experience					
Age (years)					
AIC			6,850.64		
BIC			7,656.79		
Pseudo R²			0.4510		
N			20,000		
Groups (agents)					

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
RE Variance (σ^2_u)							0.6674			
RE SD (σ_u)							0.8170			

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.9

Logistic Regression — GPT-5.2 (high reasoning)

GPT-5.2 (high)

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	0.689	0.334	[0.03, 1.34]	1.993	*	−0.648	0.445	[−1.52, 0.23]	0.523	
Infection Rate (%)	2.373	0.065	[2.25, 2.50]	10.724	***	2.321	0.063	[2.20, 2.44]	10.187	***
Infection Rate ² (%)	−0.278	0.010	[−0.30, −0.26]	0.758	***	−0.272	0.010	[−0.29, −0.25]	0.762	***
Male						0.247	0.199	[−0.14, 0.64]	1.280	
Extraverted						−0.878	0.204	[−1.28, −0.48]	0.416	***
Agreeable						1.318	0.197	[0.93, 1.70]	3.736	***
Conscientious						1.459	0.203	[1.06, 1.86]	4.304	***
Emotionally Stable						−1.760	0.206	[−2.16, −1.36]	0.172	***
Open to Experience						0.838	0.201	[0.44, 1.23]	2.311	***
Age (years)						0.016	0.007	[0.00, 0.03]	1.016	*
AIC			7,935.41					8,063.06		
BIC			8,741.56					8,150		
Pseudo R ²			0.4591					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								0.7990		
RE SD (σ_u)								0.8939		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.10

Logistic Regression — GPT-5.1

GPT-5.1

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	−2.331	0.193	[−2.71, −1.95]	0.097	***	−1.850	0.722	[−3.27, −0.43]	0.157	*
Infection Rate (%)	1.980	0.047	[1.89, 2.07]	7.240	***	1.980	0.047	[1.89, 2.07]	7.240	***
Infection Rate ² (%)	−0.150	0.008	[−0.17, −0.13]	0.860	***	−0.152	0.008	[−0.17, −0.14]	0.859	***
Male						−0.510	0.325	[−1.15, 0.13]	0.600	
Extraverted						−2.231	0.331	[−2.88, −1.58]	0.107	***
Agreeable						0.583	0.322	[−0.05, 1.21]	1.791	.
Conscientious						0.612	0.329	[−0.03, 1.26]	1.844	.
Emotionally Stable						−1.528	0.332	[−2.18, −0.88]	0.217	***
Open to Experience						2.799	0.329	[2.15, 3.44]	16.426	***

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Age (years)						-0.011	0.012	[-0.03, 0.01]	0.989	
AIC		14,105.46						14,415.9		
BIC		14,911.62						14,502.84		
Pseudo R ²		0.4867						—		
N		20,000						20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								2.4199		
RE SD (σ_u)								1.5556		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.11

Logistic Regression — GPT-4.1

GPT-4.1

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-25.099	1104.842	[-2190.59, 2140.39]	< 10 ⁻⁶		-10.245	0.946	[-12.10, -8.39]	0.000	***
Infection Rate (%)	2.158	0.082	[2.00, 2.32]	8.651	***	2.143	0.081	[1.98, 2.30]	8.526	***
Infection Rate ² (%)	-0.171	0.011	[-0.19, -0.15]	0.843	***	-0.171	0.011	[-0.19, -0.15]	0.843	***
Male						-0.082	0.413	[-0.89, 0.73]	0.921	
Extraverted						-5.271	0.445	[-6.14, -4.40]	0.005	***
Agreeable						2.484	0.412	[1.68, 3.29]	11.987	***
Conscientious						4.724	0.453	[3.84, 5.61]	112.566	***
Emotionally Stable						-4.112	0.446	[-4.99, -3.24]	0.016	***
Open to Experience						3.392	0.435	[2.54, 4.24]	29.737	***
Age (years)						0.014	0.015	[-0.02, 0.04]	1.014	
AIC		5,499.09						5,569.78		
BIC		6,305.24						5,656.72		
Pseudo R ²		0.6759						—		
N		20,000						20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								2.0837		
RE SD (σ_u)								1.4435		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.12

Logistic Regression — GPT-4o

GPT-4o

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-26.777	1717.261	[-3392.61, 3339.05]	< 10 ⁻⁶		-9.154	1.525	[-12.14, -6.16]	0.000	***
Infection Rate (%)	1.624	0.132	[1.36, 1.88]	5.071	***	1.640	0.132	[1.38, 1.90]	5.155	***
Infection Rate ² (%)	-0.055	0.016	[-0.09, -0.02]	0.947	***	-0.059	0.016	[-0.09, -0.03]	0.943	***
Male						0.529	0.694	[-0.83, 1.89]	1.697	

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Extraverted						-4.439	0.735	[-5.88, -3.00]	0.012	***
Agreeable						1.797	0.702	[0.42, 3.17]	6.031	*
Conscientious						2.261	0.696	[0.90, 3.62]	9.595	**
Emotionally Stable						-1.316	0.693	[-2.68, 0.04]	0.268	.
Open to Experience						0.295	0.701	[-1.08, 1.67]	1.343	
Age (years)						-0.036	0.026	[-0.09, 0.01]	0.964	
AIC		2,852.4						2,929.57		
BIC		3,658.56						3,016.51		
Pseudo R²		0.6176						—		
N		20,000						20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								6.3141		
RE SD (σ_u)								2.5128		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.13

Logistic Regression — GPT-3.5 Turbo

GPT-3.5 Turbo

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	0.900	0.493	[-0.07, 1.87]	2.459	.	2.354	0.766	[0.85, 3.86]	10.523	**
Infection Rate (%)	4.389	0.177	[4.04, 4.74]	80.543	***	4.301	0.173	[3.96, 4.64]	73.784	***
Infection Rate² (%)	-0.548	0.026	[-0.60, -0.50]	0.578	***	-0.537	0.025	[-0.59, -0.49]	0.585	***
Male						-0.325	0.328	[-0.97, 0.32]	0.723	
Extraverted						-3.899	0.445	[-4.77, -3.03]	0.020	***
Agreeable						1.436	0.333	[0.78, 2.09]	4.205	***
Conscientious						0.769	0.335	[0.11, 1.43]	2.158	*
Emotionally Stable						0.226	0.327	[-0.41, 0.87]	1.254	
Open to Experience						-0.446	0.334	[-1.10, 0.21]	0.640	
Age (years)						0.033	0.012	[0.01, 0.06]	1.034	**
AIC		2,493.28						2,508.2		
BIC		3,299.44						2,595.14		
Pseudo R²		0.5494						—		
N		20,000						20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								1.3539		
RE SD (σ_u)								1.1636		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.14

Logistic Regression — o3

o3

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-3.750	0.181	[-4.11, -3.39]	0.024	***	-2.208	0.457	[-3.10, -1.31]	0.110	***
Infection Rate (%)	2.121	0.043	[2.04, 2.21]	8.338	***	2.114	0.043	[2.03, 2.20]	8.285	***
Infection Rate² (%)	-0.227	0.006	[-0.24, -0.21]	0.797	***	-0.226	0.006	[-0.24, -0.21]	0.798	***
Male						0.205	0.203	[-0.19, 0.60]	1.228	
Extraverted						-3.092	0.209	[-3.50, -2.68]	0.045	***
Agreeable						0.729	0.202	[0.33, 1.12]	2.073	***
Conscientious						0.493	0.207	[0.09, 0.90]	1.637	*
Emotionally Stable						-2.056	0.209	[-2.47, -1.65]	0.128	***
Open to Experience						1.665	0.207	[1.26, 2.07]	5.284	***
Age (years)						0.024	0.007	[0.01, 0.04]	1.024	**
AIC			15,159.81					15,385.72		
BIC			15,965.97					15,472.66		
Pseudo R²			0.4520					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								0.9172		
RE SD (σ_u)								0.9577		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.15

Logistic Regression — Gemini 3 Flash

Gemini 3 Flash

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-22.512	1196.522	[-2367.70, 2322.67]	< 10 ⁻⁶		-5.840	1.236	[-8.26, -3.42]	0.003	***
Infection Rate (%)	0.989	0.067	[0.86, 1.12]	2.688	***	0.988	0.066	[0.86, 1.12]	2.686	***
Infection Rate² (%)	-0.069	0.010	[-0.09, -0.05]	0.933	***	-0.070	0.010	[-0.09, -0.05]	0.932	***
Male						-1.090	0.559	[-2.19, 0.01]	0.336	.
Extraverted						-5.837	0.627	[-7.07, -4.61]	0.003	***
Agreeable						3.253	0.567	[2.14, 4.36]	25.878	***
Conscientious						4.592	0.591	[3.43, 5.75]	98.662	***
Emotionally Stable						-4.711	0.613	[-5.91, -3.51]	0.009	***
Open to Experience						3.786	0.603	[2.60, 4.97]	44.060	***
Age (years)						-0.028	0.021	[-0.07, 0.01]	0.972	
AIC			5,822.89					5,898.41		
BIC			6,629.05					5,985.34		
Pseudo R²			0.6996					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ^2_u)								3.6634		
RE SD (σ_u)								1.9140		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.16

Logistic Regression — Gemini 3 Flash (low reasoning)

Gemini 3 Flash (low)

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-22.480	1206.646	[-2387.51, 2342.55]	< 10 ⁻⁶		-5.313	1.039	[-7.35, -3.28]	0.005	***
Infection Rate (%)	1.106	0.063	[0.98, 1.23]	3.022	***	1.098	0.062	[0.98, 1.22]	2.999	***
Infection Rate² (%)	-0.100	0.009	[-0.12, -0.08]	0.905	***	-0.099	0.009	[-0.12, -0.08]	0.905	***
Male						-0.503	0.458	[-1.40, 0.40]	0.605	
Extraverted						-6.466	0.513	[-7.47, -5.46]	0.002	***
Agreeable						3.428	0.472	[2.50, 4.35]	30.815	***
Conscientious						4.993	0.486	[4.04, 5.94]	147.392	***
Emotionally Stable						-6.263	0.527	[-7.30, -5.23]	0.002	***
Open to Experience						3.636	0.477	[2.70, 4.57]	37.933	***
Age (years)						0.010	0.017	[-0.02, 0.04]	1.010	
AIC			6,420.4					6,504.83		
BIC			7,226.55					6,591.77		
Pseudo R²			0.7333					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ² _u)								2.7263		
RE SD (σ _u)								1.6511		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.17

Logistic Regression — Gemini 3 Flash (medium reasoning)

Gemini 3 Flash (med)

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	-22.737	1197.081	[-2369.02, 2323.54]	< 10 ⁻⁶		-6.186	1.061	[-8.27, -4.11]	0.002	***
Infection Rate (%)	1.240	0.063	[1.12, 1.36]	3.454	***	1.235	0.063	[1.11, 1.36]	3.439	***
Infection Rate² (%)	-0.112	0.009	[-0.13, -0.09]	0.894	***	-0.112	0.009	[-0.13, -0.09]	0.894	***
Male						-0.535	0.468	[-1.45, 0.38]	0.586	
Extraverted						-6.235	0.516	[-7.25, -5.22]	0.002	***
Agreeable						3.364	0.479	[2.43, 4.30]	28.912	***
Conscientious						5.165	0.496	[4.19, 6.14]	175.036	***
Emotionally Stable						-6.587	0.536	[-7.64, -5.54]	0.001	***
Open to Experience						3.705	0.484	[2.76, 4.65]	40.639	***
Age (years)						0.027	0.018	[-0.01, 0.06]	1.027	
AIC			6,588.2					6,681.72		
BIC			7,394.36					6,768.66		
Pseudo R²			0.7285					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ² _u)								2.8634		
RE SD (σ _u)								1.6922		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.18

Logistic Regression — Gemini 3 Flash (high reasoning)

Gemini 3 Flash (high)

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	−22.480	1205.057	[−2384.39, 2339.43]	< 10 ^{−6}		−6.032	1.021	[−8.03, −4.03]	0.002	***
Infection Rate (%)	1.095	0.061	[0.98, 1.21]	2.989	***	1.091	0.060	[0.97, 1.21]	2.976	***
Infection Rate² (%)	−0.097	0.009	[−0.11, −0.08]	0.908	***	−0.097	0.009	[−0.11, −0.08]	0.908	***
Male						−0.614	0.441	[−1.48, 0.25]	0.541	
Extraverted						−5.756	0.482	[−6.70, −4.81]	0.003	***
Agreeable						3.363	0.454	[2.47, 4.25]	28.886	***
Conscientious						4.959	0.468	[4.04, 5.88]	142.395	***
Emotionally Stable						−6.509	0.509	[−7.51, −5.51]	0.001	***
Open to Experience						3.789	0.462	[2.88, 4.69]	44.210	***
Age (years)						0.024	0.017	[−0.01, 0.06]	1.024	
AIC			6,883.48					6,971.79		
BIC			7,689.63					7,058.73		
Pseudo R²			0.7117					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ² _u)								2.5057		
RE SD (σ _u)								1.5829		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

FIGURE B.1.19

Logistic Regression — Gemini 2.5 Flash Lite

Gemini 2.5 Flash Lite

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
Intercept	0.012	0.235	[−0.45, 0.47]	1.012		8.471	0.839	[6.83, 10.12]	4774.715	***
Infection Rate (%)	−1.863	0.044	[−1.95, −1.78]	0.155	***	−1.861	0.043	[−1.95, −1.78]	0.156	***
Infection Rate² (%)	0.219	0.007	[0.21, 0.23]	1.245	***	0.219	0.006	[0.21, 0.23]	1.244	***
Male						−0.187	0.367	[−0.91, 0.53]	0.829	
Extraverted						−4.993	0.378	[−5.73, −4.25]	0.007	***
Agreeable						0.686	0.362	[−0.02, 1.40]	1.986	.
Conscientious						0.078	0.369	[−0.64, 0.80]	1.081	
Emotionally Stable						−1.470	0.376	[−2.21, −0.73]	0.230	***
Open to Experience						−1.574	0.371	[−2.30, −0.85]	0.207	***
Age (years)						−0.039	0.014	[−0.07, −0.01]	0.962	**
AIC			13,744.38					14,045.73		
BIC			14,550.54					14,132.67		
Pseudo R²			0.5116					—		
N			20,000					20,000		
Groups (agents)								100		
RE Variance (σ² _u)								2.9474		
RE SD (σ _u)								1.7168		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

20,000 observations per configuration (100 agents $\times$ 40 infection levels $\times$ 5 repetitions). Computed in R using `glm()` and `lme4::glmer(optimizer = bobyqa)`.

FIGURE B.1.20

Logistic Regression – Gemini 2.5 Flash

Gemini 2.5 Flash

Predictor	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit					
Intercept	-21.542	1233.438	[-2439.08, 2396.00]	< 10 ⁻⁶	
Infection Rate (%)	0.531	0.055	[0.42, 0.64]	1.700	***
Infection Rate² (%)	-0.036	0.008	[-0.05, -0.02]	0.965	***
Male					
Extraverted					
Agreeable					
Conscientious					
Emotionally Stable					
Open to Experience					
Age (years)					
AIC			7,347.14		
BIC			8,153.3		
Pseudo R²			0.6272		
N			20,000		
Groups (agents)					
RE Variance (σ^2_u)					
RE SD (σ_u)					
Random-Effects Logit					
Intercept	-3.393	0.967	[-5.29, -1.50]	0.034	***
Infection Rate (%)	0.529	0.055	[0.42, 0.64]	1.697	***
Infection Rate² (%)	-0.036	0.008	[-0.05, -0.02]	0.965	***
Male	-0.066	0.435	[-0.92, 0.79]	0.936	
Extraverted	-7.353	0.561	[-8.45, -6.25]	0.001	***
Agreeable	1.630	0.437	[0.77, 2.49]	5.104	***
Conscientious	2.715	0.448	[1.84, 3.59]	15.101	***
Emotionally Stable	-2.611	0.463	[-3.52, -1.70]	0.073	***
Open to Experience	2.431	0.454	[1.54, 3.32]	11.367	***
Age (years)	-0.008	0.017	[-0.04, 0.02]	0.992	
AIC			7,430.62		
BIC			7,517.56		
Pseudo R²			—		
N			20,000		
Groups (agents)			100		
RE Variance (σ^2_u)			2.3788		
RE SD (σ_u)			1.5423		

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

20,000 observations per configuration (100 agents $\times$ 40 infection levels $\times$ 5 repetitions). Computed in R using `glm()` and `lme4::glmer(optimizer = bobyqa)`.

FIGURE B.1.21

Logistic Regression – Gemini 2.0 Flash

Gemini 2.0 Flash

Predictor	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit					
Intercept	-4.628	0.206	[-5.03, -4.22]	0.010	***
Infection Rate (%)	2.083	0.050	[1.99, 2.18]	8.025	***
Infection Rate ² (%)	-0.238	0.007	[-0.25, -0.22]	0.788	***
Male					
Extraverted					
Agreeable					
Conscientious					
Emotionally Stable					
Open to Experience					
Age (years)					
AIC			12,104.5		
BIC			12,910.66		
Pseudo R²			0.5702		
N			20,000		
Groups (agents)					
Random-Effects Logit					
Intercept	-1.669	0.759	[-3.16, -0.18]	0.188	*
Infection Rate (%)	2.074	0.049	[1.98, 2.17]	7.955	***
Infection Rate ² (%)	-0.237	0.007	[-0.25, -0.22]	0.789	***
Male	-0.279	0.339	[-0.94, 0.39]	0.756	
Extraverted	-3.165	0.343	[-3.84, -2.49]	0.042	***
Agreeable	2.130	0.336	[1.47, 2.79]	8.416	***
Conscientious	3.138	0.341	[2.47, 3.81]	23.051	***
Emotionally Stable	-3.506	0.346	[-4.18, -2.83]	0.030	***
Open to Experience	0.690	0.341	[0.02, 1.36]	1.993	*
Age (years)	-0.006	0.012	[-0.03, 0.02]	0.994	
AIC			12,369.83		
BIC			12,456.77		
Pseudo R²			—		
N			20,000		
Groups (agents)			100		

Predictor	Coef	SE	95% CI	OR	Sig	Coef	SE	95% CI	OR	Sig
Fixed-Effects Logit						Random-Effects Logit				
RE Variance (σ^2_u)						2.4789				
RE SD (σ_u)						1.5745				

Notes

Fixed-effects logit (glm) with 99 agent dummies — only infection coefficients reported. Random-effects logistic regression (lme4) with random intercepts per agent — estimates trait and demographic effects.

Dummy coding: male = 1, extraverted = 1, agreeable = 1, conscientious = 1, emotionally stable = 1, open to experience = 1. Reference categories: female, introverted, antagonistic, unconscientious, neurotic, closed to experience.

No normalization: age in raw years (18–65), infection rate as percentage (0–7%).

Significance: *** p < 0.001, ** p < 0.01, * p < 0.05

20,000 observations per configuration (100 agents × 40 infection levels × 5 repetitions). Computed in R using glm() and lme4::glmer(optimizer = bobyqa).

B.2 — RANDOM-EFFECTS LOGIT WITH MENTION FLAGS (FIGURE 17)

FIGURE B.2.1

Random-Effects Logit with Mention Flags — Claude Opus 4.5

Claude Opus 4.5

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	−15.356	1.410	< 10 ^{−6}	[−18.120, −12.592]	***	
Infection Rate (%)	2.498	0.099	12.162	[2.304, 2.692]	***	
Infection Rate ²	−0.184	0.013	0.832	[−0.210, −0.159]	***	
Male	−1.894	0.546	0.150	[−2.963, −0.825]	***	
Extraverted	−4.417	0.600	0.012	[−5.594, −3.240]	***	
Agreeable	4.813	0.602	123.041	[3.633, 5.992]	***	
Conscientious	9.197	0.782	9872.292	[7.666, 10.729]	***	
Emotionally Stable	−5.173	0.596	0.006	[−6.342, −4.004]	***	
Open to Experience	2.201	0.576	9.038	[1.072, 3.331]	***	
Age (years)	0.037	0.020	1.038	[−0.003, 0.077]	.	
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extroverted	Excluded — 87.9% (outside 15–85% window)					87.9%
Mentioned Introverted	0.749	0.154	2.115	[0.448, 1.051]	***	66.7%
Mentioned Agreeable	−0.218	0.119	0.804	[−0.451, 0.015]	.	64.6%
Mentioned Antagonistic	−1.244	0.202	0.288	[−1.639, −0.848]	***	77.9%
Mentioned Conscientious	Excluded — 100.0% (outside 15–85% window)					100.0%
Mentioned Unconscientious	Excluded — 99.6% (outside 15–85% window)					99.6%
Mentioned Neurotic	Excluded — 86.8% (outside 15–85% window)					86.8%
Mentioned Emotionally Stable	Excluded — 99.1% (outside 15–85% window)					99.1%
Mentioned Open to Experience	0.611	0.121	1.843	[0.373, 0.849]	***	41.1%
Mentioned Closed to Experience	Excluded — 90.3% (outside 15–85% window)					90.3%
Mentioned Infection	Excluded — 100.0% (outside 15–85% window)					100.0%
Mentioned Age	Excluded — 6.0% (outside 15–85% window)					6.0%

AIC: 4,417.11 BIC: 4,535.66 N: 20,000 Groups: 100 σ^2_u : 3.237

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✗ Extroverted: 87.9%
✓ Antagonistic: 77.9%
✗ Neurotic: 86.8%
✗ Closed to Experience: 90.3%

✓ Introverted: 66.7%
✗ Conscientious: 100.0%
✗ Emotionally Stable: 99.1%
✗ Infection: 100.0%

✓ Agreeable: 64.6%
✗ Unconscientious: 99.6%
✓ Open to Experience: 41.1%
✗ Age: 6.0%

FIGURE B.2.2

Random-Effects Logit with Mention Flags — Claude Sonnet 4.5

Claude Sonnet 4.5

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	−5.524	1.210	0.004	[−7.895, −3.153]	***	
Infection Rate (%)	3.298	0.094	27.072	[3.113, 3.484]	***	
Infection Rate ²	−0.327	0.013	0.721	[−0.352, −0.302]	***	

Predictor	Coef	SE	OR	95% CI	Sig	Note
Male	-0.495	0.539	0.609	[-1.553, 0.562]		
Extraverted	-5.270	0.582	0.005	[-6.411, -4.130]	***	
Agreeable	2.269	0.552	9.674	[1.188, 3.351]	***	
Conscientious	7.468	0.582	1750.519	[6.326, 8.609]	***	
Emotionally Stable	-5.576	0.583	0.004	[-6.719, -4.433]	***	
Open to Experience	-0.869	0.541	0.420	[-1.928, 0.191]		
Age (years)	-0.001	0.020	0.999	[-0.040, 0.038]		
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extroverted	-0.357	0.187	0.700	[-0.723, 0.008]	.	78.9%
Mentioned Introverted	-0.054	0.104	0.947	[-0.259, 0.150]		56.0%
Mentioned Agreeable	-0.722	0.120	0.486	[-0.958, -0.486]	***	45.6%
Mentioned Antagonistic	-1.647	0.136	0.193	[-1.914, -1.380]	***	64.5%
Mentioned Conscientious		Excluded - 100.0% (outside 15-85% window)				100.0%
Mentioned Unconscientious		Excluded - 91.0% (outside 15-85% window)				91.0%
Mentioned Neurotic		Excluded - 93.3% (outside 15-85% window)				93.3%
Mentioned Emotionally Stable		Excluded - 95.4% (outside 15-85% window)				95.4%
Mentioned Open to Experience	0.379	0.100	1.461	[0.183, 0.575]	***	39.3%
Mentioned Closed to Experience	-2.666	0.132	0.070	[-2.925, -2.406]	***	36.5%
Mentioned Infection		Excluded - 100.0% (outside 15-85% window)				100.0%
Mentioned Age		Excluded - 2.4% (outside 15-85% window)				2.4%

AIC: 5,499.39 BIC: 5,633.75 N: 20,000 Groups: 100 σ^2_u : 4.576

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

- ✓ Extroverted: 78.9%

✓ Antagonistic: 64.5%

✗ Neurotic: 93.3%

✓ Closed to Experience: 36.5%
- ✓ Introverted: 56.0%

✗ Conscientious: 100.0%

✗ Emotionally Stable: 95.4%

✗ Infection: 100.0%
- ✓ Agreeable: 45.6%

✗ Unconscientious: 91.0%

✓ Open to Experience: 39.3%

✗ Age: 2.4%

FIGURE B.2.3

Random-Effects Logit with Mention Flags — Claude Haiku 4.5

Claude Haiku 4.5

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	-9.060	0.827	0.000	[-10.682, -7.439]	***	
Infection Rate (%)	1.728	0.072	5.629	[1.588, 1.868]	***	
Infection Rate²	-0.104	0.010	0.901	[-0.123, -0.085]	***	
Male	0.585	0.320	1.795	[-0.043, 1.212]	.	
Extraverted	-4.156	0.342	0.016	[-4.827, -3.484]	***	
Agreeable	1.284	0.338	3.612	[0.622, 1.947]	***	
Conscientious	3.543	0.367	34.568	[2.824, 4.262]	***	
Emotionally Stable	0.723	0.497	2.061	[-0.251, 1.698]		
Open to Experience	-0.362	0.329	0.696	[-1.007, 0.282]		
Age (years)	0.008	0.012	1.008	[-0.015, 0.032]		
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extroverted		Excluded - 91.8% (outside 15-85% window)				91.8%
Mentioned Introverted	0.997	0.097	2.710	[0.807, 1.187]	***	16.4%
Mentioned Agreeable	-0.294	0.096	0.745	[-0.482, -0.106]	**	80.0%
Mentioned Antagonistic	-0.738	0.161	0.478	[-1.054, -0.422]	***	46.7%
Mentioned Conscientious		Excluded - 100.0% (outside 15-85% window)				100.0%
Mentioned Unconscientious	-0.720	0.178	0.487	[-1.069, -0.371]	***	81.4%
Mentioned Neurotic	1.909	0.390	6.745	[1.145, 2.673]	***	73.6%
Mentioned Emotionally Stable		Excluded - 90.8% (outside 15-85% window)				90.8%
Mentioned Open to Experience	-0.635	0.111	0.530	[-0.853, -0.418]	***	48.0%
Mentioned Closed to Experience	-0.899	0.167	0.407	[-1.226, -0.572]	***	16.1%
Mentioned Infection		Excluded - 100.0% (outside 15-85% window)				100.0%
Mentioned Age		Excluded - 9.4% (outside 15-85% window)				9.4%

AIC: 6,929.18 BIC: 7,071.44 N: 20,000 Groups: 100 σ^2_u : 1.563

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

- ✗ Extroverted: 91.8%
- ✓ Antagonistic: 46.7%
- ✓ Neurotic: 73.6%
- ✓ Closed to Experience: 16.1%
- ✓ Introverted: 16.4%
- ✗ Conscientious: 100.0%
- ✗ Emotionally Stable: 90.8%
- ✗ Infection: 100.0%
- ✓ Agreeable: 80.0%
- ✓ Unconscientious: 81.4%
- ✓ Open to Experience: 48.0%
- ✗ Age: 9.4%

FIGURE B.2.4

Random-Effects Logit with Mention Flags — Claude Sonnet 4.0

Claude Sonnet 4.0

Predictor	Coef	SE	OR	95% CI	Sig	Note	
Base Predictors							
Intercept	-1.239	1.375	0.290	[-3.935, 1.457]			
Infection Rate (%)	1.135	0.071	3.112	[0.996, 1.274]	***		
Infection Rate²	0.003	0.011	1.003	[-0.020, 0.025]			
Male	-0.747	0.593	0.474	[-1.909, 0.415]			
Extraverted	-5.019	0.621	0.007	[-6.237, -3.801]	***		
Agreeable	1.840	0.591	6.295	[0.681, 2.998]	**		
Conscientious	5.914	0.605	370.308	[4.728, 7.101]	***		
Emotionally Stable	-5.566	0.639	0.004	[-6.818, -4.314]	***		
Open to Experience	0.047	0.598	1.048	[-1.125, 1.219]			
Age (years)	0.023	0.022	1.024	[-0.020, 0.067]			
Mention Flags (rate from Figure 32; only included when between 15% and 85%)							
Mentioned Extroverted	-1.954	0.173	0.142	[-2.293, -1.615]	***	39.1%	
Mentioned Introverted	2.307	0.157	10.045	[2.000, 2.614]	***	25.0%	
Mentioned Agreeable	0.717	0.133	2.049	[0.458, 0.977]	***	44.3%	
Mentioned Antagonistic	-0.843	0.136	0.431	[-1.108, -0.577]	***	29.0%	
Mentioned Conscientious		Excluded — 87.0% (outside 15-85% window)					87.0%
Mentioned Unconscientious	-0.343	0.205	0.709	[-0.745, 0.058]	.	35.3%	
Mentioned Neurotic	2.153	0.186	8.613	[1.789, 2.517]	***	47.3%	
Mentioned Emotionally Stable	-1.118	0.172	0.327	[-1.454, -0.781]	***	72.3%	
Mentioned Open to Experience	0.141	0.096	1.152	[-0.047, 0.329]		30.0%	
Mentioned Closed to Experience		Excluded — 0.4% (outside 15-85% window)					0.4%
Mentioned Infection		Excluded — 100.0% (outside 15-85% window)					100.0%
Mentioned Age		Excluded — 7.0% (outside 15-85% window)					7.0%

AIC: 6,768.47 BIC: 6,918.64 N: 20,000 Groups: 100 σ^2_u : 6.922

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

- ✓ Extroverted: 39.1%
- ✓ Antagonistic: 29.0%
- ✓ Neurotic: 47.3%
- ✗ Closed to Experience: 0.4%
- ✓ Introverted: 25.0%
- ✗ Conscientious: 87.0%
- ✓ Emotionally Stable: 72.3%
- ✗ Infection: 100.0%
- ✓ Agreeable: 44.3%
- ✓ Unconscientious: 35.3%
- ✓ Open to Experience: 30.0%
- ✗ Age: 7.0%

FIGURE B.2.5

Random-Effects Logit with Mention Flags — Claude 3 Haiku

Claude 3 Haiku

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	0.372	0.925	1.451	[-1.440, 2.184]		
Infection Rate (%)	2.080	0.076	8.005	[1.930, 2.230]	***	
Infection Rate²	-0.242	0.011	0.785	[-0.264, -0.219]	***	
Male	0.785	0.408	2.192	[-0.014, 1.584]	.	
Extraverted	2.283	0.456	9.804	[1.390, 3.176]	***	
Agreeable	-0.540	0.426	0.583	[-1.376, 0.295]		
Conscientious	1.607	0.433	4.987	[0.758, 2.456]	***	
Emotionally Stable	0.670	0.422	1.954	[-0.157, 1.497]		
Open to Experience	-0.334	0.414	0.716	[-1.144, 0.477]		
Age (years)	-0.003	0.015	0.997	[-0.032, 0.026]		

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Mention Flags (rate from Figure 32; only included when between 15% and 85%)</i>						
Mentioned Extroverted	-5.539	0.246	0.004	[-6.021, -5.058]	***	56.3%
Mentioned Introverted	0.074	0.222	1.077	[-0.361, 0.510]		70.9%
Mentioned Agreeable	1.875	0.253	6.520	[1.379, 2.370]	***	40.0%
Mentioned Antagonistic	1.587	0.223	4.888	[1.151, 2.023]	***	31.9%
Mentioned Conscientious	0.090	0.162	1.094	[-0.228, 0.408]		51.9%
Mentioned Unconscientious				Excluded - 3.2% (outside 15-85% window)		3.2%
Mentioned Neurotic	4.928	0.365	138.144	[4.213, 5.644]	***	22.9%
Mentioned Emotionally Stable	-2.175	0.130	0.114	[-2.430, -1.920]	***	19.5%
Mentioned Open to Experience				Excluded - 1.4% (outside 15-85% window)		1.4%
Mentioned Closed to Experience				Excluded - 3.7% (outside 15-85% window)		3.7%
Mentioned Infection				Excluded - 99.5% (outside 15-85% window)		99.5%
Mentioned Age				Excluded - 1.7% (outside 15-85% window)		1.7%
AIC: 5,733.75 BIC: 5,876.02 N: 20,000 Groups: 100 σ^2_u : 3.703						

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

- ✓ Extroverted: 56.3%

✓ Antagonistic: 31.9%

✓ Neurotic: 22.9%

✗ Closed to Experience: 3.7%
- ✓ Introverted: 70.9%

✓ Conscientious: 51.9%

✓ Emotionally Stable: 19.5%

✗ Infection: 99.5%
- ✓ Agreeable: 40.0%

✗ Unconscientious: 3.2%

✗ Open to Experience: 1.4%

✗ Age: 1.7%

FIGURE B.2.6

Random-Effects Logit with Mention Flags — GPT-5.2

GPT-5.2

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Base Predictors</i>						
Intercept	-2.240	0.380	0.106	[-2.984, -1.496]	***	
Infection Rate (%)	18.873	0.809	> 10 ⁶	[17.286, 20.459]	***	
Infection Rate ²	-2.521	0.118	0.080	[-2.753, -2.290]	***	
Male	0.302	0.166	1.352	[-0.023, 0.626]	.	
Extraverted	-0.093	0.169	0.911	[-0.425, 0.238]		
Agreeable	0.589	0.166	1.802	[0.265, 0.914]	***	
Conscientious	0.858	0.180	2.358	[0.505, 1.210]	***	
Emotionally Stable	-0.536	0.171	0.585	[-0.870, -0.201]	**	
Open to Experience	0.615	0.168	1.849	[0.286, 0.944]	***	
Age (years)	0.008	0.006	1.009	[-0.003, 0.020]		
<i>Mention Flags (rate from Figure 32; only included when between 15% and 85%)</i>						
Mentioned Extroverted				Excluded - 1.1% (outside 15-85% window)		1.1%
Mentioned Introverted				Excluded - 5.1% (outside 15-85% window)		5.1%
Mentioned Agreeable				Excluded - 0.4% (outside 15-85% window)		0.4%
Mentioned Antagonistic				Excluded - 0.1% (outside 15-85% window)		0.1%
Mentioned Conscientious	-1.071	0.219	0.343	[-1.500, -0.641]	***	19.8%
Mentioned Unconscientious				Excluded - 0.4% (outside 15-85% window)		0.4%
Mentioned Neurotic				Excluded - 4.6% (outside 15-85% window)		4.6%
Mentioned Emotionally Stable				Excluded - 0.8% (outside 15-85% window)		0.8%
Mentioned Open to Experience				Excluded - 0.0% (outside 15-85% window)		0.0%
Mentioned Closed to Experience				Excluded - 0.1% (outside 15-85% window)		0.1%
Mentioned Infection				Excluded - 100.0% (outside 15-85% window)		100.0%
Mentioned Age				Excluded - 0.7% (outside 15-85% window)		0.7%
AIC: 1,548.34 BIC: 1,643.18 N: 20,000 Groups: 100 σ^2_u : 0.220						

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

- ✗ Extroverted: 1.1%

✗ Antagonistic: 0.1%

✗ Neurotic: 4.6%

✗ Closed to Experience: 0.1%
- ✗ Introverted: 5.1%

✓ Conscientious: 19.8%

✗ Emotionally Stable: 0.8%

✗ Infection: 100.0%
- ✗ Agreeable: 0.4%

✗ Unconscientious: 0.4%

✗ Open to Experience: 0.0%

✗ Age: 0.7%

FIGURE B.2.7

Random-Effects Logit with Mention Flags — GPT-5.2 (low)

GPT-5.2 (low)

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	-0.493	0.339	0.611	[-1.158, 0.171]		
Infection Rate (%)	3.107	0.085	22.352	[2.941, 3.273]	***	
Infection Rate²	-0.374	0.013	0.688	[-0.399, -0.349]	***	
Male	0.304	0.151	1.356	[0.009, 0.600]	*	
Extraverted	-0.529	0.155	0.589	[-0.833, -0.224]	***	
Agreeable	0.920	0.149	2.510	[0.628, 1.213]	***	
Conscientious	1.370	0.162	3.933	[1.052, 1.687]	***	
Emotionally Stable	-1.221	0.157	0.295	[-1.529, -0.913]	***	
Open to Experience	0.514	0.152	1.672	[0.216, 0.812]	***	
Age (years)	0.012	0.006	1.012	[0.001, 0.022]	*	
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extraverted			Excluded — 3.4% (outside 15-85% window)			3.4%
Mentioned Introverted			Excluded — 4.3% (outside 15-85% window)			4.3%
Mentioned Agreeable			Excluded — 0.3% (outside 15-85% window)			0.3%
Mentioned Antagonistic			Excluded — 1.4% (outside 15-85% window)			1.4%
Mentioned Conscientious	-0.372	0.129	0.689	[-0.626, -0.119]	**	33.4%
Mentioned Unconscientious			Excluded — 4.0% (outside 15-85% window)			4.0%
Mentioned Neurotic			Excluded — 12.9% (outside 15-85% window)			12.9%
Mentioned Emotionally Stable			Excluded — 1.3% (outside 15-85% window)			1.3%
Mentioned Open to Experience			Excluded — 0.0% (outside 15-85% window)			0.0%
Mentioned Closed to Experience			Excluded — 0.1% (outside 15-85% window)			0.1%
Mentioned Infection			Excluded — 99.7% (outside 15-85% window)			99.7%
Mentioned Age			Excluded — 1.8% (outside 15-85% window)			1.8%

AIC: 5,888.82 BIC: 5,983.66 N: 20,000 Groups: 100 σ²_u: 0.382

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

- ✗ Extroverted: 3.4%
- ✗ Antagonistic: 1.4%
- ✗ Neurotic: 12.9%
- ✗ Closed to Experience: 0.1%
- ✗ Introverted: 4.3%
- ✓ Conscientious: 33.4%
- ✗ Emotionally Stable: 1.3%
- ✗ Infection: 99.7%
- ✗ Agreeable: 0.3%
- ✗ Unconscientious: 4.0%
- ✗ Open to Experience: 0.0%
- ✗ Age: 1.8%

FIGURE B.2.8

Random-Effects Logit with Mention Flags — GPT-5.2 (med)

GPT-5.2 (med)

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	-0.568	0.414	0.566	[-1.379, 0.243]		
Infection Rate (%)	2.675	0.072	14.519	[2.535, 2.816]	***	
Infection Rate²	-0.314	0.011	0.731	[-0.336, -0.292]	***	
Male	0.301	0.184	1.352	[-0.059, 0.662]		
Extraverted	-0.704	0.189	0.495	[-1.075, -0.333]	***	
Agreeable	1.118	0.183	3.059	[0.760, 1.476]	***	
Conscientious	1.586	0.194	4.882	[1.205, 1.966]	***	
Emotionally Stable	-1.600	0.191	0.202	[-1.975, -1.225]	***	
Open to Experience	0.856	0.186	2.353	[0.491, 1.221]	***	
Age (years)	0.012	0.007	1.012	[-0.001, 0.025]	.	
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extraverted			Excluded — 2.9% (outside 15-85% window)			2.9%
Mentioned Introverted			Excluded — 4.3% (outside 15-85% window)			4.3%
Mentioned Agreeable			Excluded — 0.7% (outside 15-85% window)			0.7%
Mentioned Antagonistic			Excluded — 1.3% (outside 15-85% window)			1.3%
Mentioned Conscientious	-0.469	0.121	0.626	[-0.706, -0.232]	***	35.0%
Mentioned Unconscientious			Excluded — 3.4% (outside 15-85% window)			3.4%
Mentioned Neurotic			Excluded — 11.6% (outside 15-85% window)			11.6%
Mentioned Emotionally Stable			Excluded — 1.1% (outside 15-85% window)			1.1%

Predictor	Coef	SE	OR	95% CI	Sig	Note
Mentioned Open to Experience			Excluded — 0.0% (outside 15-85% window)			0.0%
Mentioned Closed to Experience			Excluded — 0.1% (outside 15-85% window)			0.1%
Mentioned Infection			Excluded — 99.4% (outside 15-85% window)			99.4%
Mentioned Age			Excluded — 3.3% (outside 15-85% window)			3.3%

AIC: 6,937.31 BIC: 7,032.15 N: 20,000 Groups: 100 σ^2_u : 0.652

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✗ Extroverted: 2.9%

✗ Antagonistic: 1.3%

✗ Neurotic: 11.6%

✗ Closed to Experience: 0.1%

✗ Introverted: 4.3%

✓ Conscientious: 35.0%

✗ Emotionally Stable: 1.1%

✗ Infection: 99.4%

✗ Agreeable: 0.7%

✗ Unconscientious: 3.4%

✗ Open to Experience: 0.0%

✗ Age: 3.3%

FIGURE B.2.9

Random-Effects Logit with Mention Flags — GPT-5.2 (high)

GPT-5.2 (high)

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Base Predictors</i>						
Intercept	−0.517	0.440	0.596	[−1.381, 0.346]		
Infection Rate (%)	2.309	0.063	10.064	[2.185, 2.432]	***	
Infection Rate²	−0.270	0.010	0.763	[−0.289, −0.252]	***	
Male	0.251	0.196	1.285	[−0.134, 0.635]		
Extraverted	−0.925	0.202	0.397	[−1.320, −0.529]	***	
Agreeable	1.326	0.195	3.767	[0.945, 1.708]	***	
Conscientious	1.680	0.205	5.368	[1.278, 2.083]	***	
Emotionally Stable	−1.781	0.204	0.168	[−2.181, −1.382]	***	
Open to Experience	0.843	0.198	2.324	[0.454, 1.232]	***	
Age (years)	0.014	0.007	1.014	[−0.000, 0.028]	.	

Mention Flags (rate from Figure 32; only included when between 15% and 85%)

Mentioned Extroverted			Excluded — 2.6% (outside 15-85% window)			2.6%
Mentioned Introverted			Excluded — 3.8% (outside 15-85% window)			3.8%
Mentioned Agreeable			Excluded — 1.0% (outside 15-85% window)			1.0%
Mentioned Antagonistic			Excluded — 1.3% (outside 15-85% window)			1.3%
Mentioned Conscientious	−0.654	0.117	0.520	[−0.883, −0.426]	***	27.4%
Mentioned Unconscientious			Excluded — 2.8% (outside 15-85% window)			2.8%
Mentioned Neurotic			Excluded — 7.4% (outside 15-85% window)			7.4%
Mentioned Emotionally Stable			Excluded — 1.0% (outside 15-85% window)			1.0%
Mentioned Open to Experience			Excluded — 0.0% (outside 15-85% window)			0.0%
Mentioned Closed to Experience			Excluded — 0.2% (outside 15-85% window)			0.2%
Mentioned Infection			Excluded — 99.2% (outside 15-85% window)			99.2%
Mentioned Age			Excluded — 4.8% (outside 15-85% window)			4.8%

AIC: 8,034.27 BIC: 8,129.11 N: 20,000 Groups: 100 σ^2_u : 0.774

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✗ Extroverted: 2.6%

✗ Antagonistic: 1.3%

✗ Neurotic: 7.4%

✗ Closed to Experience: 0.2%

✗ Introverted: 3.8%

✓ Conscientious: 27.4%

✗ Emotionally Stable: 1.0%

✗ Infection: 99.2%

✗ Agreeable: 1.0%

✗ Unconscientious: 2.8%

✗ Open to Experience: 0.0%

✗ Age: 4.8%

FIGURE B.2.10

Random-Effects Logit with Mention Flags — GPT-5.1

GPT-5.1

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Base Predictors</i>						
Intercept	−0.471	0.727	0.624	[−1.897, 0.954]		
Infection Rate (%)	2.304	0.057	10.013	[2.193, 2.415]	***	
Infection Rate²	−0.193	0.010	0.824	[−0.212, −0.174]	***	
Male	−0.686	0.327	0.503	[−1.327, −0.045]	*	
Extraverted	−1.654	0.335	0.191	[−2.310, −0.997]	***	

Predictor	Coef	SE	OR	95% CI	Sig	Note
Agreeable	0.323	0.324	1.381	[−0.313, 0.958]		
Conscientious	0.693	0.335	2.000	[0.036, 1.350]	*	
Emotionally Stable	−0.941	0.335	0.390	[−1.597, −0.285]	**	
Open to Experience	2.056	0.332	7.816	[1.405, 2.707]	***	
Age (years)	−0.013	0.012	0.987	[−0.037, 0.010]		
<i>Mention Flags (rate from Figure 32; only included when between 15% and 85%)</i>						
Mentioned Extroverted	−0.896	0.089	0.408	[−1.071, −0.722]	***	42.2%
Mentioned Introverted				Excluded — 14.7% (outside 15-85% window)		14.7%
Mentioned Agreeable				Excluded — 8.4% (outside 15-85% window)		8.4%
Mentioned Antagonistic	−1.041	0.150	0.353	[−1.334, −0.748]	***	16.3%
Mentioned Conscientious	−1.939	0.081	0.144	[−2.097, −1.781]	***	64.4%
Mentioned Unconscientious	−3.174	0.155	0.042	[−3.478, −2.869]	***	30.0%
Mentioned Neurotic				Excluded — 11.6% (outside 15-85% window)		11.6%
Mentioned Emotionally Stable	−3.485	0.117	0.031	[−3.715, −3.255]	***	25.2%
Mentioned Open to Experience				Excluded — 1.7% (outside 15-85% window)		1.7%
Mentioned Closed to Experience				Excluded — 8.8% (outside 15-85% window)		8.8%
Mentioned Infection				Excluded — 100.0% (outside 15-85% window)		100.0%
Mentioned Age				Excluded — 1.3% (outside 15-85% window)		1.3%
AIC: 10,221.61 BIC: 10,348.07 N: 20,000 Groups: 100 σ^2_u : 2.426						

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✓ Extroverted: 42.2% ✗ Introverted: 14.7% ✗ Agreeable: 8.4%
 ✓ Antagonistic: 16.3% ✓ Conscientious: 64.4% ✓ Unconscientious: 30.0%
 ✗ Neurotic: 11.6% ✓ Emotionally Stable: 25.2% ✗ Open to Experience: 1.7%
 ✗ Closed to Experience: 8.8% ✗ Infection: 100.0% ✗ Age: 1.3%

FIGURE B.2.11
Random-Effects Logit with Mention Flags — GPT-4.1

GPT-4.1

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Base Predictors</i>						
Intercept	−10.063	0.943	0.000	[−11.911, −8.215]	***	
Infection Rate (%)	2.178	0.086	8.827	[2.010, 2.345]	***	
Infection Rate²	−0.177	0.011	0.838	[−0.199, −0.155]	***	
Male	−0.077	0.410	0.926	[−0.881, 0.727]		
Extraverted	−4.723	0.480	0.009	[−5.664, −3.783]	***	
Agreeable	1.670	0.411	5.312	[0.864, 2.476]	***	
Conscientious	3.593	0.461	36.355	[2.691, 4.496]	***	
Emotionally Stable	−2.326	0.465	0.098	[−3.237, −1.415]	***	
Open to Experience	3.296	0.428	27.015	[2.457, 4.136]	***	
Age (years)	0.024	0.015	1.024	[−0.006, 0.054]		
<i>Mention Flags (rate from Figure 32; only included when between 15% and 85%)</i>						
Mentioned Extroverted	−0.320	0.250	0.726	[−0.811, 0.170]		75.4%
Mentioned Introverted				Excluded — 9.7% (outside 15-85% window)		9.7%
Mentioned Agreeable	0.292	0.122	1.340	[0.054, 0.531]	*	15.3%
Mentioned Antagonistic	−2.820	0.253	0.060	[−3.316, −2.325]	***	26.6%
Mentioned Conscientious				Excluded — 94.7% (outside 15-85% window)		94.7%
Mentioned Unconscientious	−0.618	0.190	0.539	[−0.990, −0.246]	**	71.5%
Mentioned Neurotic	1.729	0.152	5.633	[1.431, 2.027]	***	43.5%
Mentioned Emotionally Stable	−1.760	0.143	0.172	[−2.039, −1.480]	***	67.5%
Mentioned Open to Experience	−0.044	0.132	0.957	[−0.303, 0.216]		30.2%
Mentioned Closed to Experience				Excluded — 7.8% (outside 15-85% window)		7.8%
Mentioned Infection				Excluded — 95.1% (outside 15-85% window)		95.1%
Mentioned Age				Excluded — 0.0% (outside 15-85% window)		0.0%
AIC: 5,095.9 BIC: 5,238.16 N: 20,000 Groups: 100 σ^2_u : 2.017						

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✓ Extroverted: 75.4% ✗ Introverted: 9.7% ✓ Agreeable: 15.3%

✓ Antagonistic: 26.6%	✗ Conscientious: 94.7%	✓ Unconscientious: 71.5%
✓ Neurotic: 43.5%	✓ Emotionally Stable: 67.5%	✓ Open to Experience: 30.2%
✗ Closed to Experience: 7.8%	✗ Infection: 95.1%	✗ Age: 0.0%

FIGURE B.2.12

Random-Effects Logit with Mention Flags — GPT-4o

GPT-4o

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	−9.995	1.588	0.000	[−13.107, −6.882]	***	
Infection Rate (%)	1.026	0.146	2.789	[0.739, 1.312]	***	
Infection Rate²	0.014	0.018	1.015	[−0.021, 0.050]		
Male	0.882	0.716	2.415	[−0.522, 2.286]		
Extraverted	−2.471	0.778	0.084	[−3.995, −0.947]	**	
Agreeable	1.586	0.740	4.883	[0.136, 3.036]	*	
Conscientious	2.698	0.748	14.843	[1.231, 4.164]	***	
Emotionally Stable	0.573	0.714	1.773	[−0.828, 1.973]		
Open to Experience	0.261	0.739	1.298	[−1.188, 1.710]		
Age (years)	−0.037	0.027	0.963	[−0.089, 0.015]		
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extroverted			Excluded — 97.5% (outside 15-85% window)			97.5%
Mentioned Introverted	3.263	0.198	26.131	[2.876, 3.651]	***	24.0%
Mentioned Agreeable	1.236	0.269	3.442	[0.708, 1.764]	***	21.6%
Mentioned Antagonistic			Excluded — 1.9% (outside 15-85% window)			1.9%
Mentioned Conscientious	−1.306	0.185	0.271	[−1.669, −0.943]	***	78.5%
Mentioned Unconscientious	−4.696	1.184	0.009	[−7.016, −2.376]	***	21.9%
Mentioned Neurotic	0.597	0.193	1.816	[0.218, 0.975]	**	15.2%
Mentioned Emotionally Stable	−3.234	0.423	0.039	[−4.063, −2.405]	***	65.5%
Mentioned Open to Experience	−1.019	0.618	0.361	[−2.231, 0.193]	.	15.0%
Mentioned Closed to Experience			Excluded — 2.7% (outside 15-85% window)			2.7%
Mentioned Infection			Excluded — 85.4% (outside 15-85% window)			85.4%
Mentioned Age			Excluded — 0.0% (outside 15-85% window)			0.0%
AIC: 2,307.47 BIC: 2,449.74 N: 20,000 Groups: 100 σ² _u : 6.510						

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✗ Extroverted: 97.5%	✓ Introverted: 24.0%	✓ Agreeable: 21.6%
✗ Antagonistic: 1.9%	✓ Conscientious: 78.5%	✓ Unconscientious: 21.9%
✓ Neurotic: 15.2%	✓ Emotionally Stable: 65.5%	✓ Open to Experience: 15.0%
✗ Closed to Experience: 2.7%	✗ Infection: 85.4%	✗ Age: 0.0%

FIGURE B.2.13

Random-Effects Logit with Mention Flags — GPT-3.5 Turbo

GPT-3.5 Turbo

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	3.632	0.846	37.785	[1.974, 5.290]	***	
Infection Rate (%)	4.322	0.174	75.348	[3.980, 4.664]	***	
Infection Rate²	−0.540	0.025	0.583	[−0.590, −0.491]	***	
Male	−0.444	0.315	0.642	[−1.060, 0.173]		
Extraverted	−4.888	0.585	0.008	[−6.034, −3.742]	***	
Agreeable	1.389	0.320	4.011	[0.762, 2.016]	***	
Conscientious	0.598	0.324	1.818	[−0.038, 1.233]	.	
Emotionally Stable	0.031	0.316	1.031	[−0.588, 0.649]		
Open to Experience	−0.603	0.322	0.547	[−1.234, 0.027]	.	
Age (years)	0.034	0.011	1.035	[0.012, 0.056]	**	
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extroverted			Excluded — 0.3% (outside 15-85% window)			0.3%
Mentioned Introverted	−2.365	0.633	0.094	[−3.606, −1.124]	***	16.6%

Predictor	Coef	SE	OR	95% CI	Sig	Note
Mentioned Agreeable			Excluded — 0.0% (outside 15-85% window)			0.0%
Mentioned Antagonistic			Excluded — 0.7% (outside 15-85% window)			0.7%
Mentioned Conscientious			Excluded — 0.1% (outside 15-85% window)			0.1%
Mentioned Unconscientious			Excluded — 0.2% (outside 15-85% window)			0.2%
Mentioned Neurotic			Excluded — 0.3% (outside 15-85% window)			0.3%
Mentioned Emotionally Stable			Excluded — 0.0% (outside 15-85% window)			0.0%
Mentioned Open to Experience			Excluded — 0.0% (outside 15-85% window)			0.0%
Mentioned Closed to Experience			Excluded — 1.9% (outside 15-85% window)			1.9%
Mentioned Infection			Excluded — 0.3% (outside 15-85% window)			0.3%
Mentioned Age			Excluded — 0.0% (outside 15-85% window)			0.0%

AIC: 2,492 BIC: 2,586.84 N: 20,000 Groups: 100 σ^2_u : 1.211

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✗ Extroverted: 0.3%
 ✗ Antagonistic: 0.7%
 ✗ Neurotic: 0.3%
 ✗ Closed to Experience: 1.9%

✓ Introverted: 16.6%
 ✗ Conscientious: 0.1%
 ✗ Emotionally Stable: 0.0%
 ✗ Infection: 0.3%

✗ Agreeable: 0.0%
 ✗ Unconscientious: 0.2%
 ✗ Open to Experience: 0.0%
 ✗ Age: 0.0%

FIGURE B.2.14

Random-Effects Logit with Mention Flags — 03

03

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Base Predictors</i>						
Intercept	-4.703	0.379	0.009	[-5.446, -3.961]	***	
Infection Rate (%)	2.271	0.054	9.685	[2.165, 2.376]	***	
Infection Rate²	-0.240	0.008	0.786	[-0.256, -0.225]	***	
Male	0.548	0.164	1.729	[0.226, 0.870]	***	
Extraverted	-0.865	0.173	0.421	[-1.204, -0.526]	***	
Agreeable	0.961	0.163	2.614	[0.641, 1.281]	***	
Conscientious	1.242	0.175	3.464	[0.900, 1.585]	***	
Emotionally Stable	-1.405	0.174	0.245	[-1.746, -1.065]	***	
Open to Experience	2.121	0.168	8.341	[1.791, 2.451]	***	
Age (years)	0.016	0.006	1.016	[0.005, 0.028]	**	
<i>Mention Flags (rate from Figure 32; only included when between 15% and 85%)</i>						
Mentioned Extroverted	-2.824	0.084	0.059	[-2.989, -2.659]	***	48.4%
Mentioned Introverted	5.925	0.223	374.390	[5.488, 6.363]	***	51.7%
Mentioned Agreeable			Excluded — 8.9% (outside 15-85% window)			8.9%
Mentioned Antagonistic			Excluded — 9.8% (outside 15-85% window)			9.8%
Mentioned Conscientious	-0.443	0.070	0.642	[-0.580, -0.306]	***	79.3%
Mentioned Unconscientious	-0.837	0.114	0.433	[-1.060, -0.614]	***	47.9%
Mentioned Neurotic	3.467	0.113	32.029	[3.245, 3.688]	***	44.1%
Mentioned Emotionally Stable			Excluded — 10.9% (outside 15-85% window)			10.9%
Mentioned Open to Experience			Excluded — 1.1% (outside 15-85% window)			1.1%
Mentioned Closed to Experience			Excluded — 3.2% (outside 15-85% window)			3.2%
Mentioned Infection			Excluded — 98.4% (outside 15-85% window)			98.4%
Mentioned Age			Excluded — 9.0% (outside 15-85% window)			9.0%

AIC: 10,358.23 BIC: 10,484.69 N: 20,000 Groups: 100 σ^2_u : 0.548

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✓ Extroverted: 48.4%
 ✗ Antagonistic: 9.8%
 ✓ Neurotic: 44.1%
 ✗ Closed to Experience: 3.2%

✓ Introverted: 51.7%
 ✓ Conscientious: 79.3%
 ✗ Emotionally Stable: 10.9%
 ✗ Infection: 98.4%

✗ Agreeable: 8.9%
 ✓ Unconscientious: 47.9%
 ✗ Open to Experience: 1.1%
 ✗ Age: 9.0%

FIGURE B.2.15

Random-Effects Logit with Mention Flags — Gemini 3 Flash

Gemini 3 Flash

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Base Predictors</i>						
Intercept	-7.920	1.501	0.000	[-10.862, -4.979]	***	
Infection Rate (%)	1.029	0.071	2.799	[0.889, 1.169]	***	
Infection Rate²	-0.075	0.010	0.928	[-0.096, -0.055]	***	
Male	-0.792	0.678	0.453	[-2.120, 0.536]		
Extraverted	-6.104	0.769	0.002	[-7.610, -4.597]	***	
Agreeable	4.153	0.687	63.616	[2.805, 5.500]	***	
Conscientious	6.102	0.726	446.882	[4.680, 7.525]	***	
Emotionally Stable	-4.125	0.748	0.016	[-5.592, -2.659]	***	
Open to Experience	4.697	0.731	109.643	[3.264, 6.130]	***	
Age (years)	-0.021	0.026	0.979	[-0.071, 0.029]		
<i>Mention Flags (rate from Figure 32; only included when between 15% and 85%)</i>						
Mentioned Extraverted	-1.922	0.176	0.146	[-2.267, -1.577]	***	36.9%
Mentioned Introverted		Excluded - 9.3% (outside 15-85% window)				9.3%
Mentioned Agreeable	-1.694	0.131	0.184	[-1.950, -1.438]	***	20.4%
Mentioned Antagonistic		Excluded - 13.1% (outside 15-85% window)				13.1%
Mentioned Conscientious	-1.152	0.178	0.316	[-1.500, -0.804]	***	75.3%
Mentioned Unconscientious	-0.656	0.401	0.519	[-1.442, 0.130]		24.1%
Mentioned Neurotic	1.744	0.156	5.722	[1.439, 2.050]	***	27.5%
Mentioned Emotionally Stable	-1.833	0.123	0.160	[-2.074, -1.591]	***	37.7%
Mentioned Open to Experience	0.728	0.112	2.071	[0.510, 0.947]	***	23.3%
Mentioned Closed to Experience	-0.743	0.199	0.476	[-1.132, -0.354]	***	15.6%
Mentioned Infection	0.417	0.116	1.517	[0.189, 0.644]	***	76.3%
Mentioned Age		Excluded - 0.8% (outside 15-85% window)				0.8%

AIC: 5,200.84 BIC: 5,358.91 N: 20,000 Groups: 100 σ^2_u : 5.495

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✓ Extraverted: 36.9%

✗ Antagonistic: 13.1%

✓ Neurotic: 27.5%

✓ Closed to Experience: 15.6%

✗ Introverted: 9.3%

✓ Conscientious: 75.3%

✓ Emotionally Stable: 37.7%

✓ Infection: 76.3%

✓ Agreeable: 20.4%

✓ Unconscientious: 24.1%

✓ Open to Experience: 23.3%

✗ Age: 0.8%

FIGURE B.2.16

Random-Effects Logit with Mention Flags — Gemini 3 Flash (low)

Gemini 3 Flash (low)

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Base Predictors</i>						
Intercept	-9.588	1.252	0.000	[-12.043, -7.134]	***	
Infection Rate (%)	1.499	0.111	4.475	[1.282, 1.715]	***	
Infection Rate²	-0.152	0.017	0.859	[-0.185, -0.120]	***	
Male	-0.814	0.544	0.443	[-1.880, 0.253]		
Extraverted	-4.411	0.609	0.012	[-5.605, -3.217]	***	
Agreeable	2.113	0.553	8.274	[1.029, 3.197]	***	
Conscientious	3.919	0.574	50.333	[2.793, 5.045]	***	
Emotionally Stable	-2.747	0.624	0.064	[-3.971, -1.523]	***	
Open to Experience	2.532	0.557	12.576	[1.439, 3.624]	***	
Age (years)	0.033	0.021	1.034	[-0.007, 0.073]		
<i>Mention Flags (rate from Figure 32; only included when between 15% and 85%)</i>						
Mentioned Extraverted	-1.051	0.244	0.349	[-1.530, -0.573]	***	77.5%
Mentioned Introverted	7.769	0.374	2365.631	[7.036, 8.502]	***	35.9%
Mentioned Agreeable		Excluded - 7.6% (outside 15-85% window)				7.6%
Mentioned Antagonistic	-3.871	0.263	0.021	[-4.386, -3.357]	***	52.3%
Mentioned Conscientious		Excluded - 85.2% (outside 15-85% window)				85.2%
Mentioned Unconscientious	-3.013	0.376	0.049	[-3.750, -2.276]	***	70.3%
Mentioned Neurotic	3.135	0.255	22.999	[2.635, 3.636]	***	66.2%
Mentioned Emotionally Stable	-3.980	0.260	0.019	[-4.490, -3.471]	***	67.1%
Mentioned Open to Experience		Excluded - 1.6% (outside 15-85% window)				1.6%
Mentioned Closed to Experience	-2.723	0.228	0.066	[-3.170, -2.277]	***	36.2%

Predictor	Coef	SE	OR	95% CI	Sig	Note
Mentioned Infection	3.438	0.238	31.137	[2.972, 3.905]	***	65.1%
Mentioned Age				Excluded — 3.4% (outside 15-85% window)		3.4%

AIC: 2,271.03 BIC: 2,421.19 N: 20,000 Groups: 100 σ^2_u : 3.832

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✓ Extroverted: 77.5%
✓ Antagonistic: 52.3%
✓ Neurotic: 66.2%
✓ Closed to Experience: 36.2%

✓ Introverted: 35.9%
✗ Conscientious: 85.2%
✓ Emotionally Stable: 67.1%
✓ Infection: 65.1%

✗ Agreeable: 7.6%
✓ Unconscientious: 70.3%
✗ Open to Experience: 1.6%
✗ Age: 3.4%

FIGURE B.2.17

Random-Effects Logit with Mention Flags — Gemini 3 Flash (med)

Gemini 3 Flash (med)

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Base Predictors</i>						
Intercept	-10.672	1.286	0.000	[-13.193, -8.151]	***	
Infection Rate (%)	1.580	0.122	4.853	[1.340, 1.819]	***	
Infection Rate²	-0.161	0.018	0.851	[-0.197, -0.125]	***	
Male	-1.030	0.554	0.357	[-2.116, 0.056]	.	
Extraverted	-4.072	0.619	0.017	[-5.285, -2.859]	***	
Agreeable	1.887	0.563	6.600	[0.784, 2.990]	***	
Conscientious	4.628	0.598	102.263	[3.456, 5.799]	***	
Emotionally Stable	-2.295	0.644	0.101	[-3.556, -1.033]	***	
Open to Experience	3.032	0.564	20.744	[1.926, 4.139]	***	
Age (years)	0.044	0.021	1.045	[0.003, 0.085]	*	
<i>Mention Flags (rate from Figure 32; only included when between 15% and 85%)</i>						
Mentioned Extroverted	-1.483	0.306	0.227	[-2.084, -0.883]	***	82.3%
Mentioned Introverted	9.501	0.523	13371.711	[8.477, 10.525]	***	35.7%
Mentioned Agreeable				Excluded — 9.3% (outside 15-85% window)		9.3%
Mentioned Antagonistic	-4.630	0.299	0.010	[-5.217, -4.043]	***	59.9%
Mentioned Conscientious				Excluded — 85.3% (outside 15-85% window)		85.3%
Mentioned Unconscientious	-1.874	0.335	0.153	[-2.532, -1.217]	***	78.5%
Mentioned Neurotic	3.592	0.296	36.302	[3.012, 4.172]	***	68.2%
Mentioned Emotionally Stable	-5.897	0.421	0.003	[-6.722, -5.072]	***	73.2%
Mentioned Open to Experience				Excluded — 1.9% (outside 15-85% window)		1.9%
Mentioned Closed to Experience	-2.263	0.253	0.104	[-2.759, -1.768]	***	37.8%
Mentioned Infection	3.612	0.235	37.024	[3.151, 4.072]	***	63.5%
Mentioned Age				Excluded — 3.3% (outside 15-85% window)		3.3%

AIC: 1,943.85 BIC: 2,094.02 N: 20,000 Groups: 100 σ^2_u : 3.865

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✓ Extroverted: 82.3%
✓ Antagonistic: 59.9%
✓ Neurotic: 68.2%
✓ Closed to Experience: 37.8%

✓ Introverted: 35.7%
✗ Conscientious: 85.3%
✓ Emotionally Stable: 73.2%
✓ Infection: 63.5%

✗ Agreeable: 9.3%
✓ Unconscientious: 78.5%
✗ Open to Experience: 1.9%
✗ Age: 3.3%

FIGURE B.2.18

Random-Effects Logit with Mention Flags — Gemini 3 Flash (high)

Gemini 3 Flash (high)

Predictor	Coef	SE	OR	95% CI	Sig	Note
<i>Base Predictors</i>						
Intercept	-10.330	1.374	0.000	[-13.022, -7.638]	***	
Infection Rate (%)	1.584	0.110	4.874	[1.369, 1.799]	***	
Infection Rate²	-0.167	0.016	0.846	[-0.199, -0.135]	***	
Male	-1.333	0.592	0.264	[-2.493, -0.172]	*	
Extraverted	-4.109	0.639	0.016	[-5.362, -2.856]	***	
Agreeable	2.307	0.603	10.046	[1.126, 3.488]	***	

Predictor	Coef	SE	OR	95% CI	Sig	Note
Conscientious	4.997	0.636	147.956	[3.749, 6.244]	***	
Emotionally Stable	-2.916	0.673	0.054	[-4.235, -1.598]	***	
Open to Experience	3.233	0.613	25.350	[2.031, 4.435]	***	
Age (years)	0.044	0.022	1.045	[-0.000, 0.088]	.	
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extroverted	-0.660	0.250	0.517	[-1.150, -0.170]	**	79.3%
Mentioned Introverted	9.521	0.494	13645.907	[8.553, 10.489]	***	33.0%
Mentioned Agreeable		Excluded - 8.9% (outside 15-85% window)				8.9%
Mentioned Antagonistic	-4.254	0.260	0.014	[-4.764, -3.744]	***	57.4%
Mentioned Conscientious	-1.340	0.196	0.262	[-1.725, -0.955]	***	84.5%
Mentioned Unconscientious	-3.110	0.350	0.045	[-3.796, -2.425]	***	73.9%
Mentioned Neurotic	3.413	0.255	30.350	[2.913, 3.913]	***	65.7%
Mentioned Emotionally Stable	-4.929	0.332	0.007	[-5.580, -4.278]	***	71.6%
Mentioned Open to Experience		Excluded - 1.7% (outside 15-85% window)				1.7%
Mentioned Closed to Experience	-2.029	0.253	0.132	[-2.525, -1.532]	***	36.1%
Mentioned Infection	3.318	0.200	27.596	[2.926, 3.709]	***	64.3%
Mentioned Age		Excluded - 3.2% (outside 15-85% window)				3.2%
AIC: 2,340.41	BIC: 2,498.48	N: 20,000	Groups: 100	σ^2_u : 4.515		

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

- ✓ Extroverted: 79.3%
- ✓ Antagonistic: 57.4%
- ✓ Neurotic: 65.7%
- ✓ Closed to Experience: 36.1%
- ✓ Introverted: 33.0%
- ✓ Conscientious: 84.5%
- ✓ Emotionally Stable: 71.6%
- ✓ Infection: 64.3%
- ✗ Agreeable: 8.9%
- ✓ Unconscientious: 73.9%
- ✗ Open to Experience: 1.7%
- ✗ Age: 3.2%

FIGURE B.2.19

Random-Effects Logit with Mention Flags — Gemini 2.5 Flash Lite

Gemini 2.5 Flash Lite

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	6.637	0.941	763.132	[4.794, 8.481]	***	
Infection Rate (%)	-1.656	0.046	0.191	[-1.746, -1.565]	***	
Infection Rate²	0.193	0.007	1.213	[0.179, 0.206]	***	
Male	-0.125	0.408	0.883	[-0.924, 0.674]		
Extraverted	-5.141	0.421	0.006	[-5.966, -4.315]	***	
Agreeable	1.068	0.405	2.909	[0.275, 1.861]	**	
Conscientious	0.646	0.414	1.908	[-0.166, 1.458]		
Emotionally Stable	0.525	0.423	1.690	[-0.303, 1.353]		
Open to Experience	-1.496	0.415	0.224	[-2.309, -0.683]	***	
Age (years)	-0.045	0.015	0.956	[-0.075, -0.016]	**	
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extroverted			Excluded - 97.5% (outside 15-85% window)			97.5%
Mentioned Introverted			Excluded - 99.2% (outside 15-85% window)			99.2%
Mentioned Agreeable	-0.795	0.095	0.451	[-0.981, -0.610]	***	26.8%
Mentioned Antagonistic			Excluded - 6.0% (outside 15-85% window)			6.0%
Mentioned Conscientious			Excluded - 97.8% (outside 15-85% window)			97.8%
Mentioned Unconscientious	1.105	0.083	3.018	[0.941, 1.268]	***	53.2%
Mentioned Neurotic	2.878	0.100	17.778	[2.682, 3.074]	***	37.6%
Mentioned Emotionally Stable	-1.227	0.071	0.293	[-1.366, -1.089]	***	54.2%
Mentioned Open to Experience	-0.388	0.068	0.678	[-0.522, -0.255]	***	53.4%
Mentioned Closed to Experience			Excluded - 10.2% (outside 15-85% window)			10.2%
Mentioned Infection			Excluded - 99.2% (outside 15-85% window)			99.2%
Mentioned Age			Excluded - 2.2% (outside 15-85% window)			2.2%
AIC: 12,375.73	BIC: 12,502.19	N: 20,000	Groups: 100	σ² _u : 3.658		

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

- ✗ Extroverted: 97.5%
- ✗ Antagonistic: 6.0%
- ✗ Introverted: 99.2%
- ✗ Conscientious: 97.8%
- ✓ Agreeable: 26.8%
- ✓ Unconscientious: 53.2%

✓ Neurotic: 37.6%	✓ Emotionally Stable: 54.2%	✓ Open to Experience: 53.4%
✗ Closed to Experience: 10.2%	✗ Infection: 99.2%	✗ Age: 2.2%

FIGURE B.2.20

Random-Effects Logit with Mention Flags — Gemini 2.5 Flash

Gemini 2.5 Flash

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	-5.148	0.922	0.006	[-6.955, -3.340]	***	
Infection Rate (%)	0.559	0.060	1.749	[0.441, 0.677]	***	
Infection Rate²	-0.034	0.009	0.967	[-0.052, -0.016]	***	
Male	0.173	0.413	1.189	[-0.636, 0.982]		
Extraverted	-4.148	0.523	0.016	[-5.173, -3.122]	***	
Agreeable	-0.572	0.462	0.564	[-1.479, 0.334]		
Conscientious	1.391	0.463	4.019	[0.484, 2.298]	**	
Emotionally Stable	-1.239	0.443	0.290	[-2.107, -0.370]	**	
Open to Experience	2.201	0.432	9.035	[1.355, 3.047]	***	
Age (years)	-0.000	0.016	1.000	[-0.031, 0.031]		
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extroverted	-1.562	0.364	0.210	[-2.276, -0.849]	***	34.6%
Mentioned Introverted	2.103	0.110	8.193	[1.887, 2.319]	***	56.6%
Mentioned Agreeable	1.995	0.267	7.355	[1.473, 2.518]	***	56.2%
Mentioned Antagonistic				Excluded — 9.3% (outside 15-85% window)		9.3%
Mentioned Conscientious	0.178	0.238	1.194	[-0.289, 0.645]		76.1%
Mentioned Unconscientious	-0.265	0.201	0.767	[-0.659, 0.129]		22.9%
Mentioned Neurotic	1.533	0.164	4.631	[1.210, 1.855]	***	31.2%
Mentioned Emotionally Stable	-0.596	0.143	0.551	[-0.877, -0.316]	***	46.4%
Mentioned Open to Experience	0.010	0.104	1.010	[-0.193, 0.214]		26.6%
Mentioned Closed to Experience				Excluded — 5.5% (outside 15-85% window)		5.5%
Mentioned Infection				Excluded — 98.4% (outside 15-85% window)		98.4%
Mentioned Age	-0.588	0.179	0.556	[-0.938, -0.237]	**	21.9%
AIC: 6,606.38 BIC: 6,764.45 N: 20,000 Groups: 100 σ² _u : 2.175						

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✓ Extroverted: 34.6%	✓ Introverted: 56.6%	✓ Agreeable: 56.2%
✗ Antagonistic: 9.3%	✓ Conscientious: 76.1%	✓ Unconscientious: 22.9%
✓ Neurotic: 31.2%	✓ Emotionally Stable: 46.4%	✓ Open to Experience: 26.6%
✗ Closed to Experience: 5.5%	✗ Infection: 98.4%	✓ Age: 21.9%

FIGURE B.2.21

Random-Effects Logit with Mention Flags — Gemini 2.0 Flash

Gemini 2.0 Flash

Predictor	Coef	SE	OR	95% CI	Sig	Note
Base Predictors						
Intercept	-1.037	0.753	0.354	[-2.513, 0.438]		
Infection Rate (%)	2.101	0.052	8.177	[1.999, 2.203]	***	
Infection Rate²	-0.237	0.008	0.789	[-0.252, -0.222]	***	
Male	-0.309	0.333	0.734	[-0.962, 0.344]		
Extraverted	-3.391	0.337	0.034	[-4.052, -2.730]	***	
Agreeable	1.358	0.336	3.889	[0.699, 2.017]	***	
Conscientious	3.342	0.342	28.264	[2.672, 4.011]	***	
Emotionally Stable	-2.422	0.347	0.089	[-3.101, -1.743]	***	
Open to Experience	0.535	0.337	1.708	[-0.125, 1.195]		
Age (years)	-0.001	0.012	0.999	[-0.025, 0.023]		
Mention Flags (rate from Figure 32; only included when between 15% and 85%)						
Mentioned Extroverted				Excluded — 96.4% (outside 15-85% window)		96.4%
Mentioned Introverted				Excluded — 98.9% (outside 15-85% window)		98.9%
Mentioned Agreeable	0.129	0.101	1.138	[-0.068, 0.326]		44.5%

Predictor	Coef	SE	OR	95% CI	Sig	Note
Mentioned Antagonistic	-1.246	0.099	0.288	[-1.441, -1.051]	***	34.4%
Mentioned Conscientious	Excluded — 100.0% (outside 15-85% window)					100.0%
Mentioned Unconscientious	0.673	0.089	1.959	[0.497, 0.848]	***	75.8%
Mentioned Neurotic	Excluded — 96.8% (outside 15-85% window)					96.8%
Mentioned Emotionally Stable	-1.813	0.093	0.163	[-1.996, -1.631]	***	70.7%
Mentioned Open to Experience	-0.836	0.072	0.433	[-0.977, -0.696]	***	40.9%
Mentioned Closed to Experience	-1.142	0.095	0.319	[-1.328, -0.956]	***	24.3%
Mentioned Infection	Excluded — 99.7% (outside 15-85% window)					99.7%
Mentioned Age	Excluded — 1.9% (outside 15-85% window)					1.9%
AIC: 11,352.4	BIC: 11,486.76	N: 20,000	Groups: 100	σ^2_u : 2.377		

Inclusion gate (Figure 32 mention rates). A mention flag enters only when its Figure 32 rate is between 15% and 85%. Below 15% almost no responses mention it; above 85% nearly all of them do — either way there is no usable variation.

✗ Extroverted: 96.4%	✗ Introverted: 98.9%	✓ Agreeable: 44.5%
✓ Antagonistic: 34.4%	✗ Conscientious: 100.0%	✓ Unconscientious: 75.8%
✗ Neurotic: 96.8%	✓ Emotionally Stable: 70.7%	✓ Open to Experience: 40.9%
✓ Closed to Experience: 24.3%	✗ Infection: 99.7%	✗ Age: 1.9%

APPENDIX C • MODEL DIAGNOSTICS

Diagnostics for the Agent-Level Logistic Regressions

These diagnostics validate the three logit specifications that underlie the regression-based results in this paper. C.1 tests the fixed-effects logit (infection curve shape only); C.2 tests the random-effects logit (infection + traits + random intercepts, used in Figures 12, 13, 14); C.3 tests the random-effects logit with mention flags (adds yes/no flags for whether the response mentioned each trait label or context, used in Figure 17). Each diagnostic panel is a 3×7 grid of small multiples, one per LLM configuration.

Assumption → Diagnostic mapping

#	Assumption	Diagnostic	C.1	C.2	C.3
1	Outcome is Bernoulli given the linear predictor (variance matches)	DHARMa Q-Q — specifically the <code>disp =</code> ratio in the top-right	✓	✓	✓
2	Logit link correctly specified	DHARMa Q-Q — overall shape of curve	✓	✓	✓
3a	Infection enters as a correct quadratic function of log-odds	Residuals vs infection	✓	✓	✓
3b	Age enters linearly in log-odds	Residuals vs age	—	✓	✓
4	Observations are independent given the predictors (within-agent rows)	DHARMa Q-Q — underdispersion flags rep clustering	✓	✓	✓
5	Random intercepts follow Normal(0, σ^2)	Random-intercept Normal Q-Q	—	✓	✓
6	Random intercepts uncorrelated with fixed-effect predictors (no endogeneity)	BLUPs vs predictors	—	✓	✓
Overall fit	Predictions match observed aggregate rates	Calibration (pred vs obs)	✓	✓	✓

C.1 (fixed-effects) validates only Assumptions 1, 2, 3a, 4 plus overall fit; agent dummies absorb between-agent variation so Assumptions 3b, 5, 6 don't apply. C.2 and C.3 cover the full set.

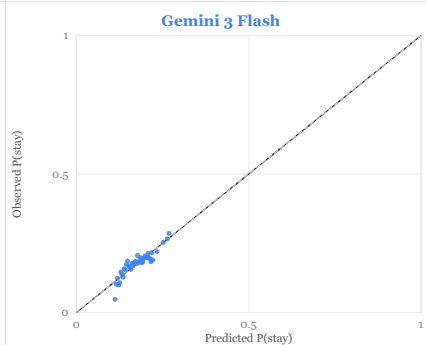
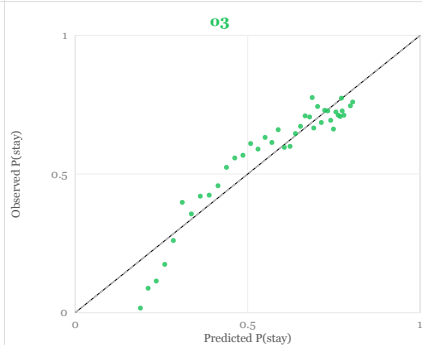
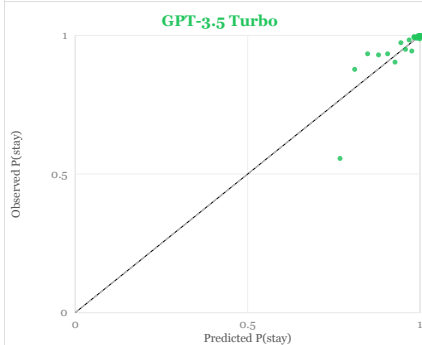
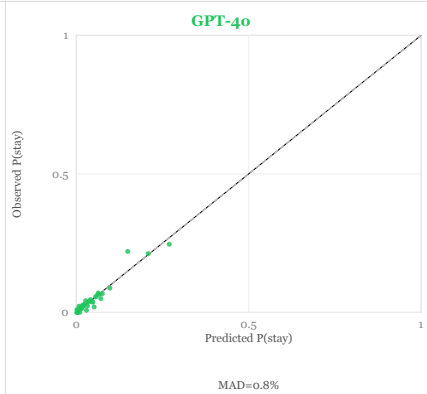
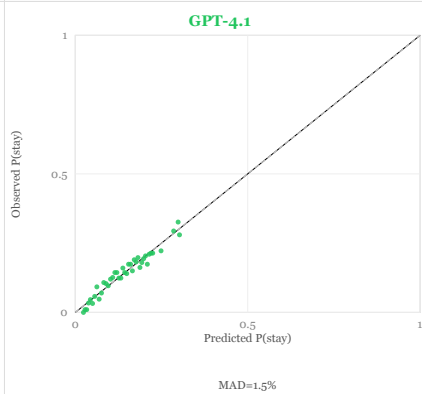
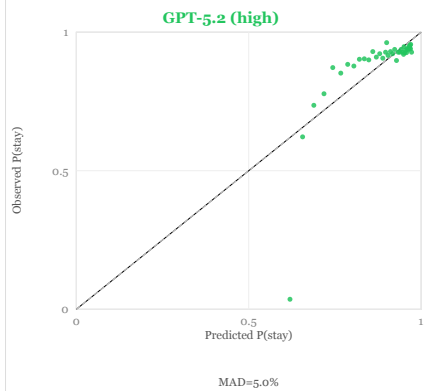
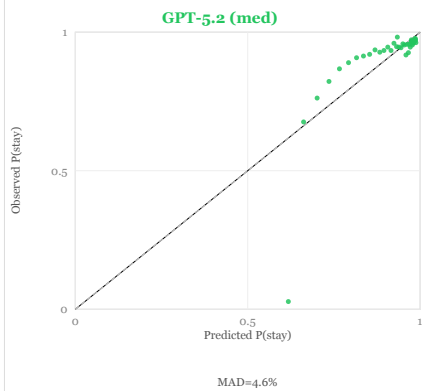
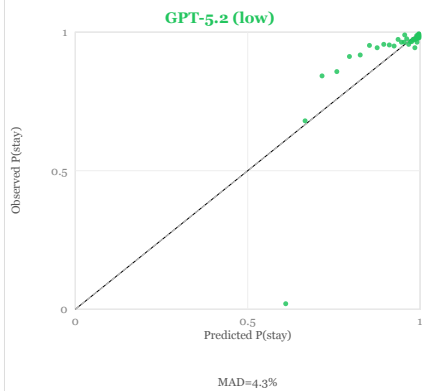
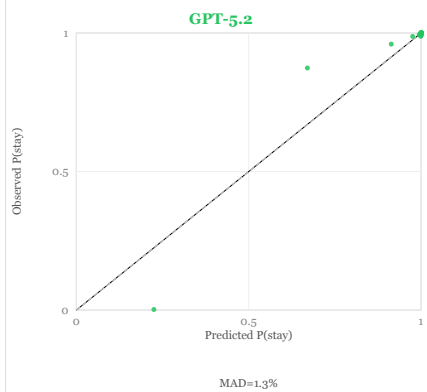
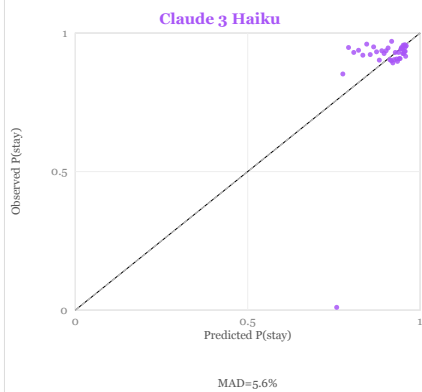
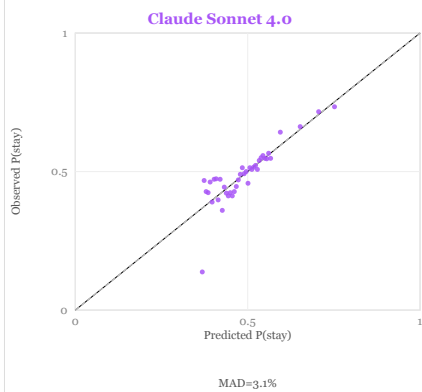
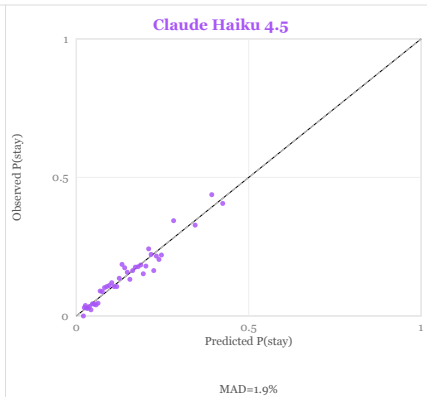
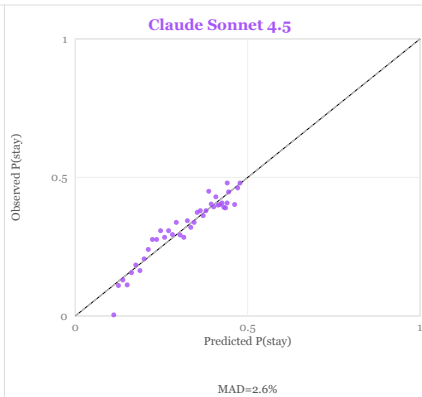
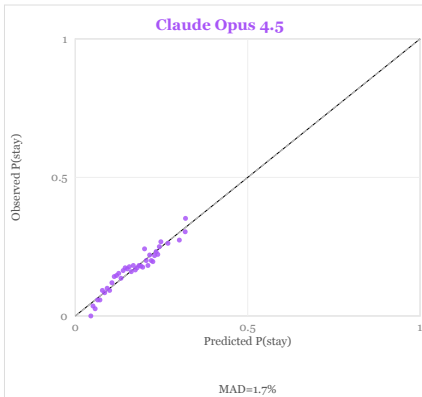
C.1 — FIXED-EFFECTS LOGIT VALIDATION

Diagnostics for the fixed-effects logit (`glm(stay_home ~ infection_pct + infection_pct^2 + factor(agent_id))`): per-agent dummies absorb all between-agent variation, so the only structural thing to check is whether the quadratic infection form is adequate. If this model looks clean but the random-effects model (C.2) doesn't, the issue lives in the trait or random-intercept specification. If both look off, the infection curve itself is the culprit.

FIGURE C.1.1

Calibration — Fixed-Effects Logit

Mean predicted $P(\text{stay-home})$ per infection bin vs observed rate, pooled across 100 agents (500 observations per bin). A well-calibrated quadratic tracks the diagonal. MAD = mean absolute deviation. 3×7 grid of small multiples, one per LLM configuration.



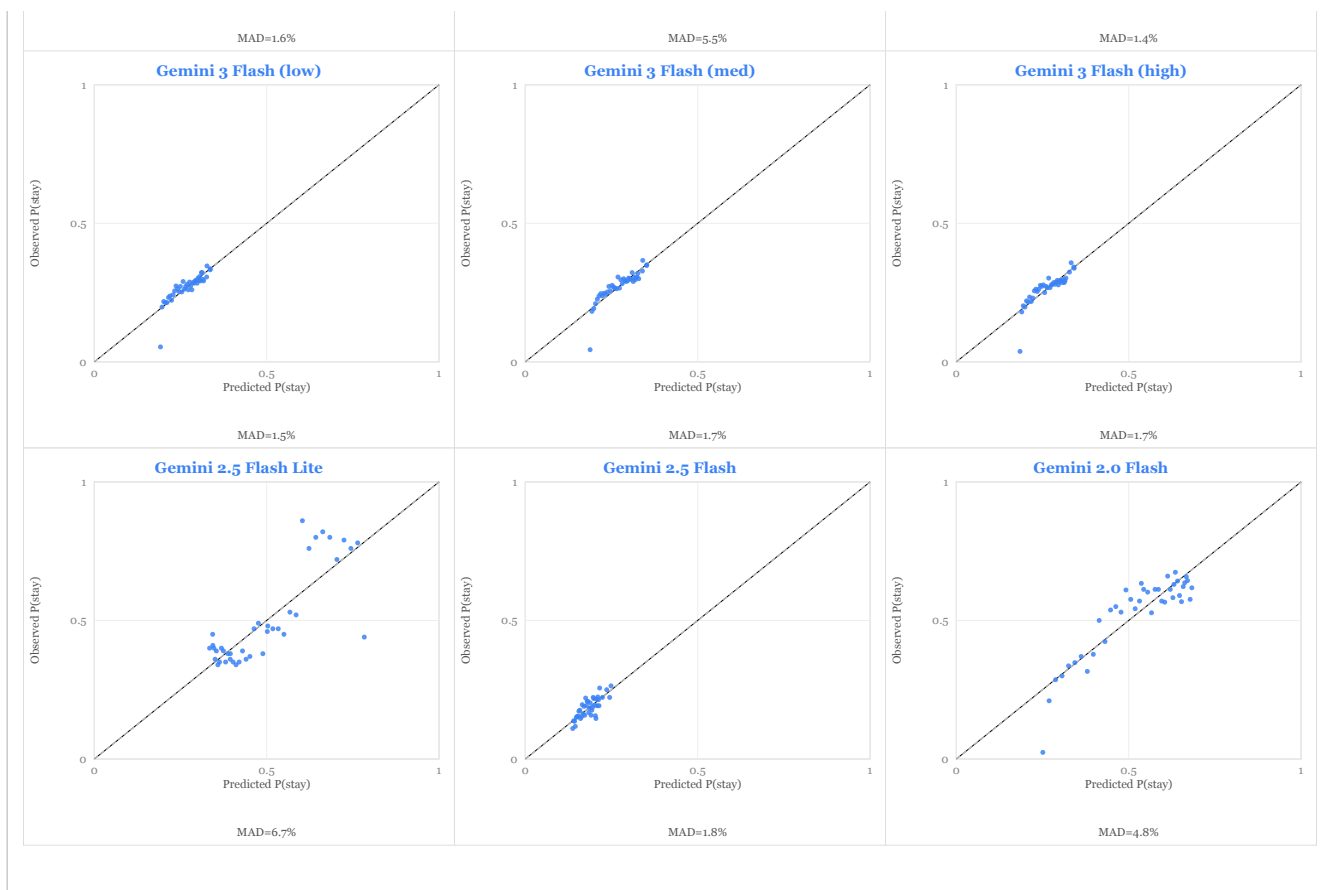
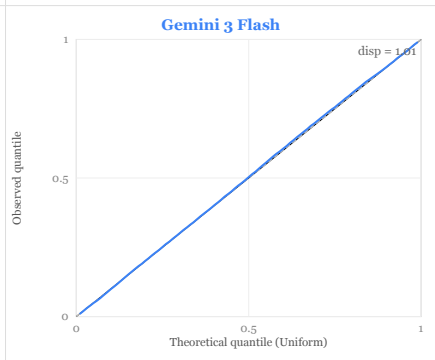
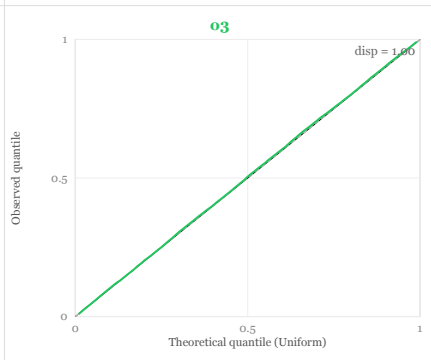
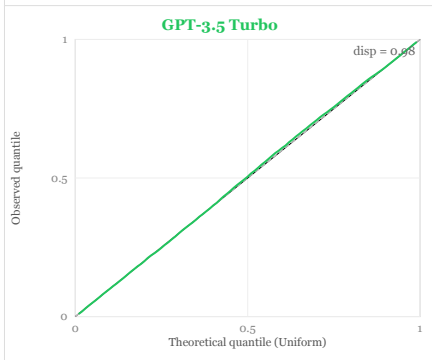
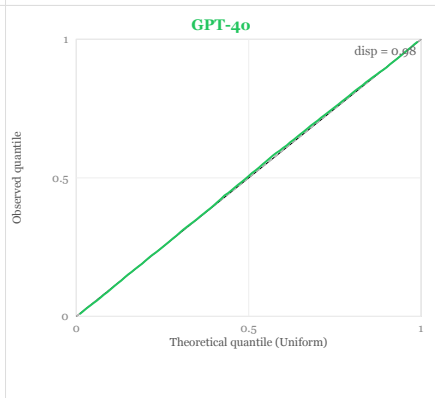
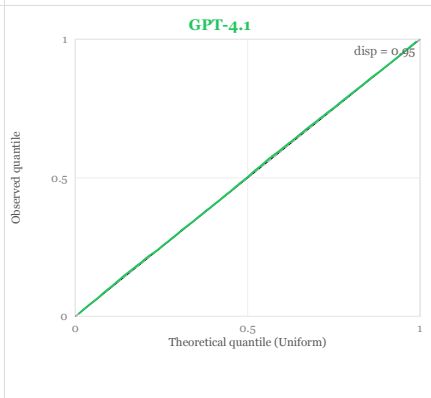
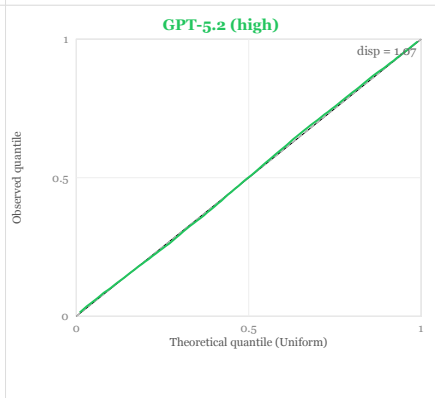
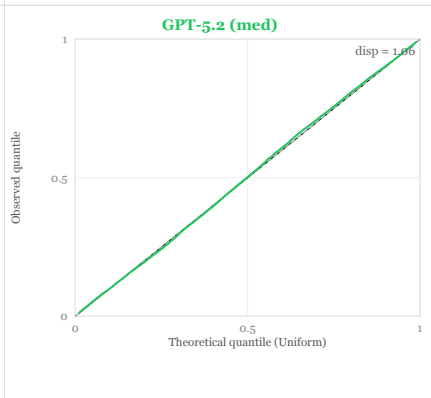
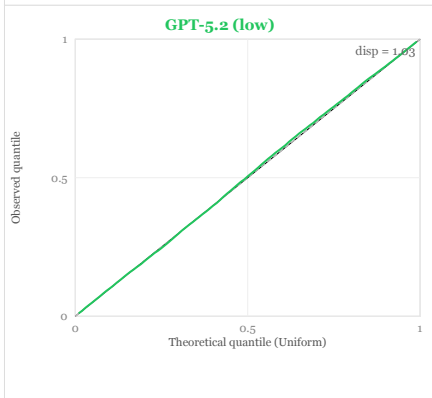
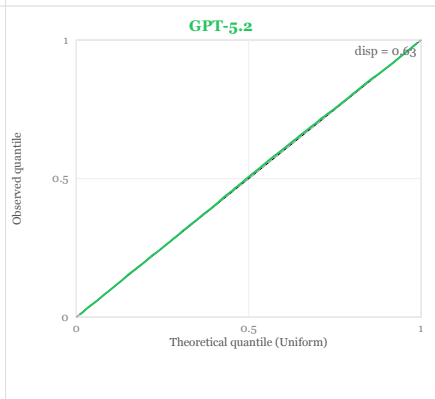
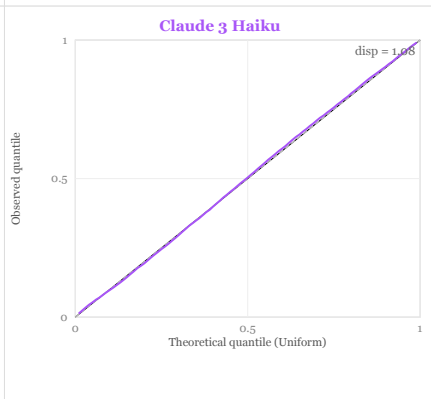
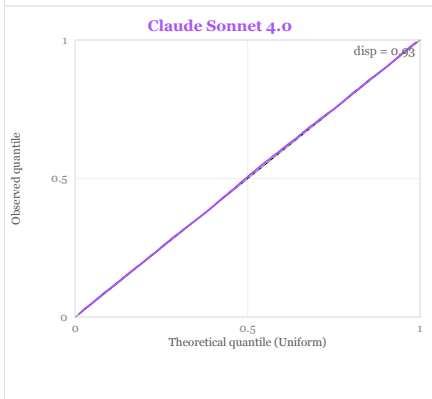
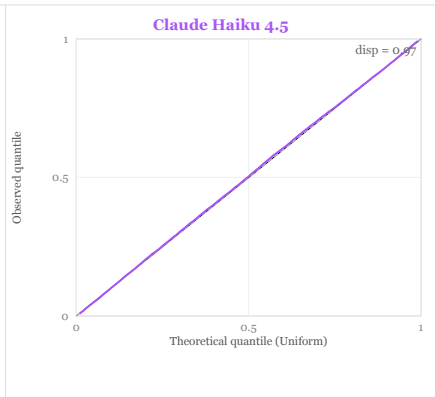
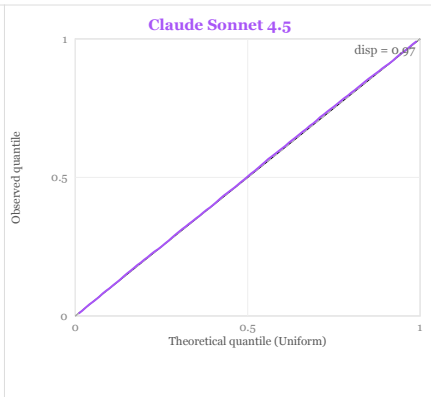
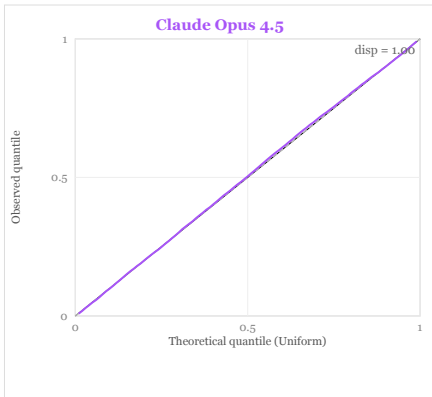


FIGURE C.1.2

DHARMA Residual Q-Q — Fixed-Effects Logit

Scaled residuals from `DHARMA::simulateResiduals` (250 sims) vs theoretical $Uniform(0,1)$; diagonal = correctly specified. Dispersion ratio in the top-right of each panel summarizes the variance check. 3×7 grid of small multiples, one per LLM configuration.



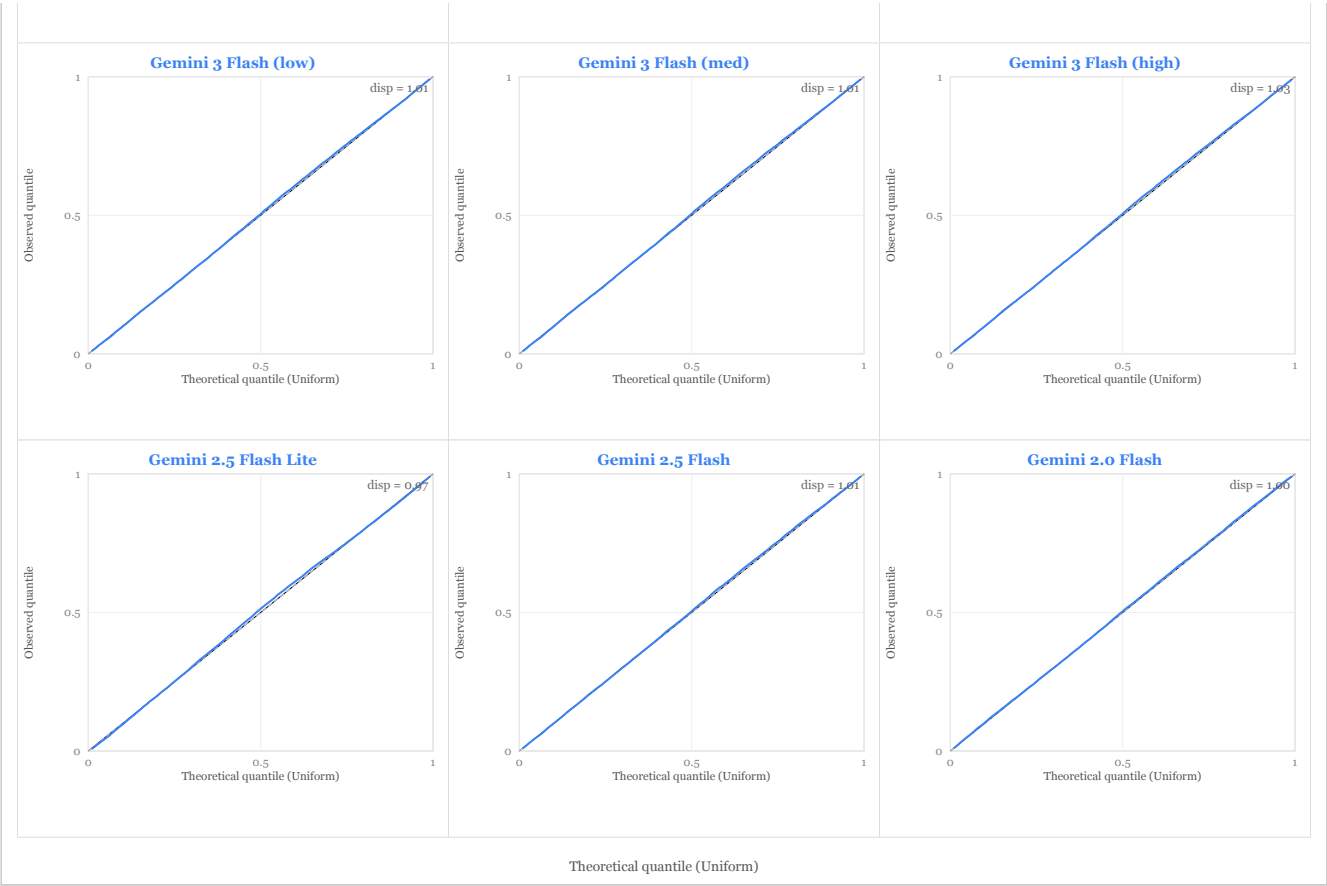
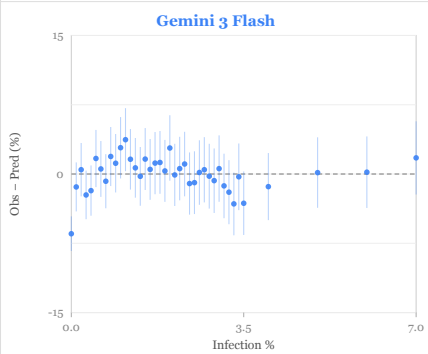
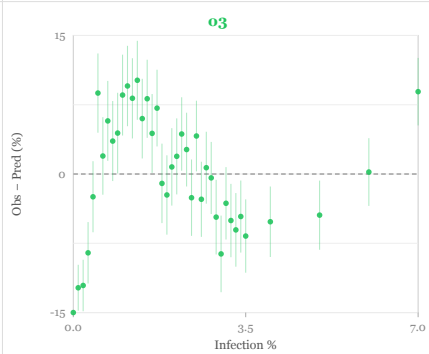
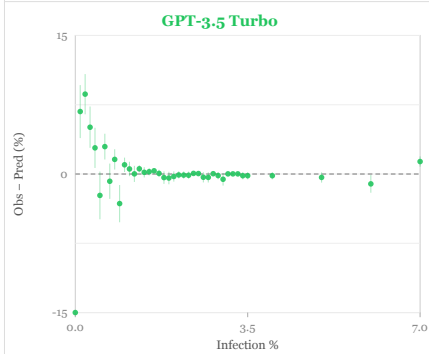
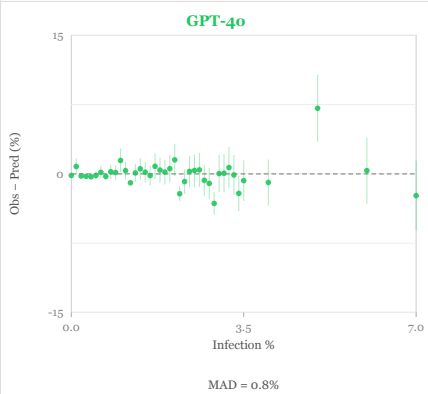
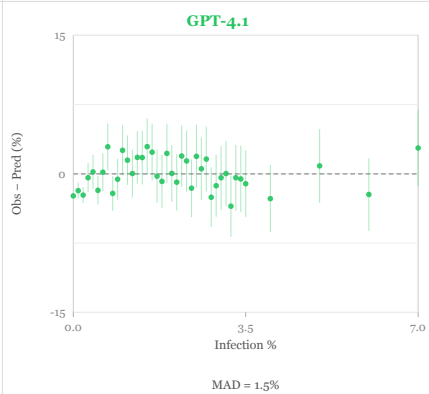
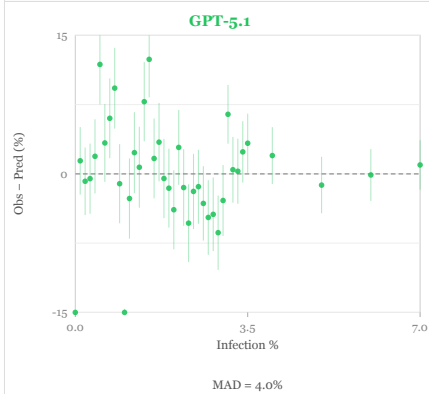
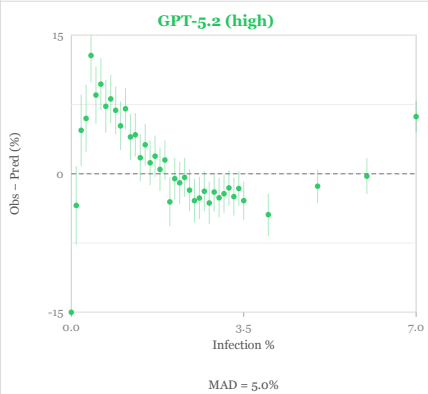
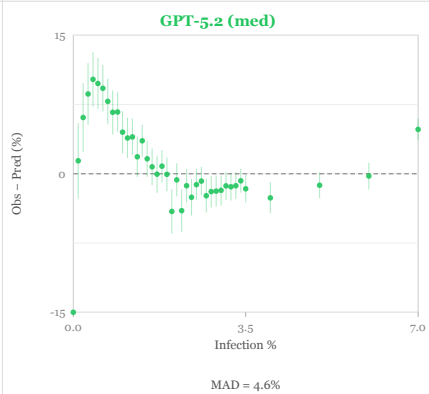
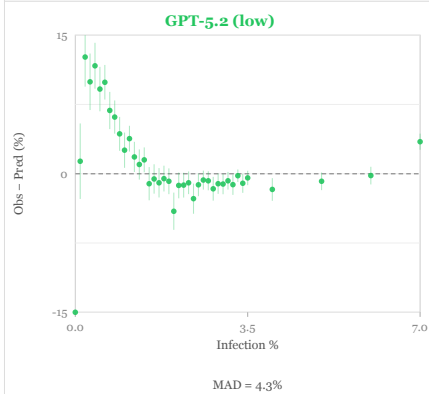
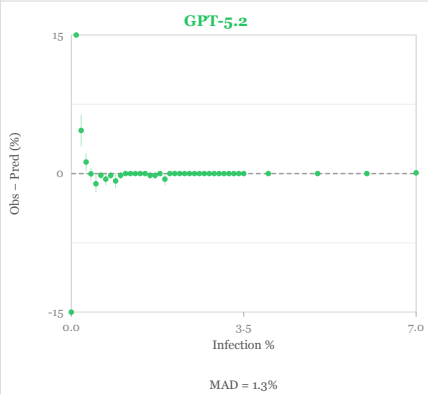
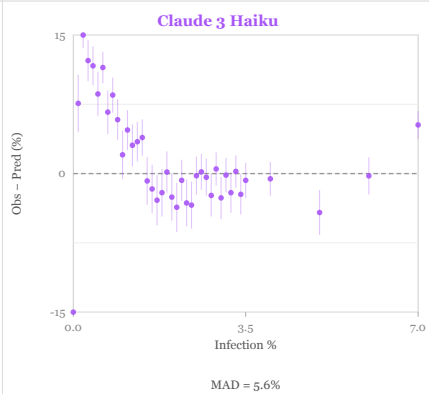
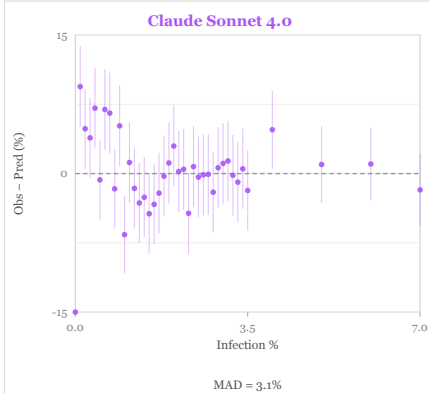
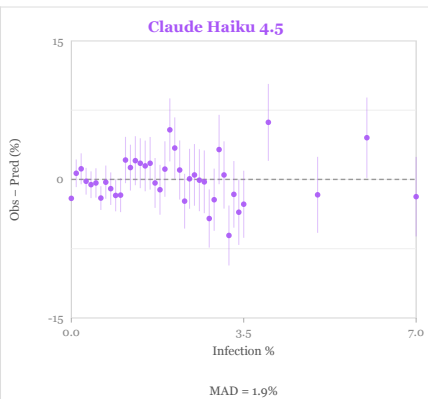
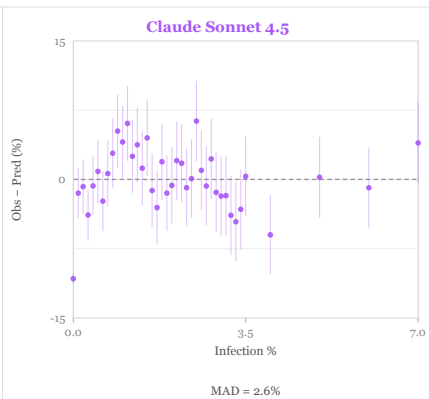
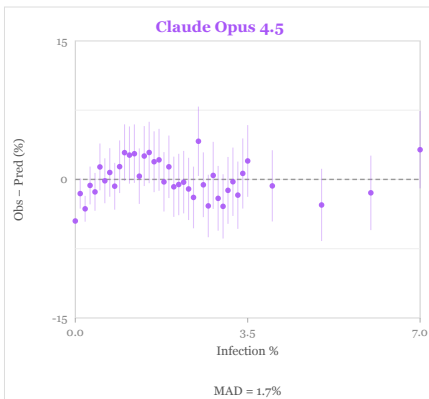
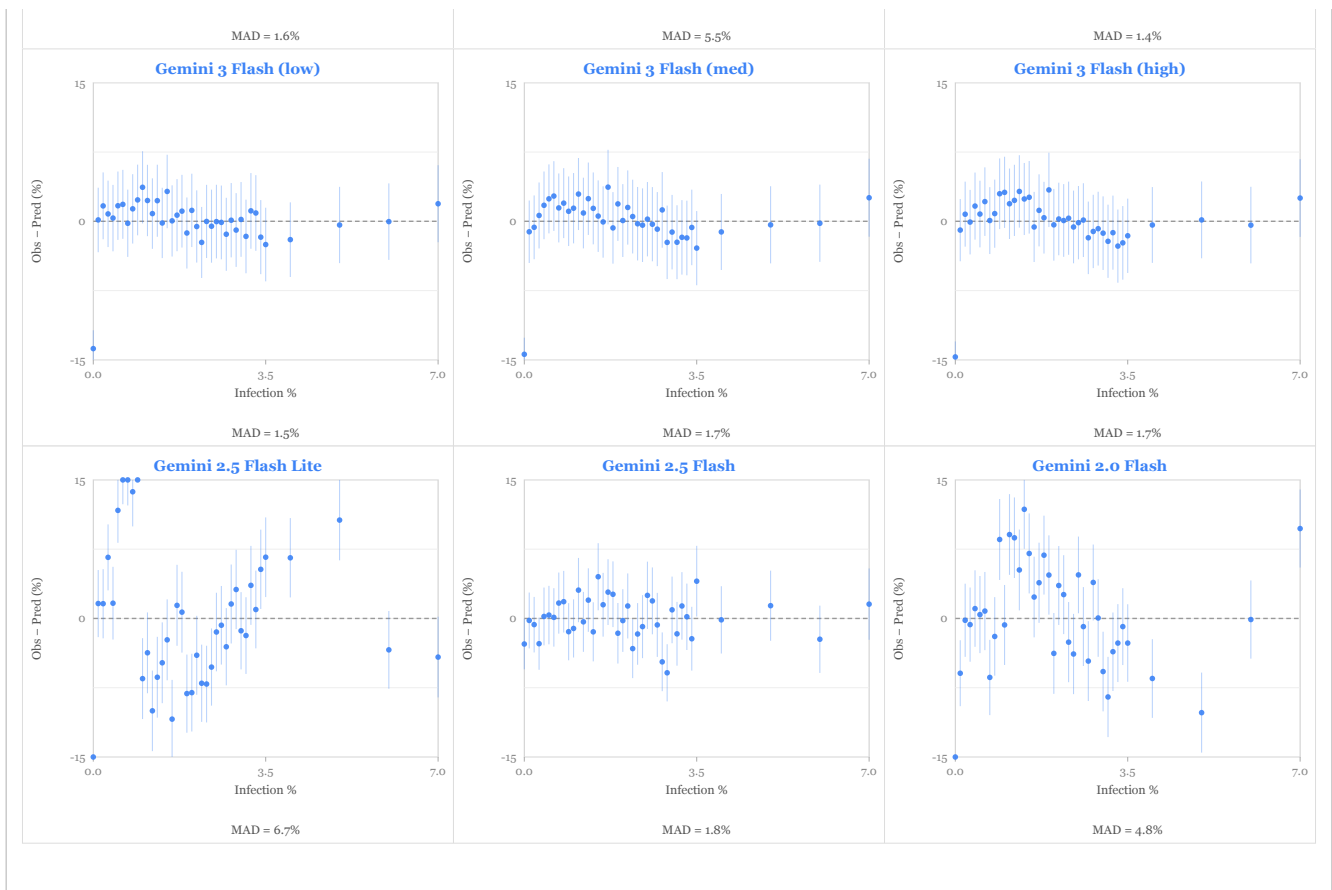


FIGURE C.1.3

Residuals vs Infection — Fixed-Effects Logit

Observed stay-home rate minus predicted rate (as a percent) per infection bin. Reference line at 0. Points near 0 indicate the quadratic captures the trend at that infection level. 3×7 grid of small multiples, one per LLM configuration.





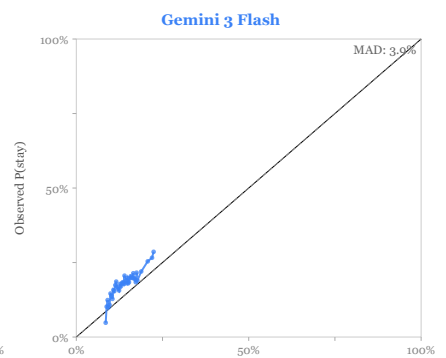
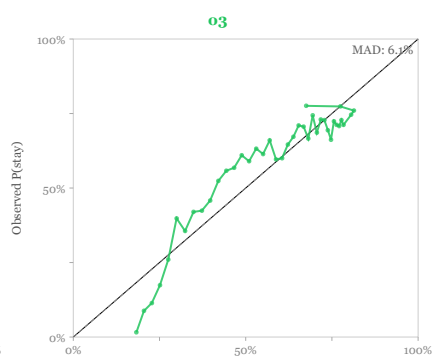
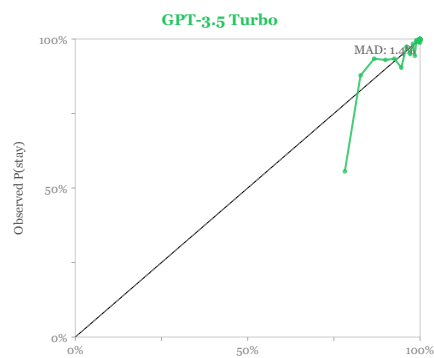
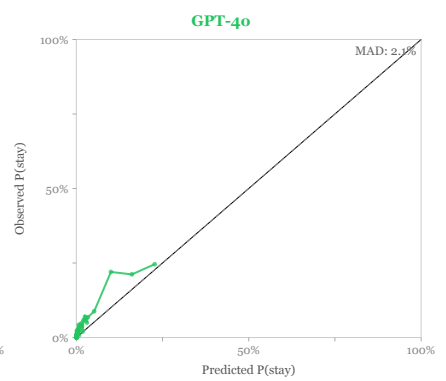
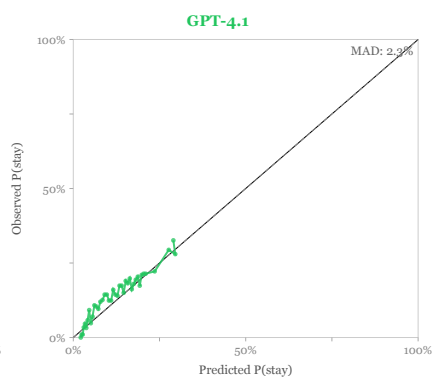
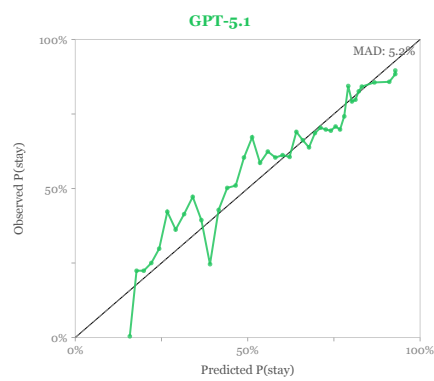
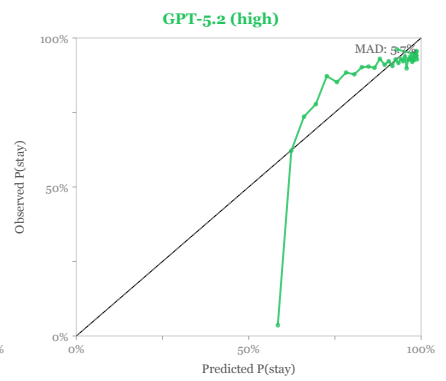
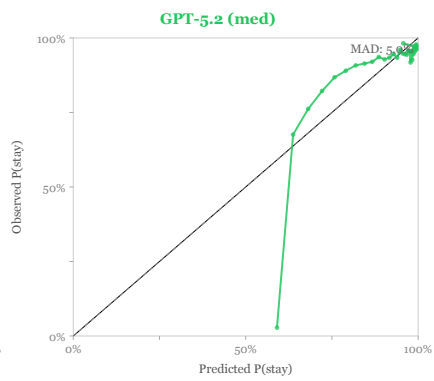
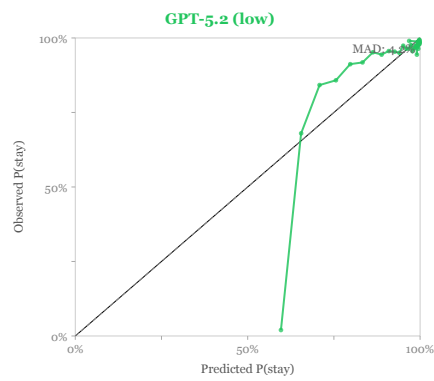
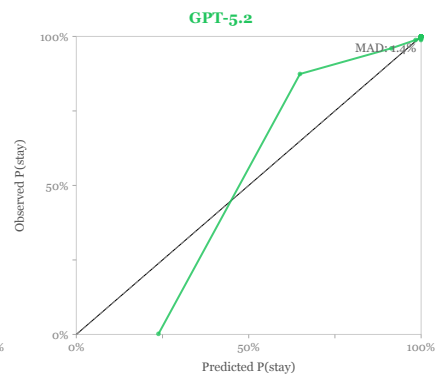
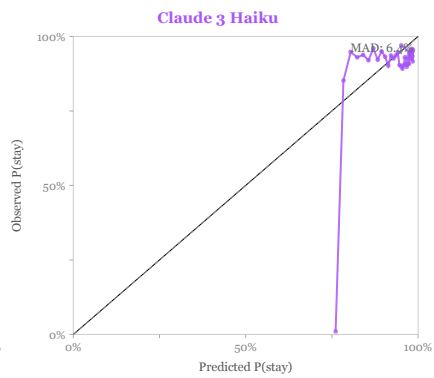
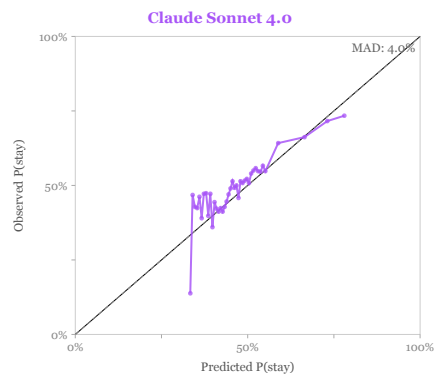
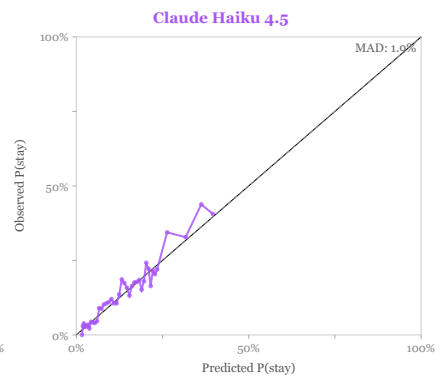
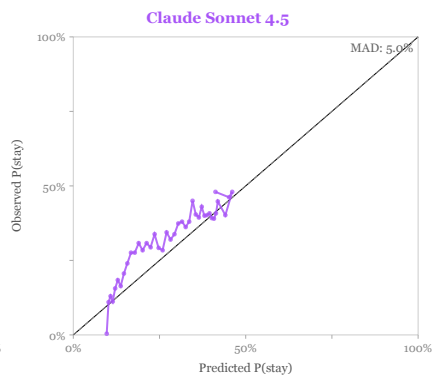
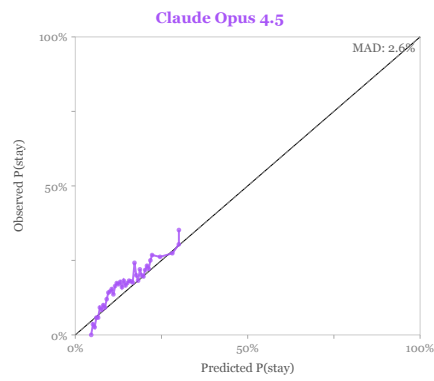
C.2 — RANDOM-EFFECTS LOGIT VALIDATION

Diagnostics for the random-effects logit (`glmer(... + (1 | agent_id))`). Each figure tests a specific model assumption or fit property; under a correctly specified model all checks track their reference line (diagonal or 0). Dispersion ratio is the primary quantitative diagnostic. KS p-values are over-powered at $N=20,000$ and not displayed. Notable departures from dispersion = 1.0: Sonnet 4.0 (disp ≈ 0.74) and GPT-5.2 off (disp ≈ 0.62), both underdispersed due to near-deterministic rep-to-rep behavior.

FIGURE C.2.1

Calibration — Random-Effects Logit

Aggregated predicted vs observed stay-home rate across 40 infection bins. A well-calibrated model tracks the diagonal. MAD = mean absolute deviation, as a percent. 3×7 grid of small multiples, one per LLM configuration.



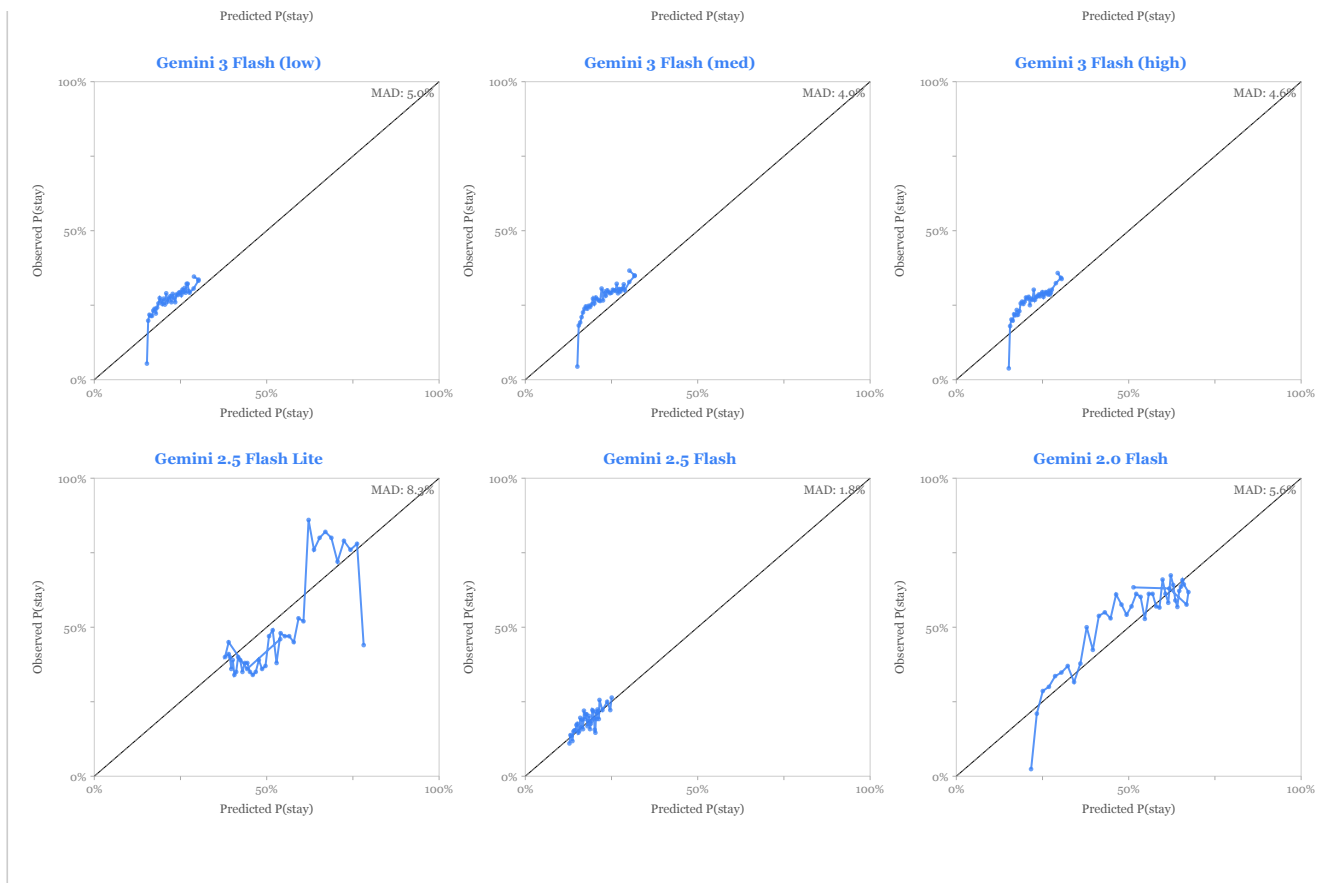


FIGURE C.2.2

DHARMA Residual Q-Q — Random-Effects Logit

Scaled residuals from `DHARMA::simulateResiduals` (250 sims) vs theoretical $Uniform(0,1)$. Diagonal = model correctly specified. Dispersion ratio in the top-right of each panel summarizes the variance check.



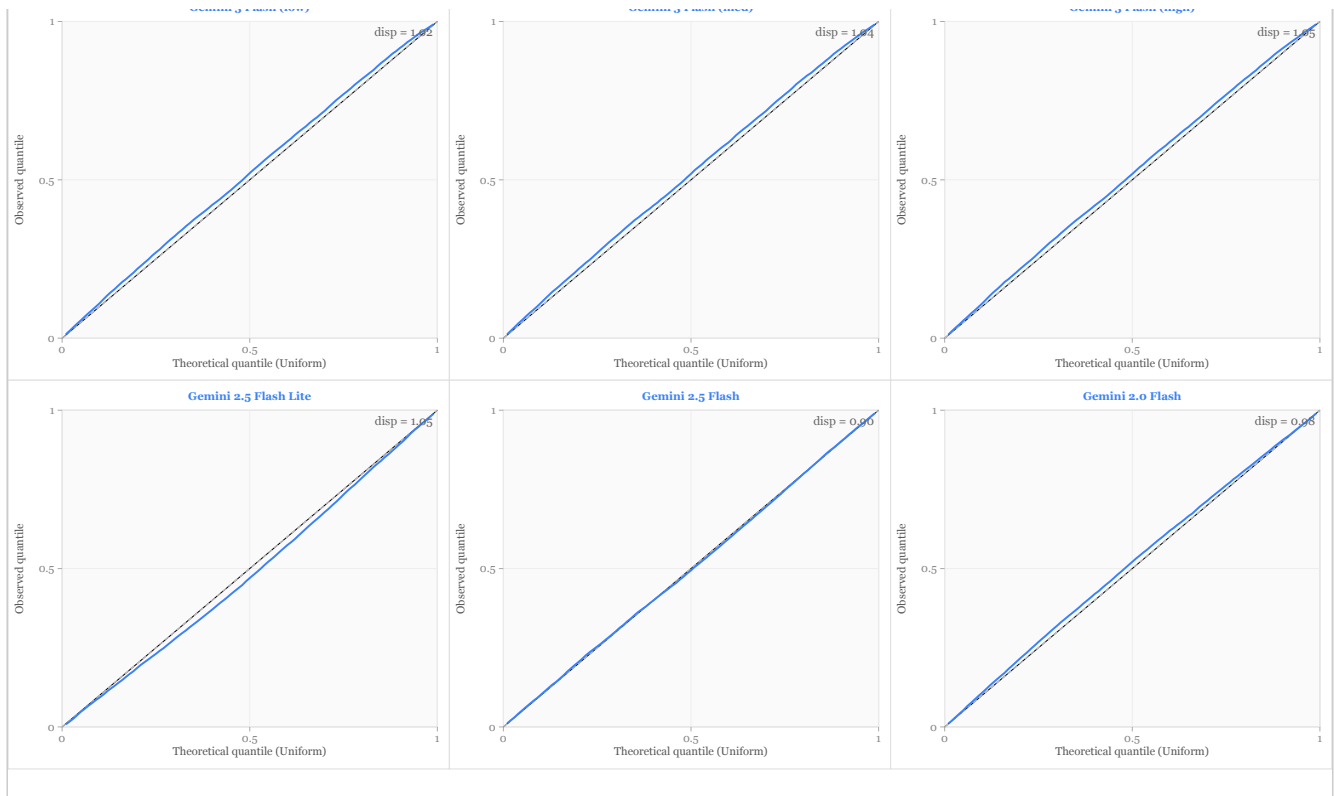


FIGURE C.2.3

Random-Intercept Normal Q-Q — Random-Effects Logit

Sorted agent-level BLUPs vs theoretical Normal quantiles, standardized by the estimated σ . Straight line = Normal random-intercept assumption holds.



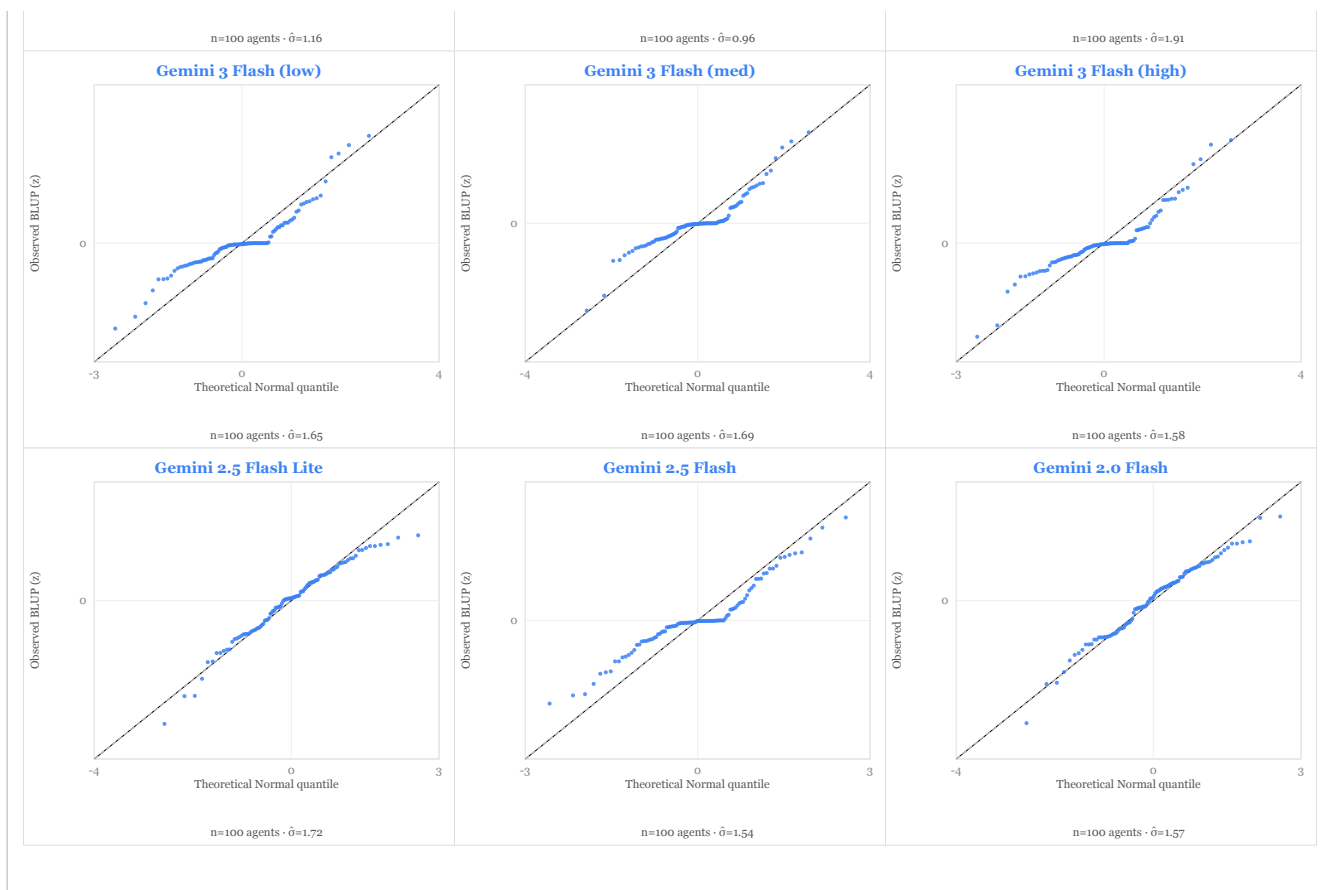


FIGURE C.2.4

BLUPs vs Predictors — Random-Effects Logit

For each fixed-effect predictor, a 95% CI of the Δ in mean BLUP between groups (for dummies) or across the range (for age). All intervals overlapping 0 indicate random intercepts aren't absorbing systematic variation that belongs in the fixed effects (Assumption 6).

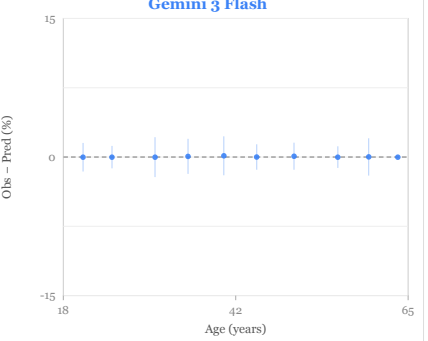
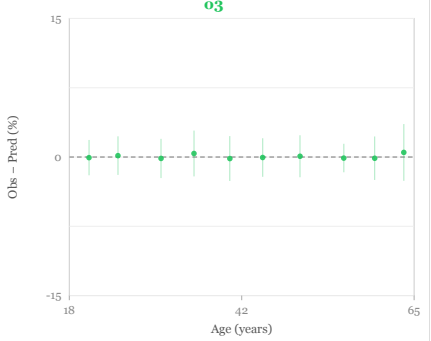
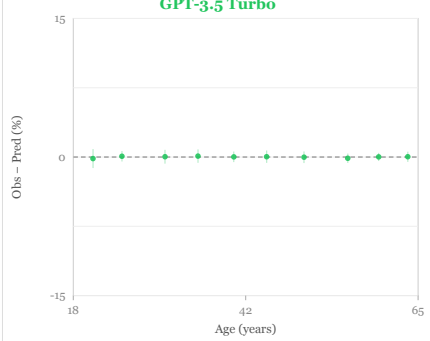
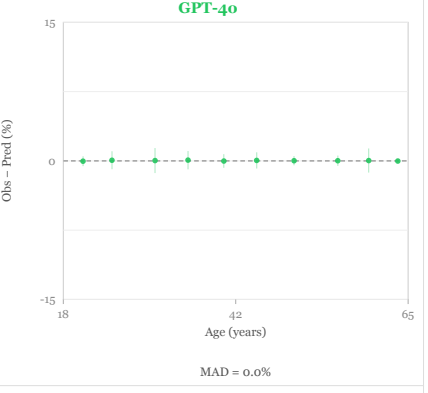
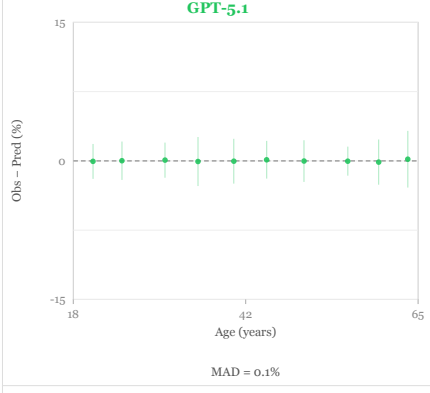
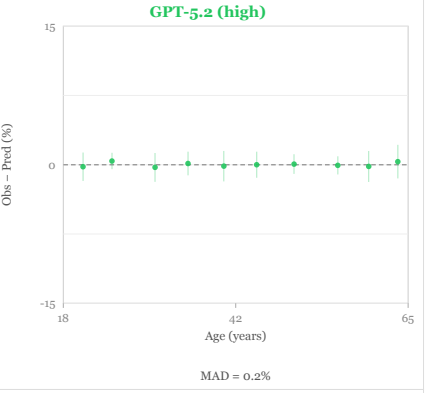
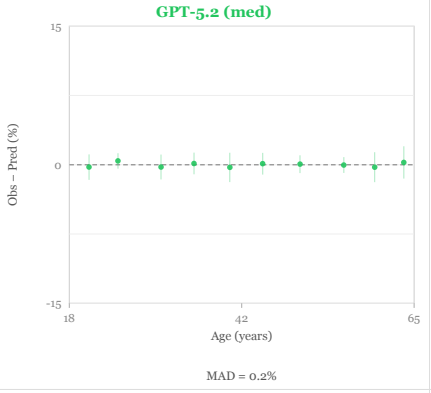
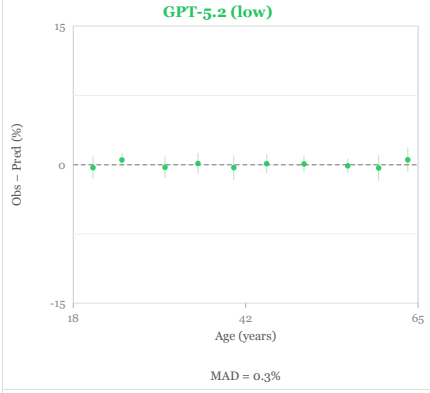
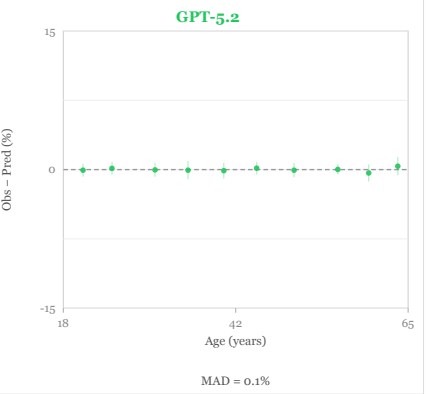
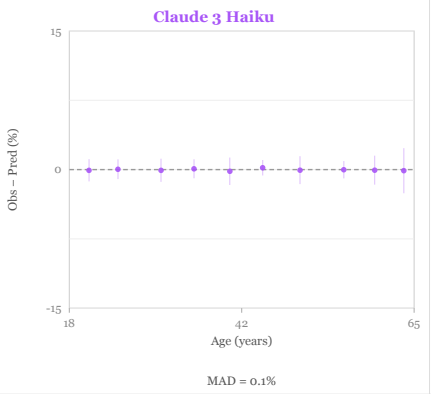
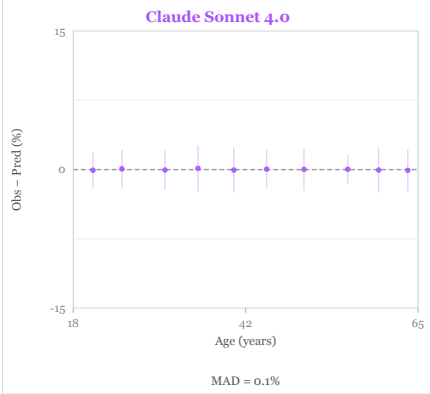
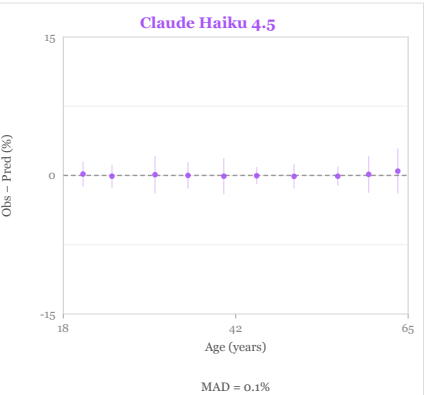
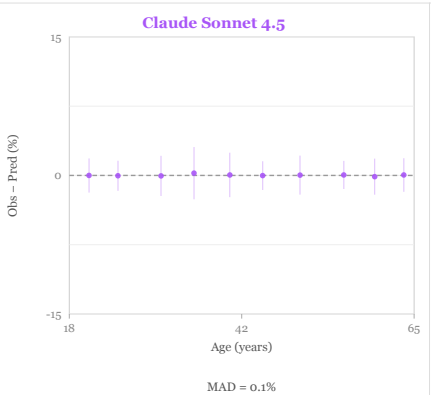
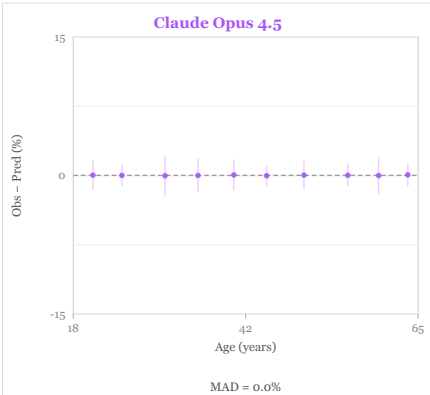


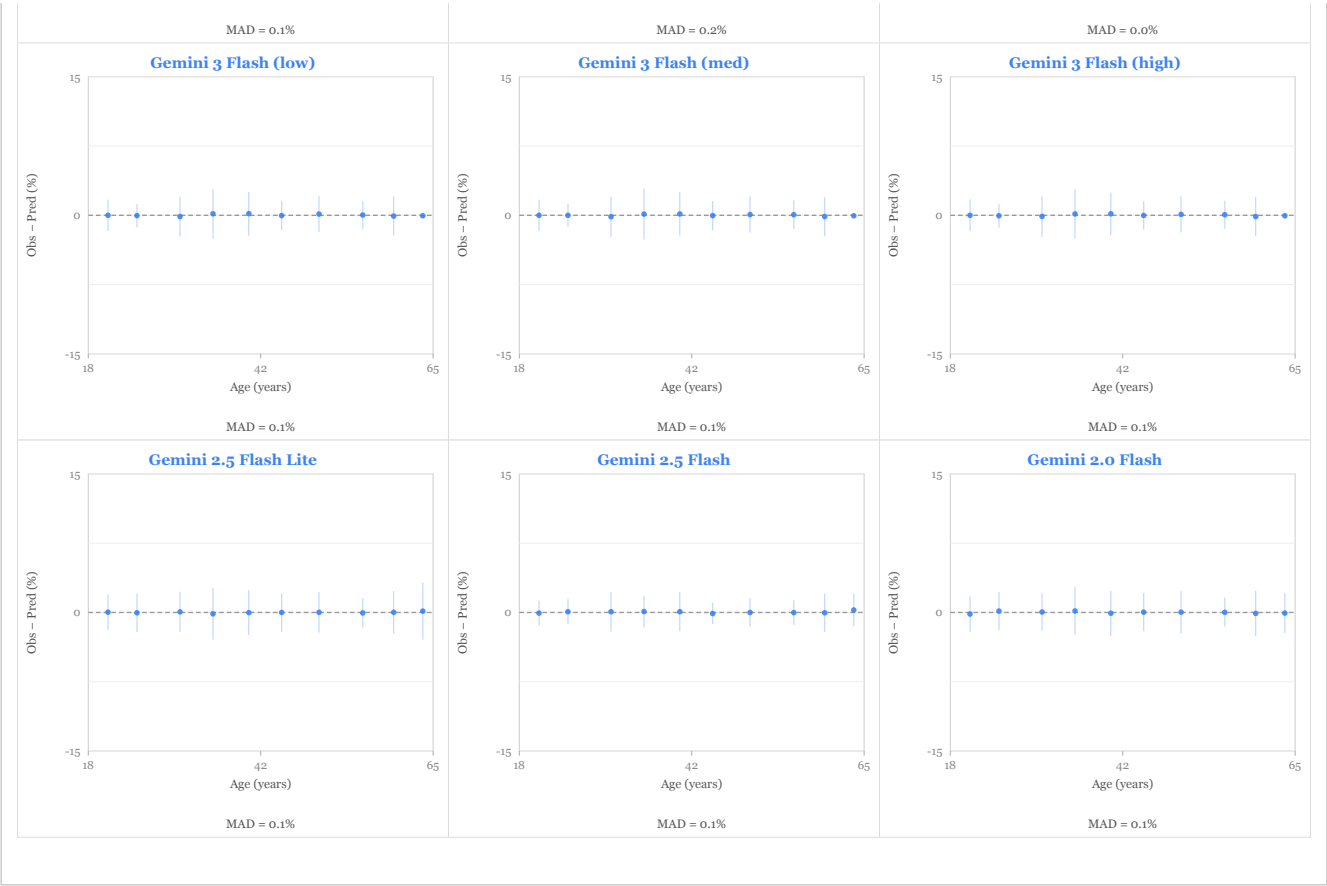


FIGURE C.2.5
Residuals vs Predictors — Random-Effects Logit

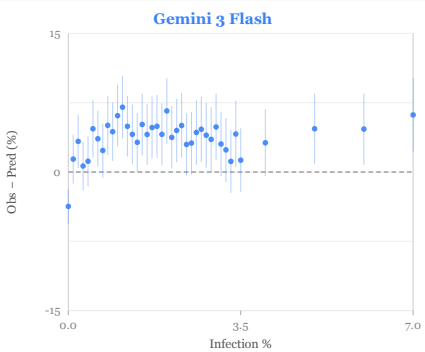
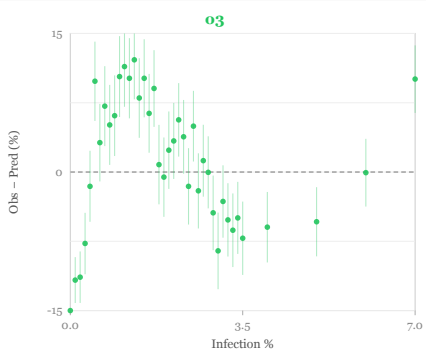
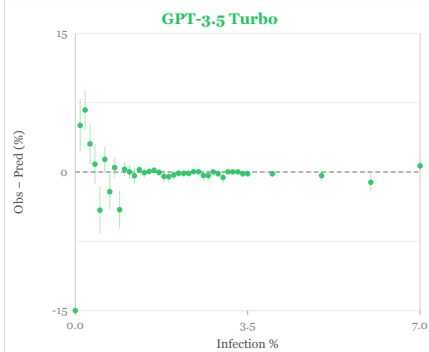
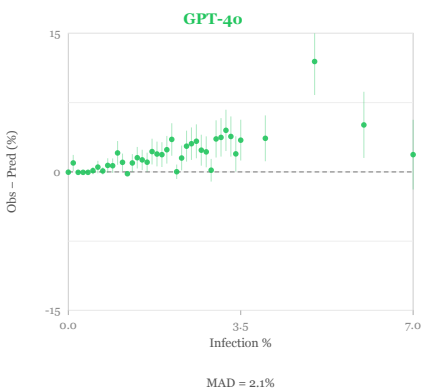
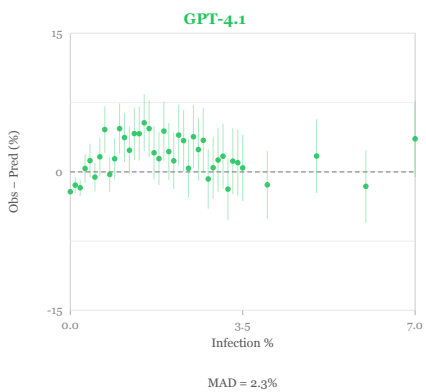
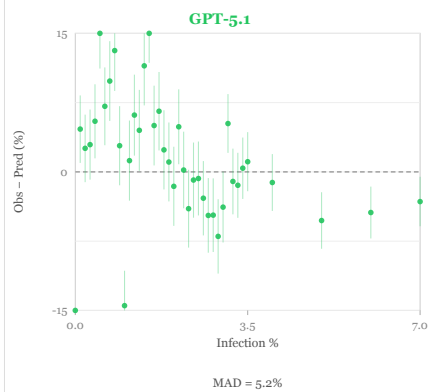
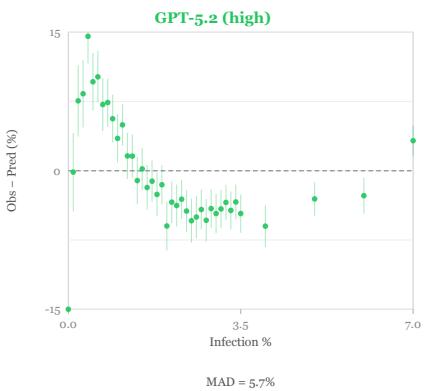
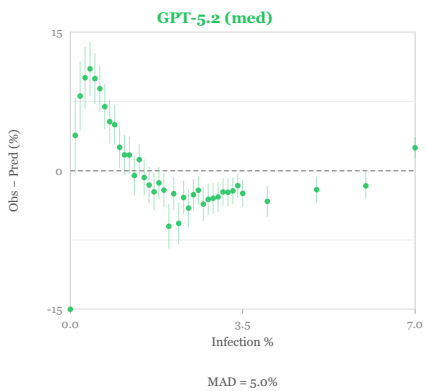
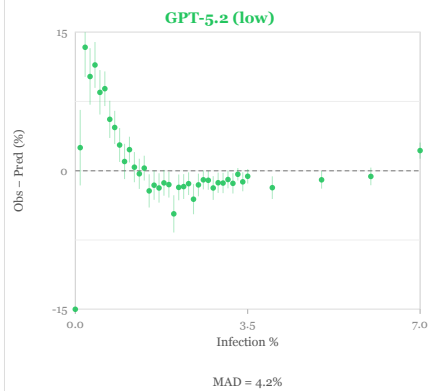
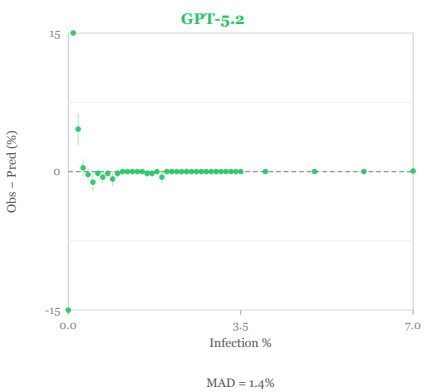
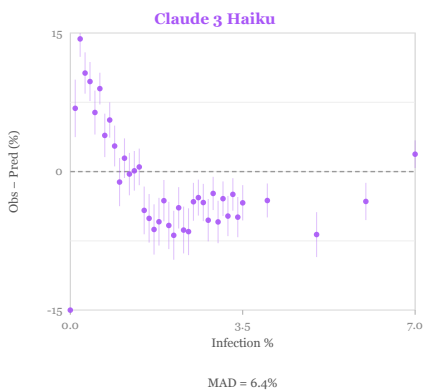
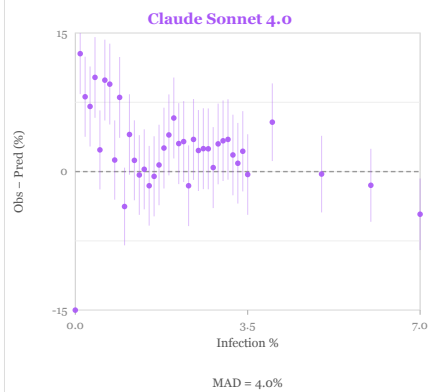
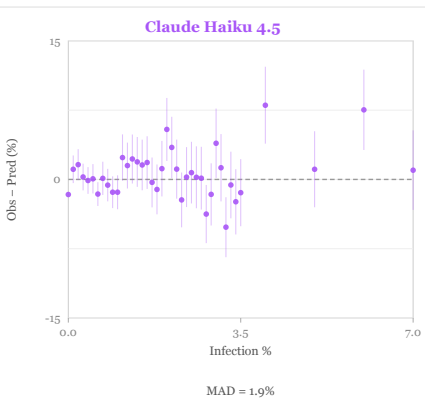
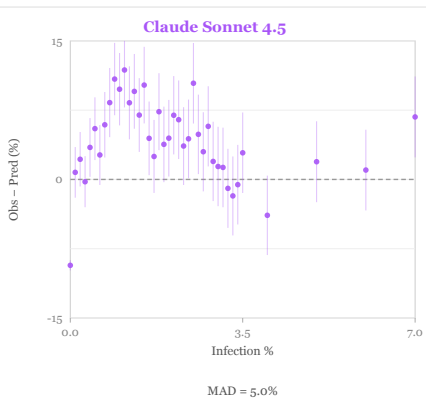
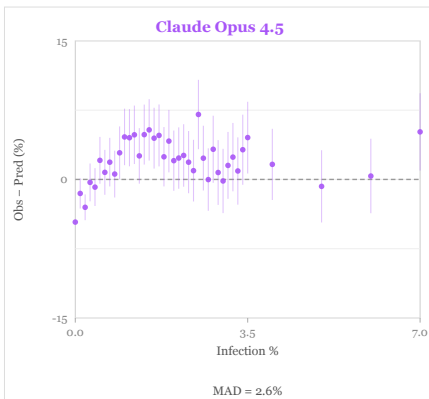
Observed stay-home rate minus predicted rate (as a percent) per bin, with 95% binomial CI whiskers. Reference line at 0. Points near 0 indicate the model captures the trend at that predictor value. MAD reported in each panel is the mean absolute deviation across all bins. (a) Residuals vs age; (b) Residuals vs infection.

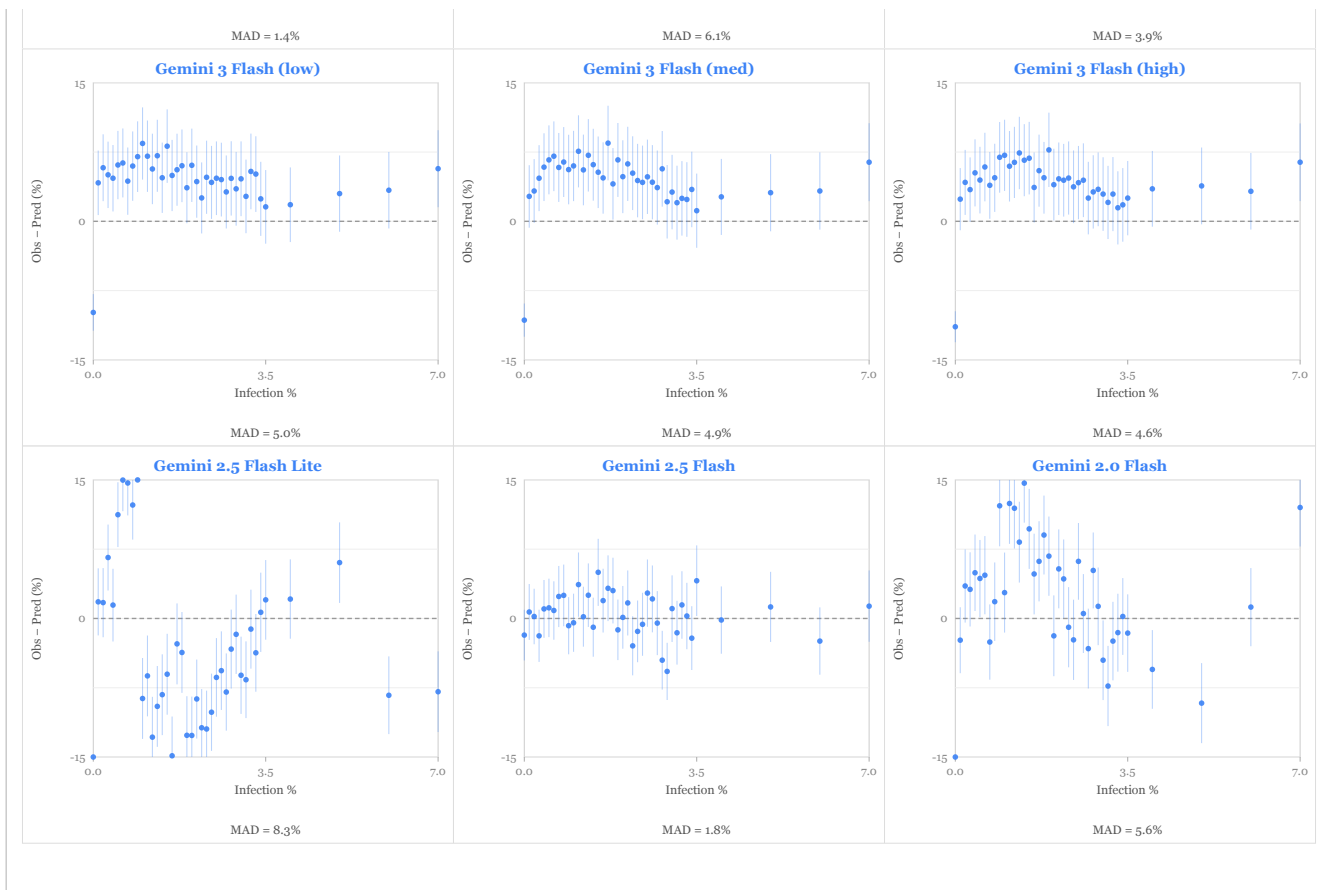
(a) Residuals vs age





(b) Residuals vs infection





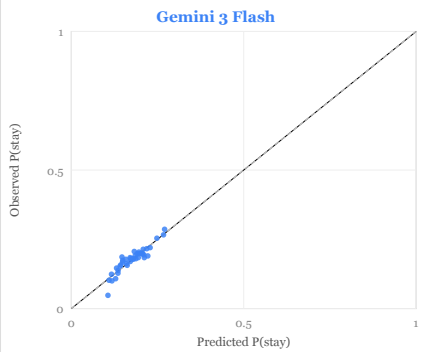
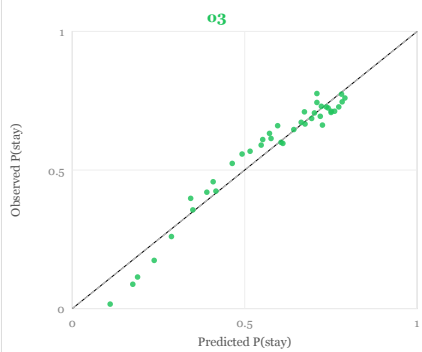
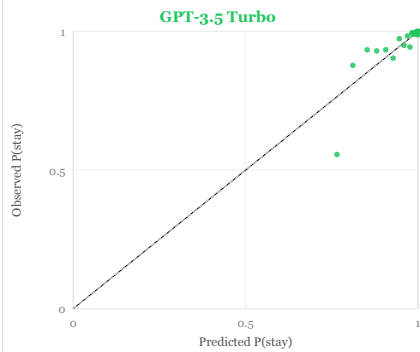
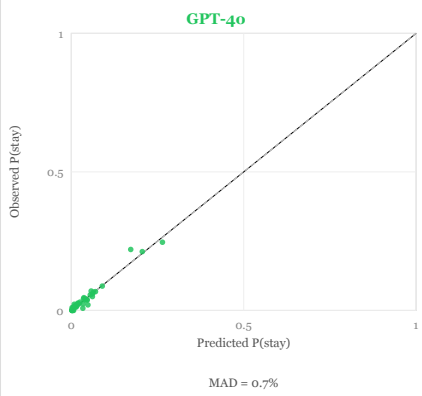
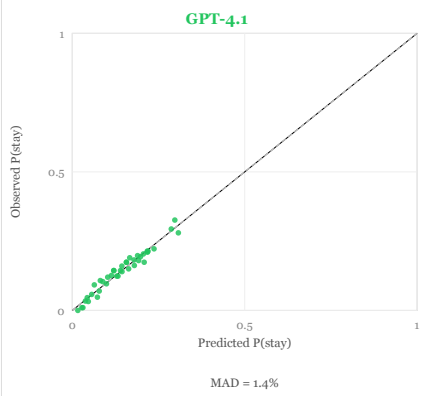
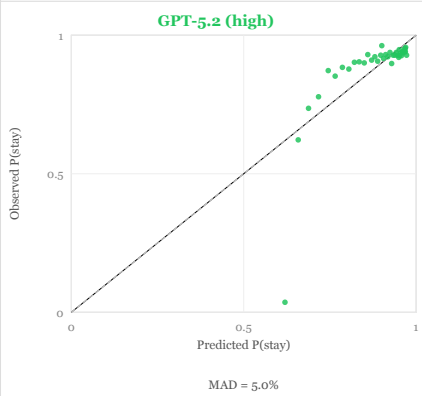
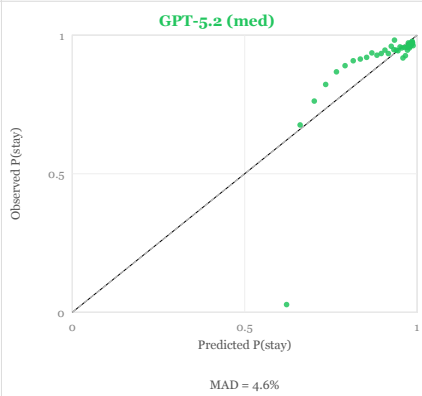
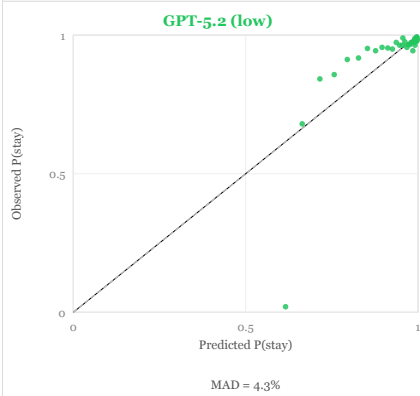
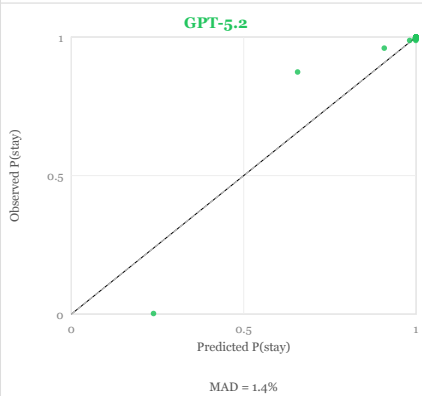
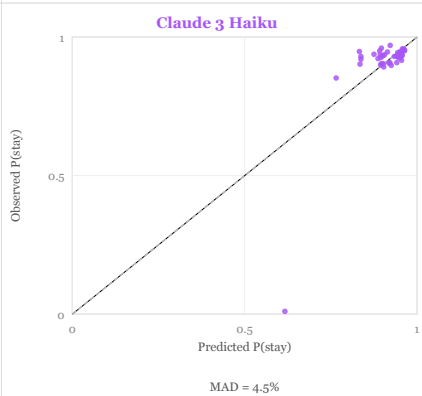
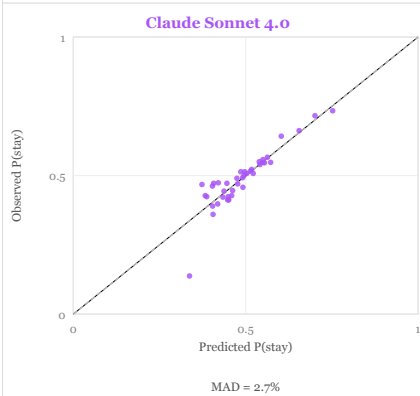
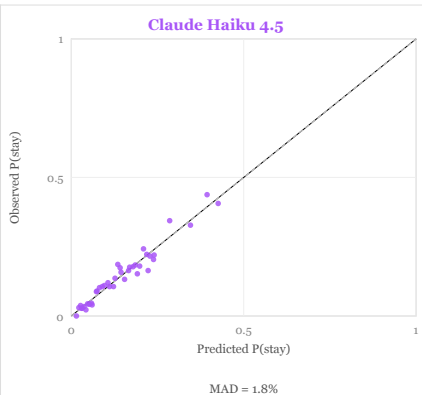
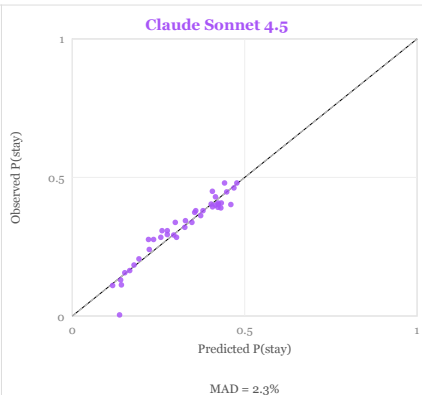
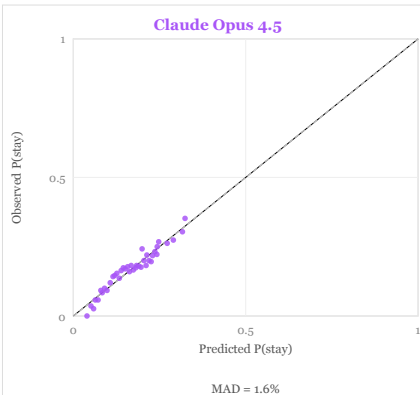
C.3 — RANDOM-EFFECTS LOGIT WITH MENTION FLAGS VALIDATION

Diagnostics for the random-effects logit with mention flags (`glmer(stay_home ~ ... + mention_flags + (1 | agent_id))`). Same diagnostic suite as C.2; the additional mention-flag terms test whether explicitly invoking a trait label or context shifts the agent's decision. Identification rests on the same assumptions as C.2 plus: no confounding between mention occurrence and unmodeled predictors. A well-validated specification supports the mention-shift claims in Figure 17; a poorly-validated specification would warrant flagging those claims as exploratory rather than inferential.

FIGURE C.3.1

Calibration — Mention-Flag Logit

Aggregated predicted vs observed stay-home rate across 40 infection bins. A well-calibrated model tracks the diagonal. MAD = mean absolute deviation, as a percent.



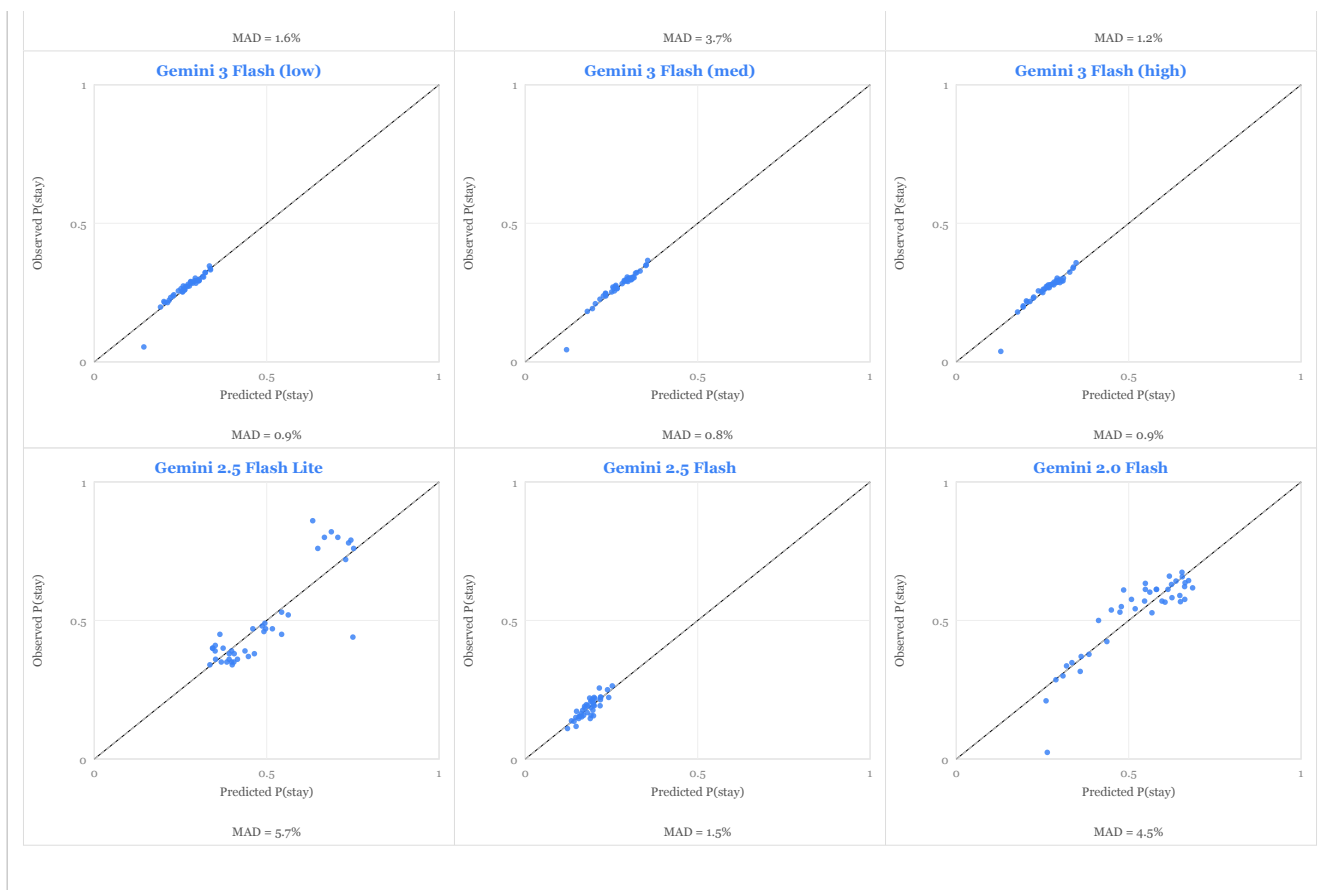


FIGURE C.3.2

DHARMA Residual Q-Q — Mention-Flag Logit

Scaled residuals from `DHARMA::simulateResiduals` (250 sims) vs theoretical $Uniform(0,1)$. Diagonal = model correctly specified. Dispersion ratio in the top-right of each panel summarizes the variance check.



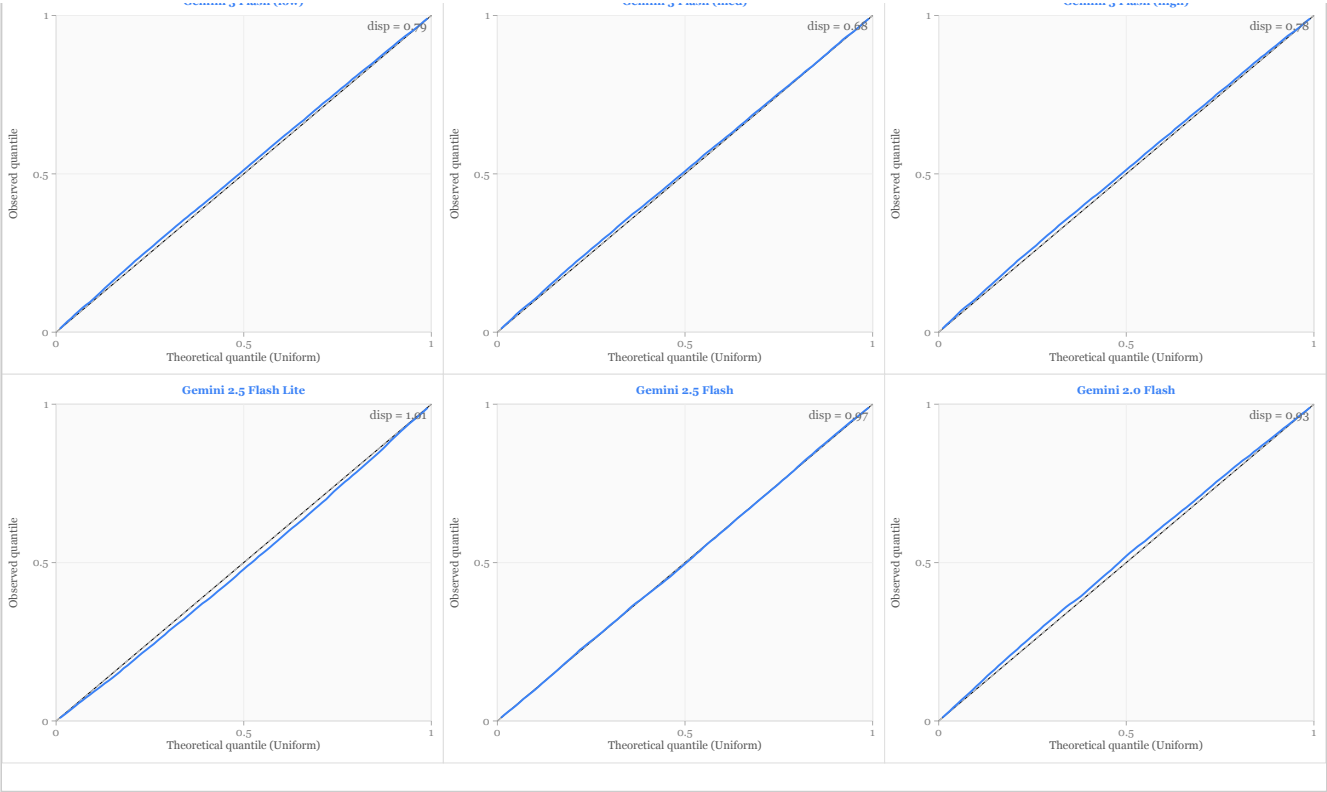
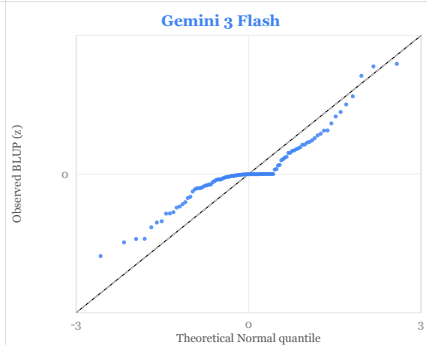
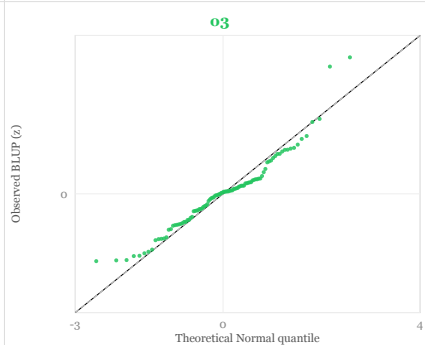
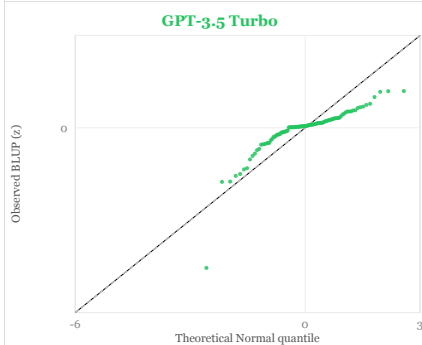
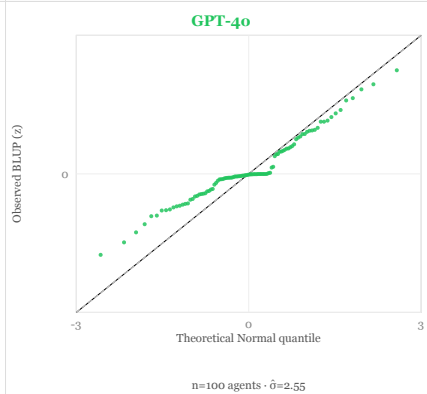
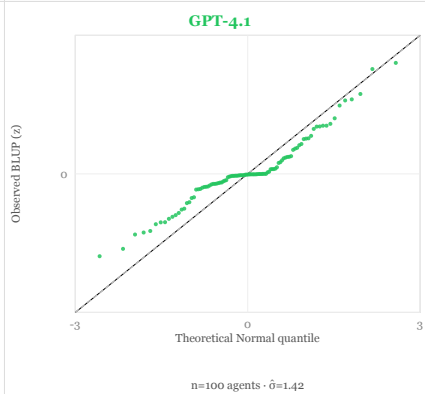
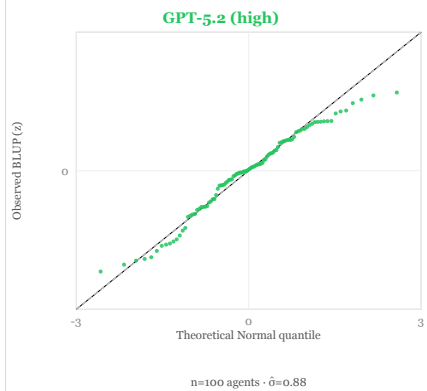
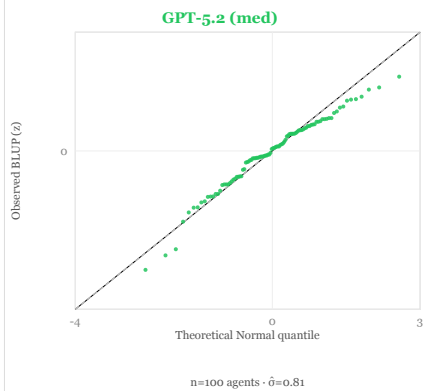
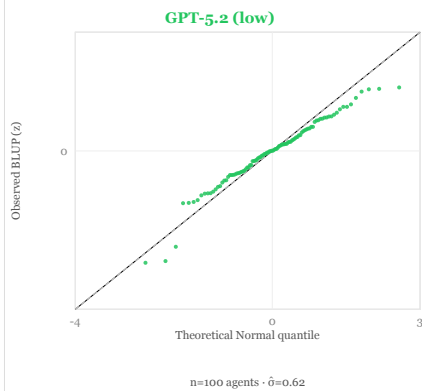
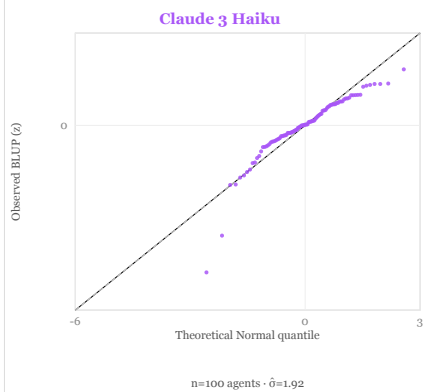
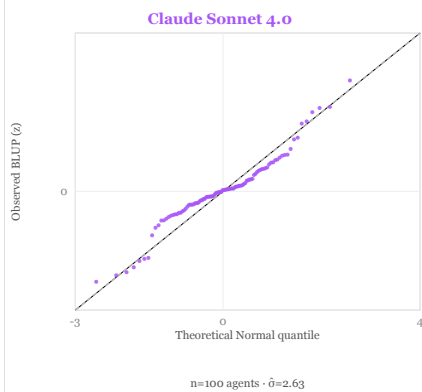
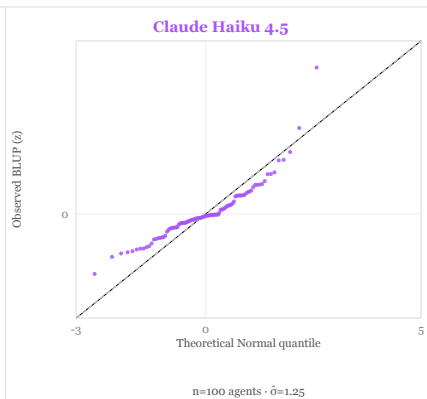
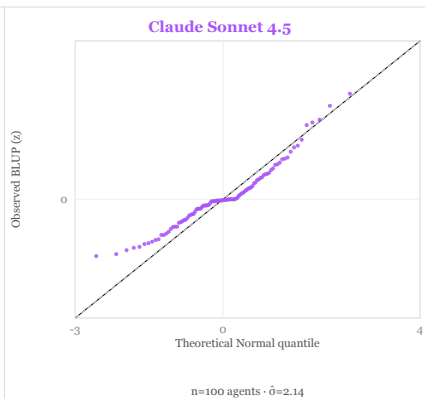
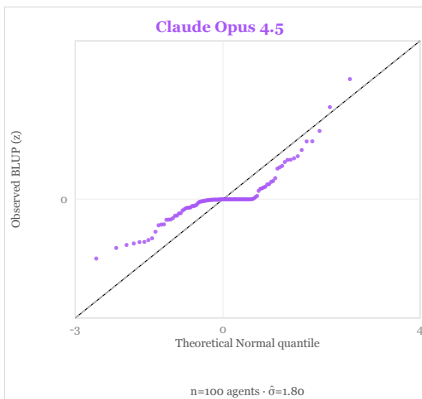


FIGURE C.3.3
Random-Intercept Normal Q-Q — Mention-Flag Logit

Sorted agent-level BLUPs vs theoretical Normal quantiles, standardized by the estimated σ . Straight line = Normal random-intercept assumption holds.



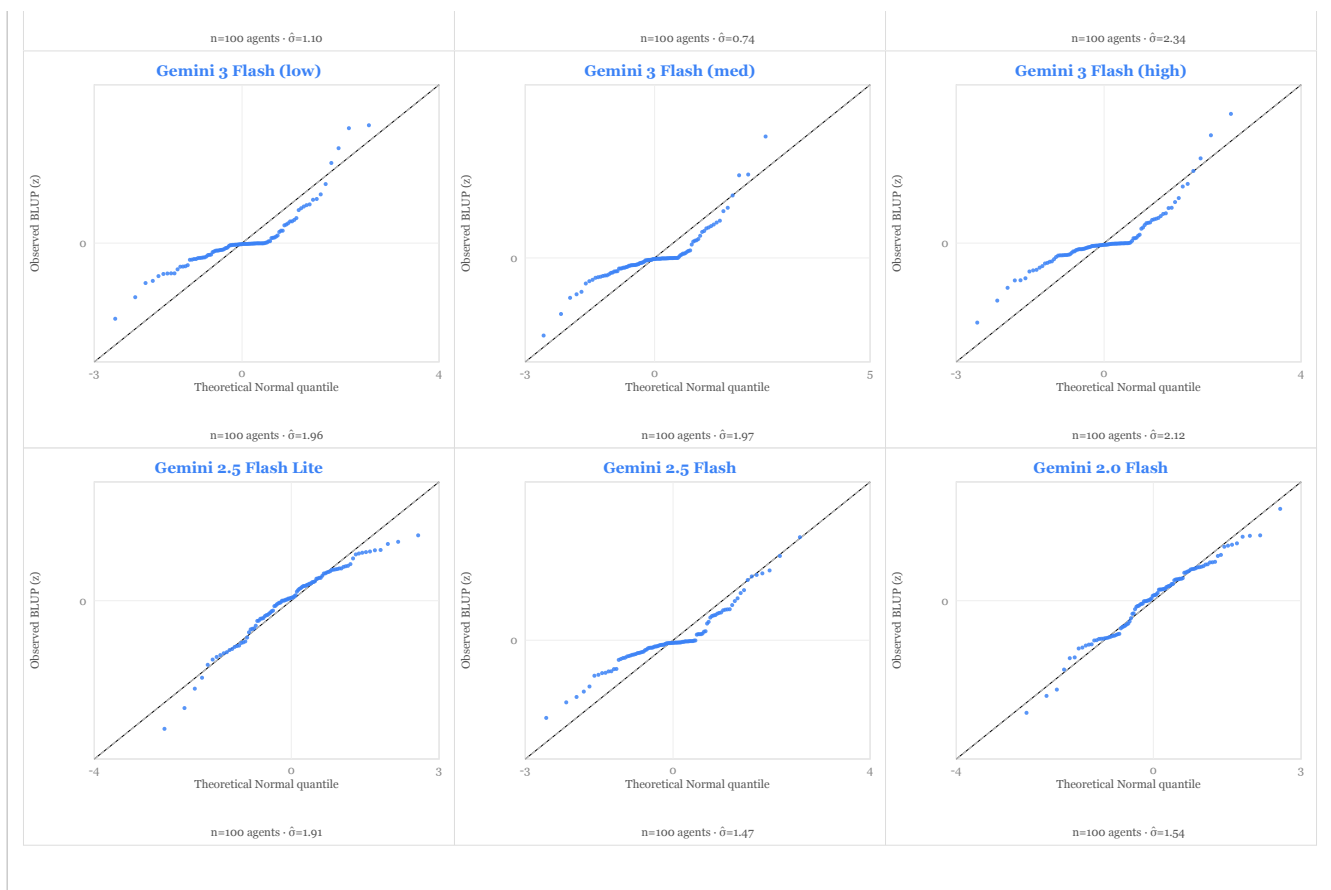


FIGURE C.3.4

BLUPs vs Predictors — Mention-Flag Logit

For each fixed-effect predictor, a 95% CI of the Δ in mean BLUP between groups (for dummies) or across the range (for age). All intervals overlapping 0 indicate random intercepts aren't absorbing systematic variation that belongs in the fixed effects.

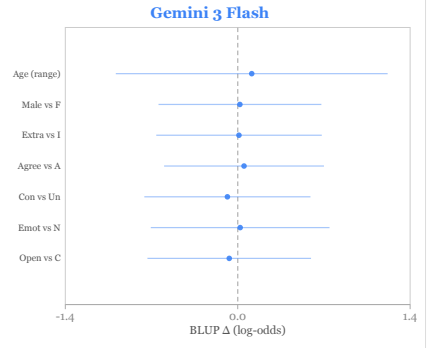
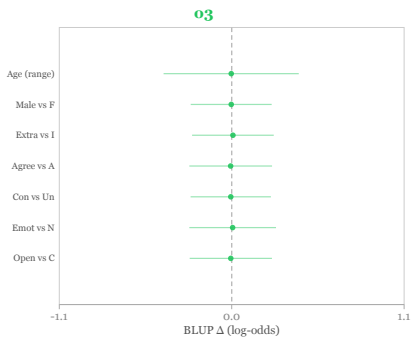
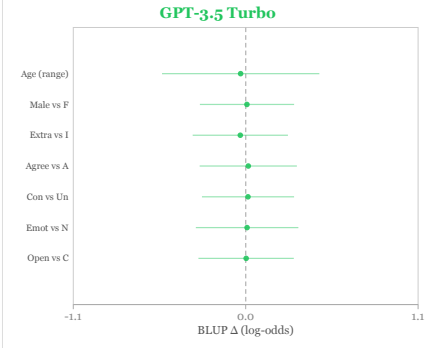
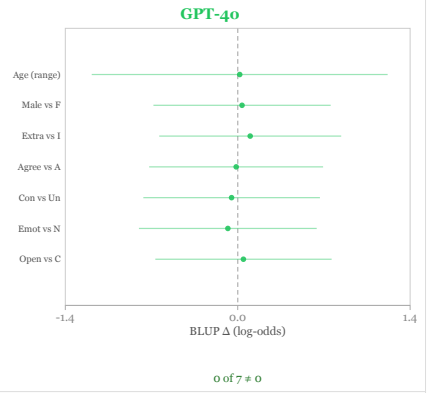
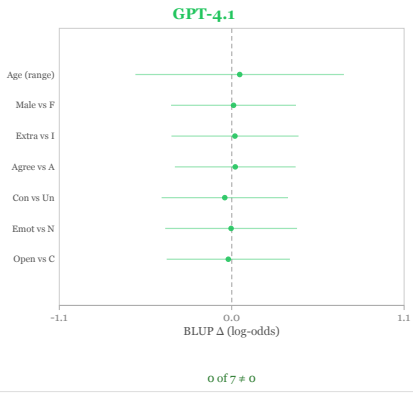
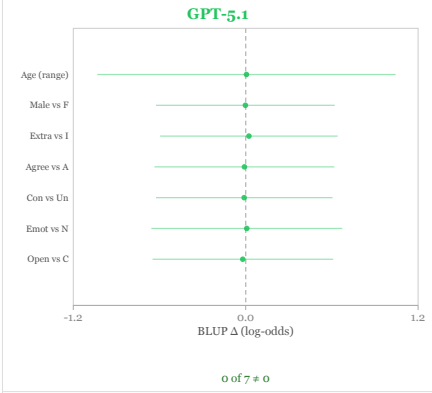
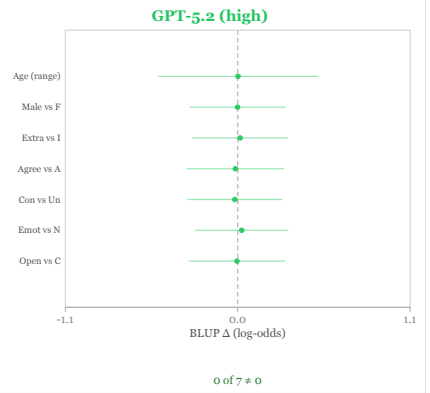
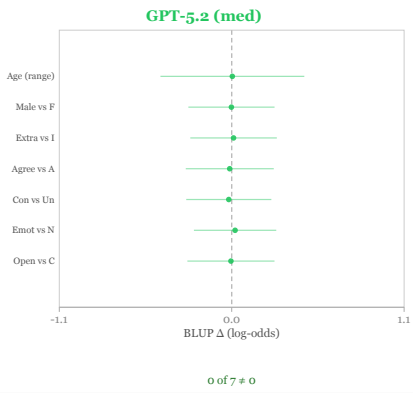
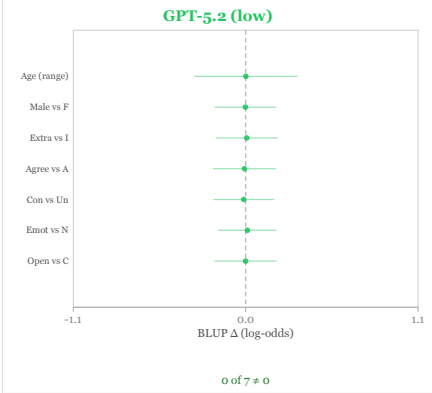
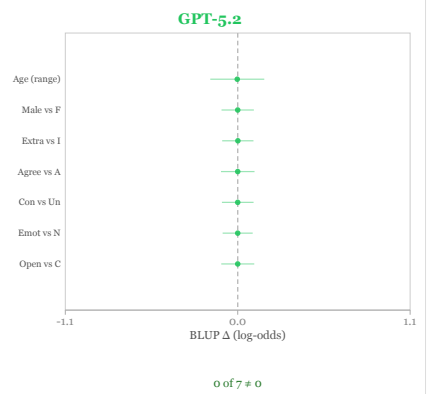
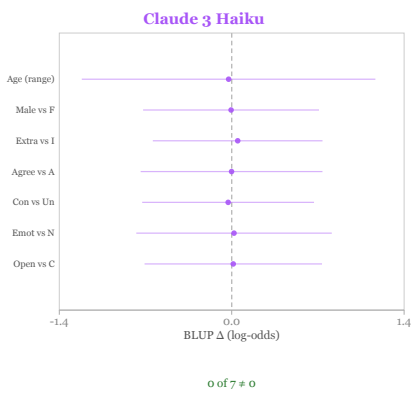
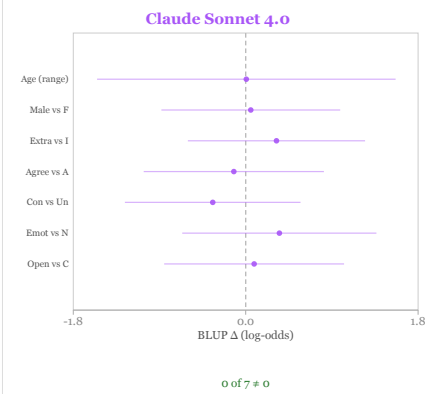
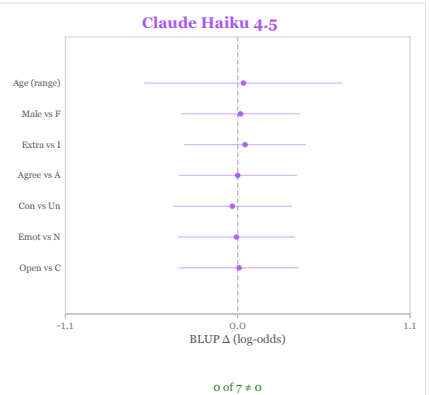
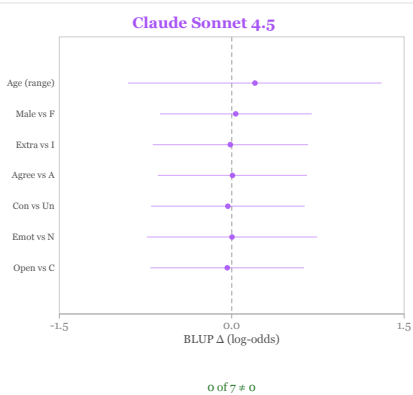
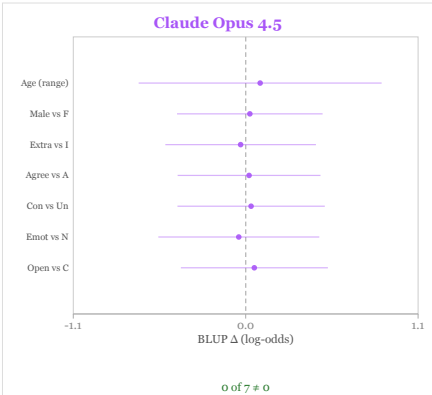
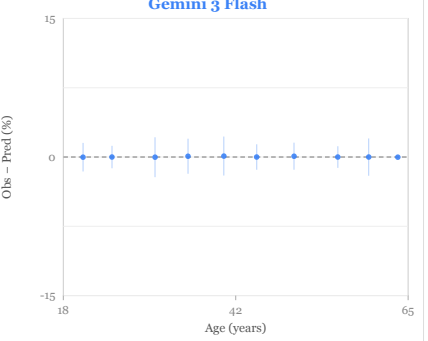
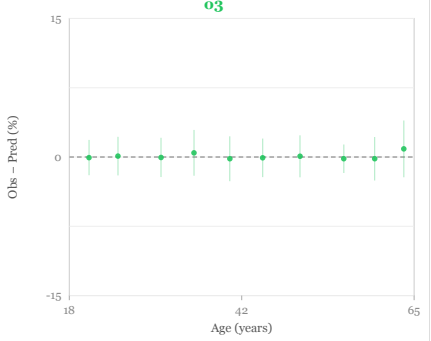
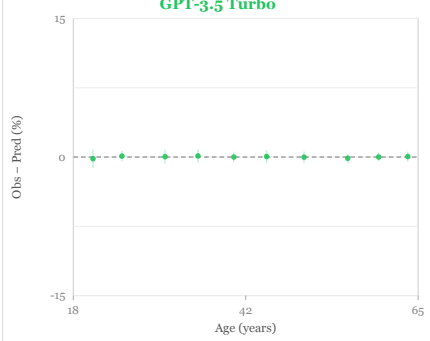
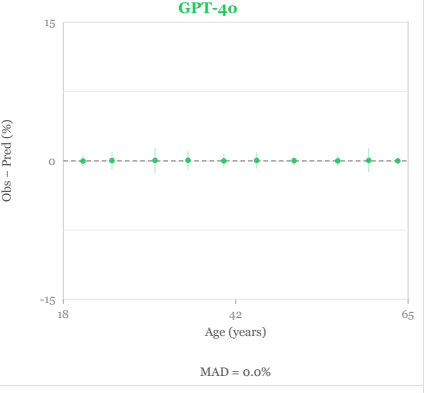
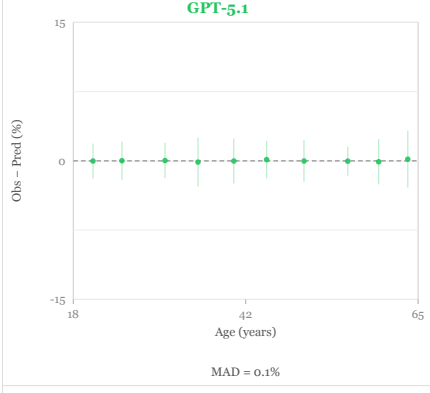
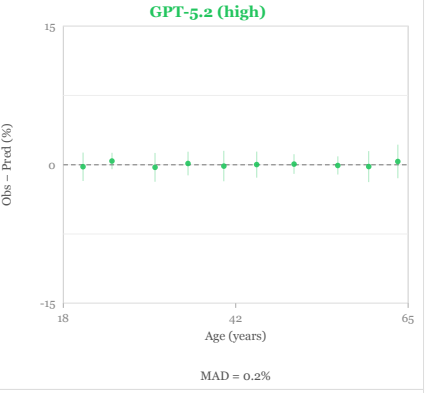
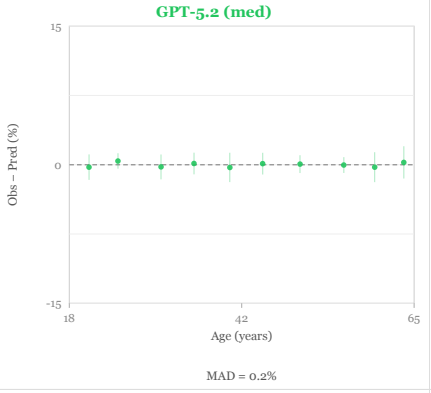
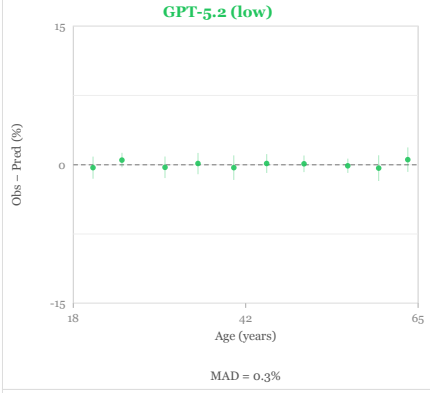
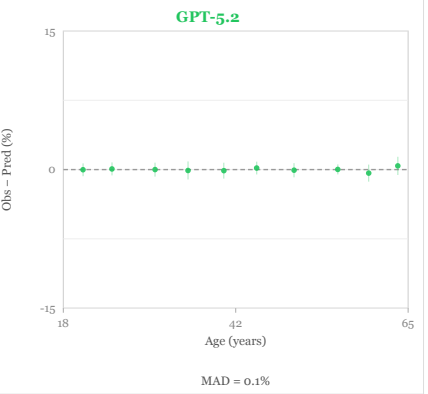
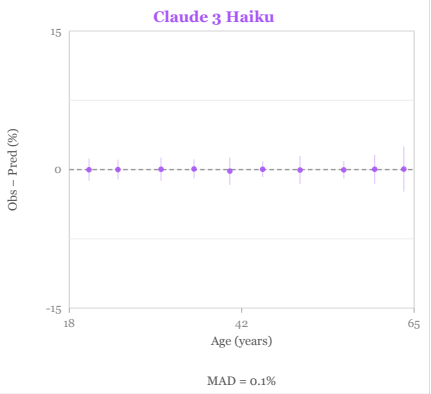
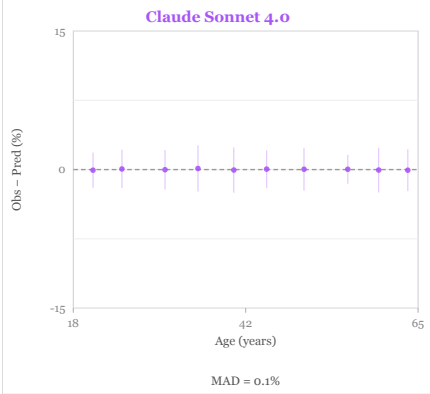
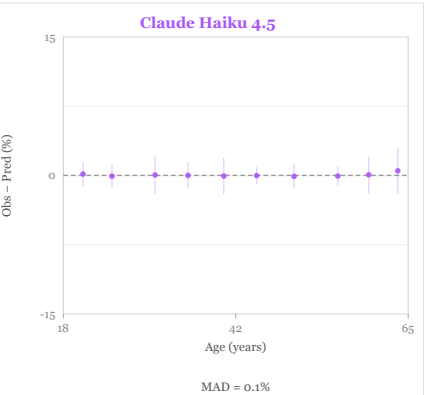
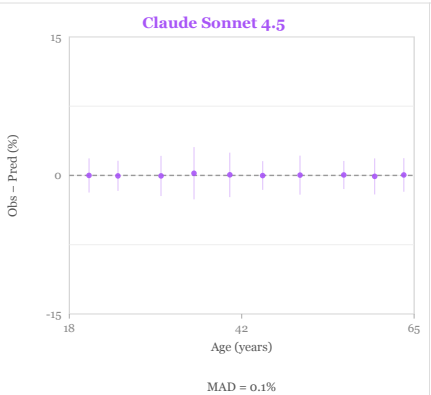
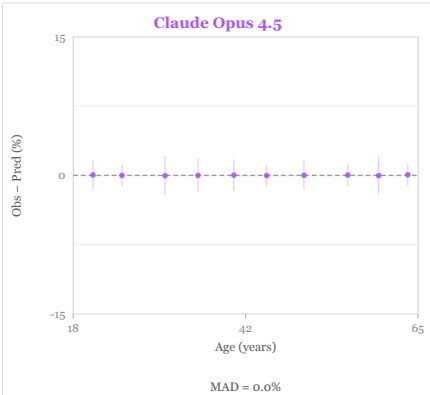


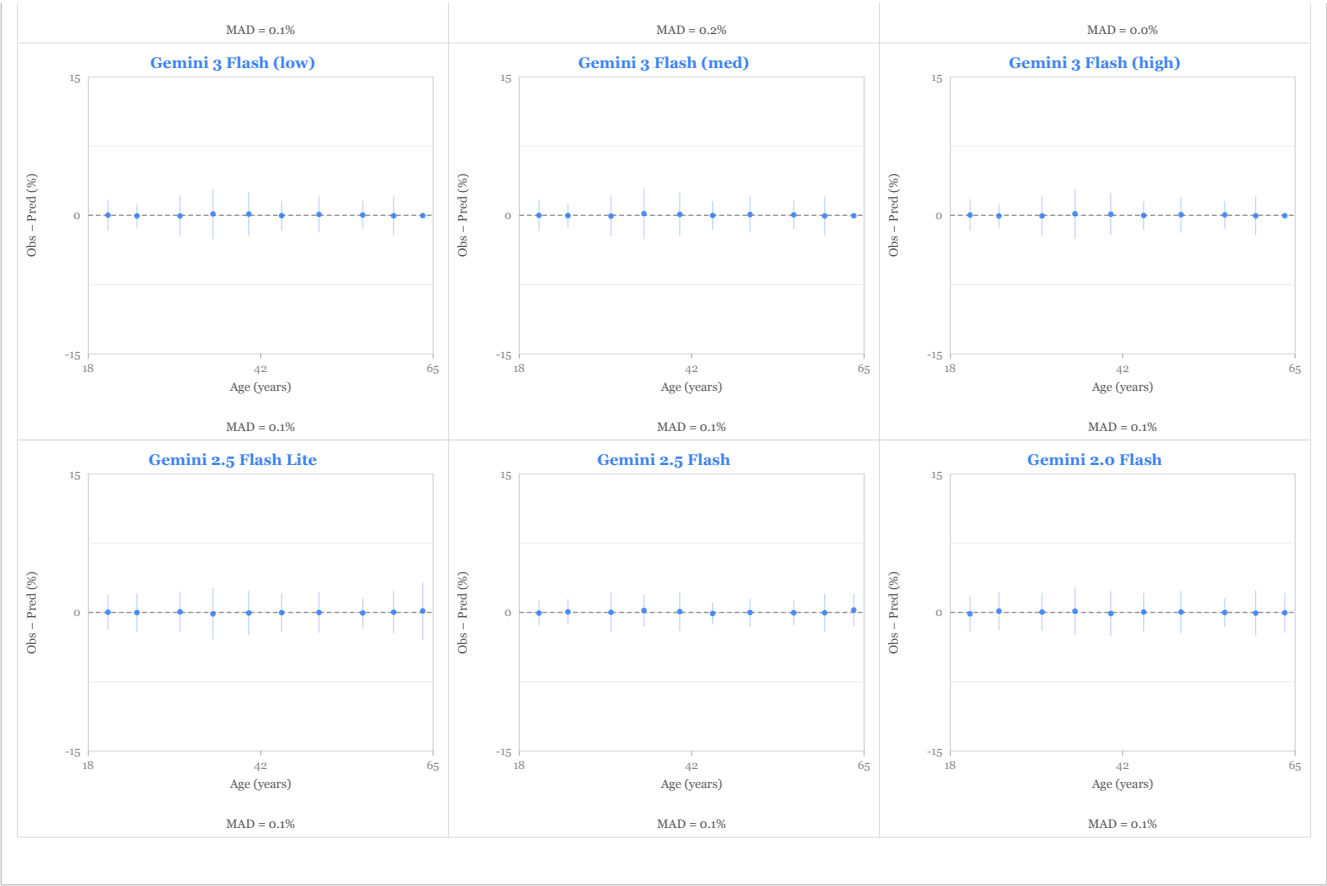


FIGURE C.3.5
Residuals vs Predictors — Mention-Flag Logit

Observed stay-home rate minus predicted rate (as a percent) per bin, with 95% binomial CI whiskers. Reference line at 0. Points near 0 indicate the model captures the trend at that predictor value. (a) Residuals vs age; (b) Residuals vs infection.

(a) Residuals vs age





(b) Residuals vs infection

